

Project Data Page:

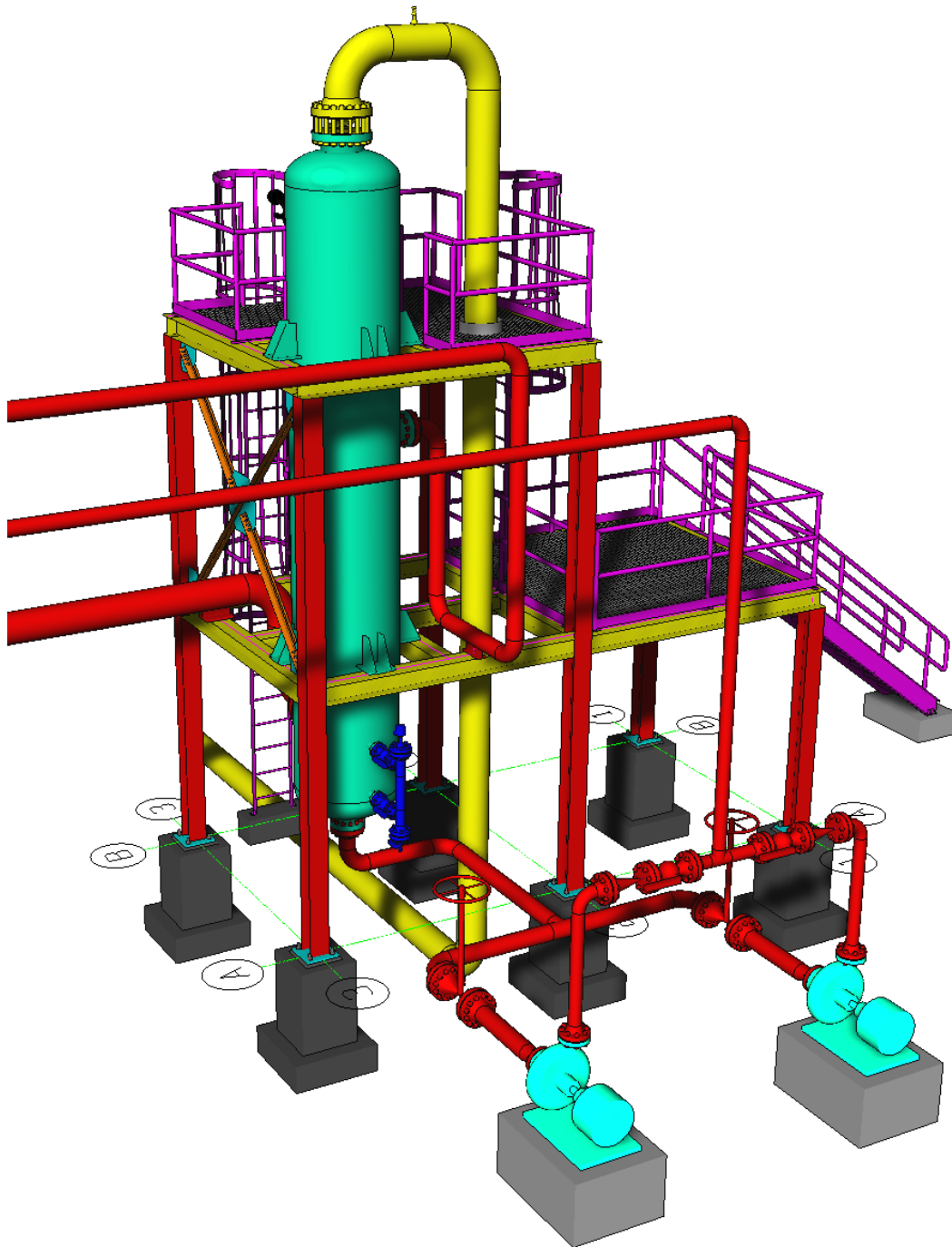




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Ammonia Convertor (105-D)

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : C:\Users\20384248\Downloads\1-105-D.pvdb

Date of Analysis : Sep 16, 2026 3:56pm

PV Elite 28 U1, March 2026



Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 3400.000 mm. ID
Vessel Design Length, Tangent to Tangent 35044.50 mm.
Specified Datum Line Distance 0.00 mm.
Internal Design Temperature 285 °C
Internal Design Pressure 184.720 bars
External Design Temperature 285 °C
Maximum Allowable Working Pressure 185 bars
Shop Test Pressure 288.625 bars
Required Minimum Design Metal Temperature -12.2 °C
Warmest Computed Minimum Design Metal Temperature 18.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	
Shell	SA-336 F11	3	...	K11572	
Head	SA-387 11	2	...	K11789	
Flange	SA-336 F11	3	...	K11572	
Nozzle	SA-182 F11	2	...	K11572	+IT
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F11	2	...	K11572	
Flg Bolting	SA-540 B23	3	<= 240	H43400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-387 11	2	...	K11789	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Left head	184.730	0.00	187.000	3.0000	N/A	No
Shell-1	184.730	0.00	186.000	3.0000	N/A	No
Shell-2	184.730	0.00	186.000	3.0000	N/A	No
Shell-3	184.730	0.00	186.000	3.0000	N/A	No
Shell-4	184.730	0.00	186.000	3.0000	N/A	No
Shell-5	184.730	0.00	186.000	3.0000	N/A	No
Shell-6	184.730	0.00	186.000	3.0000	N/A	No
Shell-7	184.730	0.00	186.000	3.0000	N/A	No
Shell-8	184.730	0.00	186.000	3.0000	N/A	No
Shell-10	184.730	0.00	186.000	3.0000	N/A	No
Shell-11	184.730	0.00	186.000	3.0000	N/A	No
Shell-12	184.730	0.00	186.000	3.0000	N/A	No
Flange-1	184.730	0.00	186.000	3.0000	N/A	No
Flange-2	184.730	0.00	186.000	3.0000	N/A	No
Shell-13	184.730	0.00	186.000	3.0000	N/A	No
Right head	184.730	0.00	187.000	3.0000	N/A	No



Liquid Level: 3500.00 mm. Dens.: 0.000 kg./cm³ Sp. Gr.: 0.030

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Sphere	0.0	0.0	124.0	105.0	103.8	...	1.00	1.00
Cylinder	3000.0	3000.0	206.0	206.0	204.5	...	1.00	1.00
Cylinder	6000.0	3000.0	206.0	206.0	204.5	...	1.00	1.00
Cylinder	9000.0	3000.0	206.0	206.0	204.5	...	1.00	1.00
Cylinder	12000.0	3000.0	206.0	206.0	204.5	...	1.00	1.00
Cylinder	15000.0	3000.0	206.0	206.0	204.5	...	1.00	1.00
Cylinder	18000.0	3000.0	206.0	206.0	204.5	...	1.00	1.00
Cylinder	21000.0	3000.0	206.0	206.0	204.5	...	1.00	1.00
Cylinder	24000.0	3000.0	206.0	206.0	204.5	...	1.00	1.00
Cylinder	27000.0	3000.0	206.0	206.0	204.5	...	1.00	1.00
Cylinder	30000.0	3000.0	206.0	206.0	204.5	...	1.00	1.00
Cylinder	33000.0	3000.0	206.0	206.0	204.5	...	1.00	1.00
Body Flg	33924.5	920.0	524.0	520.0	515.2	515.2	1.00	1.00
Body Flg	34844.5	920.0	524.0	520.0	515.2	515.2	1.00	1.00
Cylinder	35044.5	200.0	206.0	206.0	204.5	...	1.00	1.00
Sphere	35044.5	0.0	124.0	105.0	103.8	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Trunnion	WRC-107/537	0.036	n/a	Passed
Trunnion	WRC-107/537	0.044	n/a	Passed
A1_600	WRC-107/537	0.989	n/a	Passed
A2_450	WRC-107/537	0.989	n/a	Passed
B_600	WRC-107/537	0.989	n/a	Passed
C_350	WRC-107/537	0.989	n/a	Passed

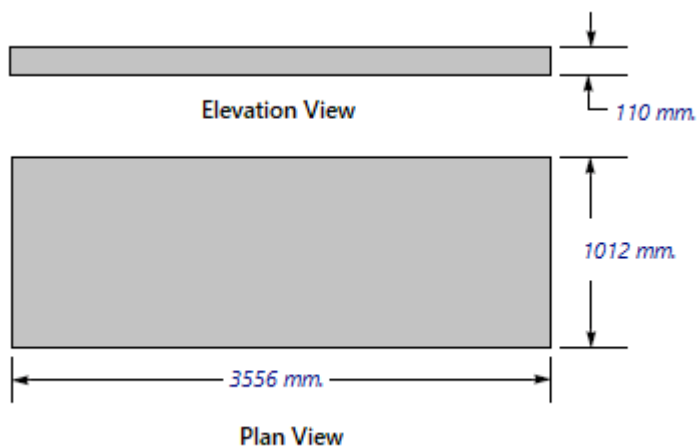
Foundation/Support Summary

Saddle Parameters:

Saddle Width 300.000 mm.
Saddle Bearing Angle 120.000 deg.
Centerline Dimension 2590.000 mm.
Wear Pad Width 350.000 mm.
Wear Pad Thickness 26.000 mm.
Wear Pad Bearing Angle 132.000 deg.
Distance from Saddle to Tangent 7455.000 mm.

Baseplate Length 3556.000 mm.
Baseplate Thickness 110.000 mm.
Baseplate Width 1012.000 mm.
Number of Ribs (including outside ribs) 9
Rib Thickness 30.000 mm.
Web Thickness 30.000 mm.
Height of Center Web 547.000 mm.
Number of Bolts in Baseplate 16

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	933030.44	Kgf
Maximum Transverse Saddle Shear Load	47655.05	Kgf
Maximum Longitudinal Saddle Shear Load	95310.10	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	971864.19	Kgf
Maximum Transverse Saddle Shear Load	2013.51	Kgf
Maximum Longitudinal Saddle Shear Load	782.17	Kgf

Weights:

Fabricated - Bare w/o removable internals	778636.6	kg.
Shop Test - Fabricated + water (full)	1119199.6	kg.
Shipping - Fab. + rem. intls.+ shipping app.	778636.6	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	795694.9	kg.
Empty - Fab. + intls. + details + wghts.	1495975.9	kg.
Operating - Empty + operating liquid (no ca)	1506192.6	kg.

**Nozzle Calculation Summary (bars & mm.):**

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
A1_600	185	...	...	OK	OK	Passed
A2_450	187	...	...	OK	OK	Passed
B_600	185	...	...	OK	OK	Passed
T3_100	187	...	...	...	OK	Passed
M1_600	186	...	...	OK	OK	Passed
M2_600	186	...	...	OK	OK	Passed
C_350	187	...	...	OK	OK	Passed
T1_100	187	...	...	...	OK	Passed
T2_100	187	...	...	...	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 185 Nozzle : B600
 Minimum MAWP Shells/Flanges : 185 Element : Flange-2
 Minimum MAPnc Shells/Flanges : 186 Element : Flange-2

Computed Vessel M.A.W.P. : 185 bars

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Inside Radius mm.
10	A1_600	0.000	0.000	307.800
10	A2_450	0.000	180.000	231.600
10	B_600	0.000	0.000	307.800
10	T3_100	0.000	0.000	53.800
20	M1_600	1500.000	0.000	307.800
120	M2_600	29300.004	0.000	307.800
170	C_350	0.000	0.000	180.800
170	T1_100	0.000	270.000	53.800
170	T2_100	0.000	90.000	53.800

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

4.5.13 spacing violation detected between : B600 and A1600.

The Centerline Dist. is 1699.59 while the sum of the inside Radii is 1755.43 mm..

4.5.13 spacing violation detected between : T3100 and A1600.

The Centerline Dist. is 635.47 while the sum of the inside Radii is 1156.11 mm..

4.5.13 spacing violation detected between : T3100 and B600.

The Centerline Dist. is 1064.12 while the sum of the inside Radii is 1156.11 mm..

4.5.13 spacing violation detected between : T1100 and C350.

The Centerline Dist. is 774.98 while the sum of the inside Radii is 1017.80 mm..

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
T3_100	4.000 in	Actual	RTJWn	4.945	12.000	...	...	913.35	...
T1_100	4.000 in	Actual	RTJWn	5.260	16.000	...	...	1008.42	1500
T2_100	4.000 in	Actual	RTJWn	4.945	12.000	...	...	1174.75	1500
C_350	14.000 in	Actual	None	16.362	30.000	...	...	946.26	...
A2_450	18.000 in	Actual	None	20.835	36.000	...	...	940.06	...
A1_600	24.000 in	Actual	None	27.622	46.000	...	...	965.11	...
B_600	24.000 in	Actual	None	27.622	46.000	...	...	965.11	...
M1_600	24.000 in	Actual	RTJWn	27.780	48.000	...	...	1102.57	1500
M2_600	24.000 in	Actual	RTJWn	27.780	48.000	...	...	1102.57	1500

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
T3_100	SA-182 F11	102.000	10.000	...	...	...
T1_100	SA-182 F11	102.000	10.000	...	...	...
T2_100	SA-182 F11	102.000	10.000	...	...	...
C_350	SA-182 F11	102.000	10.000	...	...	...
A2_450	SA-182 F11	102.000	10.000	...	...	...
A1_600	SA-182 F11	102.000	10.000	...	...	...
B_600	SA-182 F11	102.000	10.000	...	...	...
M1_600	SA-182 F11	200.000	10.000	...	...	...
M2_600	SA-182 F11	200.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
T3_100	...	0.0	800.00	0.00	Left head
T1_100	...	270.0	800.00	0.00	Right head
T2_100	...	90.0	800.00	0.00	Right head
C_350	...	0.0	800.00	0.00	Right head
A2_450	...	180.0	800.00	0.00	Left head
A1_600	...	0.0	800.00	0.00	Left head
B_600	...	0.0	800.00	0.00	Left head
M1_600	1500.000	0.0	800.00	0.00	Shell-1
M2_600	29300.002	0.0	800.00	0.00	Shell-11

**Bill of Materials:**

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 124.0mm. THK X 3501.0mm. ID X 0.0mm.	SA-387 11
11	CYLINDER: 206.0mm. THK X 3400.0mm. ID X 3000.0mm.	SA-387 11
2	WELD NECK FLANGE: 524.0mm. THK X 4700.0mm. OD	SA-336 F11
1	CYLINDER: 206.0mm. THK X 3400.0mm. ID X 200.0mm.	SA-336 F11
2	INSULATION: 1750mm X 100mm THK	
11	INSULATION: 3000mm X 100mm THK	
2	SADDLE: 300.0mm X 120 DEG	SA-387 11
8	PACKING: 3000.000mm	...
4	TRUNNIONS:	SA-387 11
4	TRUNNION PADS: 1600.0mm OD X 1200.0mm ID X 50.0mm THK	
2	INSULATION: 920mm X 100mm THK	
1	INSULATION: 200mm X 100mm THK	
1	GASKET: 3440.0mm. OD X 3410.0mm. ID	...
44	BODY FLANGE BOLTS: 170.0mm. DIA	SA-540 B23
88	NUTS FOR BODY FLANGE BOLTS: 170.0mm. DIA	...
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm.	E*
Notes							
Flange-1 Bolting	[11][17]	Impact Testing reqd		Yes			
	[17]			No			Impact Testing Required
Flange-2	[11][17]	Impact Testing reqd		Yes			
Left head	[9][17]	Impact Testing reqd		Yes			
Shell-1	[8][17]	Impact Testing reqd		Yes			
Shell-2	[8][17]	Impact Testing reqd		Yes			
Shell-3	[8][17]	Impact Testing reqd		Yes			
Shell-4	[8][17]	Impact Testing reqd		Yes			
Shell-5	[8][17]	Impact Testing reqd		Yes			
Shell-6	[8][17]	Impact Testing reqd		Yes			
Shell-7	[8][17]	Impact Testing reqd		Yes			
Shell-8	[8][17]	Impact Testing reqd		Yes			
Shell-10	[8][17]	Impact Testing reqd		Yes			
Shell-11	[8][17]	Impact Testing reqd		Yes			
Shell-12	[8][17]	Impact Testing reqd		Yes			
Shell-13	[8][17]	Impact Testing reqd		Yes			
Right head	[9][17]	Impact Testing reqd		Yes			
New Saddle	[24]	A	18	18	Yes	1.000	26.000 1.00
New Saddle	[24]	A	18	18	Yes	1.000	26.000 1.00
A1600	[1]			Yes			
A2450	[1]			Yes			
B600	[1]			Yes			
T3100	[1]	A	3	-18	Yes	0.631	12.000 1.00
M1600	[1]	A	8	-7	Yes	0.722	48.000 1.00
Nozzle Flg	[5]	B	-18	-34	Yes		
M2600	[1]	A	29	14	Yes	0.722	48.000 1.00
Nozzle Flg	[5]	A	-18	-34	Yes		
C350	[1]			Yes			
T1100	[1]	A	-13	-61	Yes	0.437	16.000 1.00
Nozzle Flg	[5]	B	-18	-34	Yes		
T2100	[1]	A	-18	-39	Yes	0.631	12.000 1.00
Nozzle Flg	[5]	B	-18	-34	Yes		
Warmest MDMT:		29	18				
Required Minimum Design Metal Temperature					-12.2	°C	
Warmest Computed Minimum Design Metal Temperature					18.0	°C	

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld





Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Ammonia Convertor (105-D)

Design Internal Pressure (for Hydrotest)	184.72	bars
Design Internal Temperature	285.0	°C
Projection of Nozzle from Vessel Top	500	mm.
Projection of Nozzle from Vessel Bottom	500	mm.
Minimum Design Metal Temperature	-12.2	°C
Type of Construction	Welded	
Special Service	Lethal	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW



Load Case 19	FS+BS+EP+OW
Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.
Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	0.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	C
Component Elevation Ratio z/h	1.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000
M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	No
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/16
Element From Node	10
Element To Node	20
Element Type	Spherical
Description	Left head
Distance "FROM" to "TO"	0 mm.
Inside Diameter	3501 mm.
Element Thickness	105 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	124 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	184.72 bars



Design Temperature Internal Pressure	285	°C
Design External Pressure	0	bars
Design Temperature External Pressure	285	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Allowable Stress, Ambient	207	N./mm ²
Allowable Stress, Operating	165.2	N./mm ²
Allowable Stress, Hydrotest	294.49	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm.
Yield Stress, Operating	247.8	N./mm ²
MDMT Curve	A	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	3500	mm.
Liquid Density	0.29952E-04	kg./cm ³

Element From Node	10	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	-1750.5	mm.
Height/Length of Insulation	1750.5	mm.
Thickness of Insulation	100	mm.
Density	0.000352	kg./cm ³

Element From Node	10	
Detail Type	Nozzle	
Detail ID	A1600	
Dist. from "FROM" Node / Offset dist	1445.7	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-182 F11 +IT	

Element From Node	10	
Detail Type	Nozzle	
Detail ID	A2450	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	18	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-182 F11 +IT	

Element From Node	10	
Detail Type	Nozzle	
Detail ID	B600	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	0.0	



Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange None
 Nozzle Matl SA-182 F11 +IT

Element From Node 10
 Detail Type Nozzle
 Detail ID T3100
 Dist. from "FROM" Node / Offset dist 1000 mm.
 Nozzle Diameter 4 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange None
 Nozzle Matl SA-182 F11 +IT

Element From Node 10
 Detail Type Weight
 Detail ID Cutback wt
 Dist. from "FROM" Node / Offset dist 0 mm.
 Miscellaneous Weight 5500 Kgf
 Offset from Element Centerline 0 mm.

Element# 2/16
 Element From Node 20
 Element To Node 30
 Element Type Cylinder
 Description Shell-1
 Distance "FROM" to "TO" 3000 mm.
 Inside Diameter 3400 mm.
 Element Thickness 206 mm.
 Internal Corrosion Allowance 3 mm.
 Nominal Thickness 206 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 184.72 bars
 Design Temperature Internal Pressure 285 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 285 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-387 11
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 20
 Detail Type Liquid
 Detail ID Liquid 20
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 3400 mm.
 Liquid Density 0.29950E-04 kg./cm³

Element From Node 20
 Detail Type Insulation
 Detail ID Ins: 10
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Insulation 3000 mm.
 Thickness of Insulation 100 mm.
 Density 0.000352 kg./cm³

Element From Node 20
 Detail Type Nozzle
 Detail ID M1600
 Dist. from "FROM" Node / Offset dist 1500 mm.
 Nozzle Diameter 24 in.
 Nozzle Schedule None



Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-182 F11 +IT

Element From Node	20
Detail Type	Weight
Detail ID	manway davit-1
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	60 Kgf
Offset from Element Centerline	0 mm.

Element#	3/16
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	3400 mm.
Element Thickness	206 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	206 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	184.72 bars
Design Temperature Internal Pressure	285 °C
Design External Pressure	0 bars
Design Temperature External Pressure	285 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3400 mm.
Liquid Density	0.29952E-04 kg./cm ³

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.
Thickness of Insulation	100 mm.
Density	0.000352 kg./cm ³

Element From Node	30
Detail Type	Weight
Detail ID	MTI 1 2 3 4 5 6
Dist. from "FROM" Node / Offset dist	1352 mm.
Miscellaneous Weight	30 Kgf
Offset from Element Centerline	0 mm.

Element#	4/16
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	3400 mm.
Element Thickness	206 mm.



Internal Corrosion Allowance	3	mm.
Nominal Thickness	206	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	184.72	bars
Design Temperature Internal Pressure	285	°C
Design External Pressure	0	bars
Design Temperature External Pressure	285	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	1455	mm.
Width of Saddle	300	mm.
Height of Saddle at Bottom	2590	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	350	mm.
Thickness of Wear Plate	26	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.

Element From Node	40	
Detail Type	Packing	
Detail ID	Pack: [1 of 1]	
Dist. from "FROM" Node / Offset dist	0	mm.
Height of Packed Section	3000	mm.
Density	0.00321	kg./cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	3400	mm.
Liquid Density	0.29952E-04	kg./cm ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	3000	mm.
Thickness of Insulation	100	mm.
Density	0.000352	kg./cm ³

Element#	5/16	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Shell-4	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	3400	mm.
Element Thickness	206	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	206	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	184.72	bars



Design Temperature Internal Pressure	285	°C
Design External Pressure	0	bars
Design Temperature External Pressure	285	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Packing	
Detail ID	Pack: [1 of 1]	
Dist. from "FROM" Node / Offset dist	0	mm.
Height of Packed Section	3000	mm.
Density	0.00321	kg./cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid 50	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	3400	mm.
Liquid Density	0.29950E-04	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	3000	mm.
Thickness of Insulation	100	mm.
Density	0.000352	kg./cm ³

Element From Node	50	
Detail Type	Weight	
Detail ID	MTI 7 8 9 10 11	
Dist. from "FROM" Node / Offset dist	2456	mm.
Miscellaneous Weight	30	Kgf
Offset from Element Centerline	0	mm.

Element#	6/16	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	Shell-5	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	3400	mm.
Element Thickness	206	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	206	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	184.72	bars
Design Temperature Internal Pressure	285	°C
Design External Pressure	0	bars
Design Temperature External Pressure	285	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Packing	
Detail ID	Pack: [1 of 1]	
Dist. from "FROM" Node / Offset dist	0	mm.
Height of Packed Section	3000	mm.
Density	0.00321	kg./cm ³



Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.0
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3400 mm.
Liquid Density	0.29950E-04 kg./cm ³
Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.
Thickness of Insulation	100 mm.
Density	0.000352 kg./cm ³

Element#	7/16
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	Shell-6
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	3400 mm.
Element Thickness	206 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	206 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	184.72 bars
Design Temperature Internal Pressure	285 °C
Design External Pressure	0 bars
Design Temperature External Pressure	285 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Packing
Detail ID	Pack: [1 of 1]
Dist. from "FROM" Node / Offset dist	0 mm.
Height of Packed Section	3000 mm.
Density	0.00321 kg./cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.0

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid 70
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3400 mm.
Liquid Density	0.29952E-04 kg./cm ³

Element From Node	70
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.
Thickness of Insulation	100 mm.
Density	0.000352 kg./cm ³

Element From Node	70
Detail Type	Weight
Detail ID	clip weight
Dist. from "FROM" Node / Offset dist	0 mm.



Miscellaneous Weight	300	Kgf
Offset from Element Centerline	0	mm.

Element#	8/16	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Shell-7	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	3400	mm.
Element Thickness	206	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	206	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	184.72	bars
Design Temperature Internal Pressure	285	°C
Design External Pressure	0	bars
Design Temperature External Pressure	285	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Packing	
Detail ID	Pack: [1 of 1]	
Dist. from "FROM" Node / Offset dist	0	mm.
Height of Packed Section	3000	mm.
Density	0.00321	kg./cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid 80	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	3400	mm.
Liquid Density	0.29952E-04	kg./cm ³

Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	3000	mm.
Thickness of Insulation	100	mm.
Density	0.000352	kg./cm ³

Element From Node	80	
Detail Type	Weight	
Detail ID	Misc weight	
Dist. from "FROM" Node / Offset dist	1312	mm.
Miscellaneous Weight	10000	Kgf
Offset from Element Centerline	0	mm.

Element#	9/16	
Element From Node	90	
Element To Node	110	
Element Type	Cylinder	
Description	Shell-8	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	3400	mm.
Element Thickness	206	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	206	mm.



External Corrosion Allowance	0	mm.
Design Internal Pressure	184.72	bars
Design Temperature Internal Pressure	285	°C
Design External Pressure	0	bars
Design Temperature External Pressure	285	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	90	
Detail Type	Packing	
Detail ID	Pack: [1 of 1]	
Dist. from "FROM" Node / Offset dist	0	mm.
Height of Packed Section	3000	mm.
Density	0.00321	kg./cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	
Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid 90	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	3400	mm.
Liquid Density	0.29952E-04	kg./cm ³
Element From Node	90	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	3000	mm.
Thickness of Insulation	100	mm.
Density	0.000352	kg./cm ³
Element From Node	90	
Detail Type	Weight	
Detail ID	MTI 13 14 15 16	
Dist. from "FROM" Node / Offset dist	1714	mm.
Miscellaneous Weight	30	Kgf
Offset from Element Centerline	0	mm.

Element#	10/16	
Element From Node	110	
Element To Node	120	
Element Type	Cylinder	
Description	Shell-10	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	3400	mm.
Element Thickness	206	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	206	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	184.72	bars
Design Temperature Internal Pressure	285	°C
Design External Pressure	0	bars
Design Temperature External Pressure	285	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	110	
Detail Type	Packing	
Detail ID	Pack: [1 of 1]	
Dist. from "FROM" Node / Offset dist	0	mm.



Height of Packed Section	3000	mm.
Density	0.00321	kg./cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	110	
Detail Type	Liquid	
Detail ID	Liquid 110	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	3400	mm.
Liquid Density	0.29952E-04	kg./cm ³

Element From Node	110	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	3000	mm.
Thickness of Insulation	100	mm.
Density	0.000352	kg./cm ³

Element#	11/16	
Element From Node	120	
Element To Node	130	
Element Type	Cylinder	
Description	Shell-11	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	3400	mm.
Element Thickness	206	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	206	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	184.72	bars
Design Temperature Internal Pressure	285	°C
Design External Pressure	0	bars
Design Temperature External Pressure	285	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	120	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	755	mm.
Width of Saddle	300	mm.
Height of Saddle at Bottom	2590	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	350	mm.
Thickness of Wear Plate	26	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.40000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.

Element From Node	120	
Detail Type	Packing	
Detail ID	Pack: [1 of 1]	
Dist. from "FROM" Node / Offset dist	0	mm.
Height of Packed Section	3000	mm.
Density	0.00321	kg./cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	



Element From Node	120
Detail Type	Liquid
Detail ID	Liquid 120
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3400 mm.
Liquid Density	0.29952E-04 kg./cm ³

Element From Node	120
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.
Thickness of Insulation	100 mm.
Density	0.000352 kg./cm ³

Element From Node	120
Detail Type	Nozzle
Detail ID	M2600
Dist. from "FROM" Node / Offset dist	2300 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-182 F11 +IT

Element From Node	120
Detail Type	Weight
Detail ID	MTI 19 20 21 22
Dist. from "FROM" Node / Offset dist	1570 mm.
Miscellaneous Weight	35 Kg
Offset from Element Centerline	0 mm.

Element From Node	120
Detail Type	Weight
Detail ID	manway davit-2
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	60 Kg
Offset from Element Centerline	0 mm.

Element#	12/16
Element From Node	130
Element To Node	140
Element Type	Cylinder
Description	Shell-12
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	3400 mm.
Element Thickness	206 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	206 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	184.72 bars
Design Temperature Internal Pressure	285 °C
Design External Pressure	0 bars
Design Temperature External Pressure	285 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	130
Detail Type	Liquid
Detail ID	Liquid 130



Dist. from "FROM" Node / Offset dist 0 mm.
Height/Length of Liquid 3400 mm.
Liquid Density 0.29952E-04 kg./cm³

Element From Node 130
Detail Type Insulation
Detail ID Ins: 10
Dist. from "FROM" Node / Offset dist 0 mm.
Height/Length of Insulation 3000 mm.
Thickness of Insulation 100 mm.
Density 0.000352 kg./cm³

Element# 13/16
Element From Node 140
Element To Node 150
Element Type Flange
Description Flange-1
Distance "FROM" to "TO" 920 mm.
Flange Inside Diameter 3400 mm.
Element Thickness 520 mm.
Internal Corrosion Allowance 3 mm.
Nominal Thickness 524 mm.
External Corrosion Allowance 0 mm.
Design Internal Pressure 184.72 bars
Design Temperature Internal Pressure 285 °C
Design External Pressure 0 bars
Design Temperature External Pressure 285 °C
Effective Diameter Multiplier 1.2
Material Name SA-336 F11
Allowable Stress, Ambient 207 N./mm²
Allowable Stress, Operating 165.2 N./mm²
Allowable Stress, Hydrotest 294.49 N./mm²
Material Density 0.00775 kg./cm³
P Number Thickness 0 mm.
Yield Stress, Operating 247.8 N./mm²
MDMT Curve B
External Pressure Chart Name CS-3
UNS Number K11572
Class / Thickness / Grade 3 ::
Product Form Forgings
Perform Flange Stress Calculation (Y/N) Y
Weld is pre-Heated No

Element From Node 140
Detail Type Liquid
Detail ID Liquid 140
Dist. from "FROM" Node / Offset dist 0 mm.
Height/Length of Liquid 3400 mm.
Liquid Density 0.29952E-04 kg./cm³

Element From Node 140
Detail Type Insulation
Detail ID Ins: 10
Dist. from "FROM" Node / Offset dist 0 mm.
Height/Length of Insulation 920 mm.
Thickness of Insulation 100 mm.
Density 0.000352 kg./cm³

Element# 14/16
Element From Node 150
Element To Node 160
Element Type Flange
Description Flange-2
Distance "FROM" to "TO" 920 mm.
Flange Inside Diameter 3400 mm.



Element Thickness	520	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	524	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	184.72	bars
Design Temperature Internal Pressure	285	°C
Design External Pressure	0	bars
Design Temperature External Pressure	285	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-336 F11	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	150	
Detail Type	Liquid	
Detail ID	Liquid 150	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	3400	mm.
Liquid Density	0.29952E-04	kg./cm ³

Element From Node	150	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	920	mm.
Thickness of Insulation	100	mm.
Density	0.000352	kg./cm ³

Element#	15/16	
Element From Node	160	
Element To Node	170	
Element Type	Cylinder	
Description	Shell-13	
Distance "FROM" to "TO"	200	mm.
Inside Diameter	3400	mm.
Element Thickness	206	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	206	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	184.72	bars
Design Temperature Internal Pressure	285	°C
Design External Pressure	0	bars
Design Temperature External Pressure	285	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-336 F11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	160	
Detail Type	Liquid	
Detail ID	Liquid 160	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	3400	mm.
Liquid Density	0.29952E-04	kg./cm ³

Element From Node	160	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	200	mm.
Thickness of Insulation	100	mm.
Density	0.000352	kg./cm ³



Element From Node	170	
Element To Node	180	
Element Type	Spherical	
Description	Right head	
Distance "FROM" to "TO"	0	mm.
Inside Diameter	3501	mm.
Element Thickness	105	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	124	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	184.72	bars
Design Temperature Internal Pressure	285	°C
Design External Pressure	0	bars
Design Temperature External Pressure	285	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Allowable Stress, Ambient	207	N./mm ²
Allowable Stress, Operating	165.2	N./mm ²
Allowable Stress, Hydrotest	294.49	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm.
Yield Stress, Operating	247.8	N./mm ²
MDMT Curve	A	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	170	
Detail Type	Liquid	
Detail ID	Liquid 170	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	3500	mm.
Liquid Density	0.29952E-04	kg./cm ³
Element From Node	170	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	1750.5	mm.
Thickness of Insulation	100	mm.
Density	0.000352	kg./cm ³
Element From Node	170	
Detail Type	Nozzle	
Detail ID	C350	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	14	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-182 F11 +IT	
Element From Node	170	
Detail Type	Nozzle	
Detail ID	T1100	
Dist. from "FROM" Node / Offset dist	750	mm.
Nozzle Diameter	4	in.
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf



Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-182 F11 +IT
Element From Node	170
Detail Type	Nozzle
Detail ID	T2100
Dist. from "FROM" Node / Offset dist	1300 mm.
Nozzle Diameter	4 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-182 F11 +IT
Element From Node	170
Detail Type	Weight
Detail ID	Cutback-2 wt
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	5500 Kgf
Offset from Element Centerline	0 mm.

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Left head		...	...	...	...
Shell-1		3000	...	3000	...
Shell-2		6000	...	3000	...
Shell-3		9000	...	3000	...
Shell-4		12000	...	3000	...
Shell-5		15000	...	3000	...
Shell-6		18000	...	3000	...
Shell-7		21000	...	3000	...
Shell-8		24000	...	3000	...
Shell-10		27000	...	3000	...
Shell-11		30000	...	3000	...
Shell-12		33000	...	3000	...
Flange-1		33924.5	...	920	...
Flange-2		34844.5	...	920	...
Shell-13		35044.5	...	200	...
Right head		35044.5	...	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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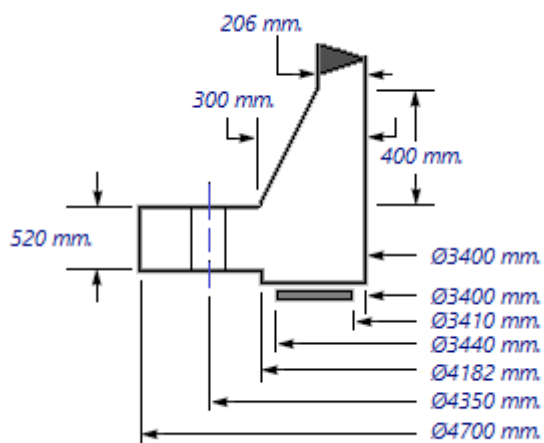


Flange Input Data Values

Description: FLANGE :

Flange-1

Flange Type	Integral Weld Neck		
Design Pressure	P	184.73	bars
Design Temperature		285	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	3400.000	mm.
Flange Outside Diameter	A	4700.000	mm.
Flange Thickness	t	520.0000	mm.
Thickness of Hub at Small End	go	206.0000	mm.
Thickness of Hub at Large End	gl	300.0000	mm.
Length of Hub	h	400.0000	mm.
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	165.20	N./mm ²
Flange Allowable Stress At Ambient	Sfa	207.00	N./mm ²
Bolt Material	SA-540 B23		
Bolt Allowable Stress At Temperature	Sb	200.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	200.00	N./mm ²
Diameter of Bolt Circle	C	4350.000	mm.
Nominal Bolt Diameter	a	170.0000	mm.
Type of Threads	Special, Root Area		
Root Area of a Single Bolt		206.6800	cm ²
Number of Bolts		44	
Flange Face Outside Diameter	Fod	4182.000	mm.
Flange Face Inside Diameter	Fid	3400.000	mm.
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	3440.000	mm.
Gasket Inside Diameter	Gi	3410.000	mm.
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm.
Raised face depth	Fd	0.0000	mm.



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Hub Small End Required Thickness due to Internal Pressure:



$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$

$$= (185(3400 + 3) / (165 - 0.5 \cdot 185)) + (3)$$

$$= 204.7246 \text{ mm.}$$

Small End Hub MAWP:

$$= S \cdot \log(2 \cdot T + R) / R$$

$$= 165 \cdot \log(2 \cdot 203 + 1703) / 1703$$

$$= 186.029 \text{ bars}$$

Small End Hub MAPnc:

$$= S_a \cdot \log(2.0 \cdot T + R) / R$$

$$= 207 \cdot \log(2.0 \cdot 206 + 1700) / 1700$$

$$= 236.746 \text{ bars}$$

Corroded Flange ID,	$B_{cor} = B + 2 \cdot c_i$	3406.000 mm.
Corroded Large Hub,	$g1_{cor} = g1 - c_i$	297.000 mm.
Corroded Small Hub,	$g0_{cor} = g0 - c_i$	203.000 mm.
Code R Dimension,	$R = ((C - B_{cor}) / 2) - g1_{cor}$	175.000 mm.
Gasket Contact Width,	$N = (G_o - G_i) / 2$	15.000 mm.
Basic Gasket Width,	$b_o = N / 2$	7.500 mm.
Effective Gasket Width,	$b = C_b \cdot \sqrt{b_o}$	6.901 mm.
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	3426.198 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 3426^2 \cdot 185$$

$$= 17367958.000 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.9 \cdot \pi \cdot 3426 \cdot 3 \cdot 185$$

$$= 839587.188 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 3406^2 \cdot 185$$

$$= 17163792.000 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 17367958 - 17163792$$

$$= 204167.172 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(17367958 + 839587 + 0, 0)$$

$$= 18207546.000 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.9 \cdot \pi \cdot 3426 + 0 \cdot 0 \cdot 0$$

$$= 522259.250 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a)$$

$$= \max((18207546 + 0 + 4 \cdot 0 / 3426) / 200, 522259 / 200)$$

$$= 8928.077 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 170 + 6 \cdot 520 / (3 + 0.5)$$

$$= 1231.428 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 4350 \cdot \sin(\pi / 44)$$



= 310.325 mm.

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{310}{(2 \cdot 170 + 520)}}, 1)$$

$$= 1.0000$$

Bolting Information for User Defined Bolts (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	8928.077	9093.919	
Circ. Spacing between the Bolts:	0.000	310.325	1231.428

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 9094 \cdot 200 / (68.9 \cdot \pi(3440 + 3410))$$

$$= 122.574 \text{ mm. [Note: Exceeds actual gasket width, 15.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm

Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 200 \cdot 9094$$

$$= 18545756.00 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 18207546 - 17367958$$

$$= 839586.38 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (4350 - 3426) / 2$$

$$= 461.9011 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((4350 - 3406) / 2 + 462)$$

$$= 466.9505 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (4350 - 3406 - 297) / 2$$

$$= 323.4999 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	17163792.	323.4999	1.0000	5552594.
Face Pressure, Mt	204167.	466.9505	1.0000	95338.
Gasket Load, Mg	839586.	461.9011	1.0000	387814.
Gasket Seating, Matm	18545756.	461.9011	1.0000	8566474.

Total Moment for Operation, Mo 6035746. Kg-m.

Total Moment for Gasket seating, Mg 8566474. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 831.516 mm.

Hub Ratio, h/h0 = HL / H0 0.481

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.463

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.852



Factor V per 4.16.5 0.326
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
= 4700/3406
= 1.380

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
= $(0.66845 + (5.7169(1.38^2 \cdot \log_{10}(1.38))) / (1.38^2 - 1)) / (1.38 - 1)$
= 6.191

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
= $(1.38^2(1 + 8.55246 \cdot \log_{10}(1.38)) - 1) / ((1.0472 + 1.9448 \cdot 1.38^2) \cdot (1.38 - 1))$
= 1.763

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
= $(1.38^2(1 + 8.55246 \cdot \log_{10}(1.38)) - 1) / (1.36136(1.38^2 - 1) \cdot (1.38 - 1))$
= 6.804

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
= $(1.38^2 + 1) / (1.38^2 - 1)$
= 3.212

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g_0^2 \cdot h_0 / V$
= $6.8 \cdot 203^2 \cdot 832 / 0.33$
= 714949760.000 mm.³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h₀
= 0.85/832
= 0.001 1/mm.

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / (T + t^3/d)$
= $(520 \cdot 0.001 + 1) / (1.76 + 520^3/714949760)$
= 1.066

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g_1^2)$
= $(1 \cdot 6035745 / 3406) / (1.07 \cdot 297^2)$
= 184.78 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g_1^2)$
= $(1 \cdot 8566474 / 3406) / (1.07 \cdot 297^2)$
= 262.25 N./mm²

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
= $(1.71 \cdot 6035745 / 3406) / (1.07 \cdot 520^2)$
= 103.11 N./mm²

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
= $(1.71 \cdot 8566474 / 3406) / (1.07 \cdot 520^2)$
= 146.35 N./mm²

Tangential Flange Stress, Operating [STo]:

= $(Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
= $(6.19 \cdot 6035745 / (520^2 \cdot 3406)) - 3.21 \cdot 103$
= 66.72 N./mm²



Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (6.19 \cdot 8566474 / (520^2 \cdot 3406)) - 3.21 \cdot 146$$

$$= 94.70 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (185 + \max(103.66.7)) / 2$$

$$= 143.94 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (262 + \max(146.94.7)) / 2$$

$$= 204.30 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 18207546 / 9094$$

$$= 196.35 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (522259 / 9094)$$

$$= 5.63 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	184.78	247.80	262.25	310.49
Radial Flange	103.11	165.20	146.35	207.00
Tangential Flange	66.72	165.20	94.70	207.00
Maximum Average	143.94	165.20	204.30	207.00
Bolting	196.35	200.00	5.63	200.00

Minimum Required Flange Thickness 515.214 mm.

Estimated M.A.W.P. (Operating) 186 bars
 Estimated Finished Weight of Flange at given Thk. 42332.6 kg.
 Estimated Unfinished Weight of Forging at given Thk 58970.6 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 8566474 / 1 \cdot 9807 \cdot 0.33 / (1.06 \cdot 204092 \cdot 203^2 \cdot 832 \cdot 0.3)$$

$$= 0.643 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 6035745 / 1 \cdot 9807 \cdot 0.33 / (1.06 \cdot 187373 \cdot 203^2 \cdot 832 \cdot 0.3)$$

$$= 0.493 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 8566474 / 1 \cdot 9807 \cdot 0.33 / (1.07 \cdot 204092 \cdot 203^2 \cdot 832 \cdot 0.3)$$

$$= 0.638 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 6035745 / 1 \cdot 9807 \cdot 0.33 / (1.07 \cdot 187373 \cdot 203^2 \cdot 832 \cdot 0.3)$$

$$= 0.490 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Warning:



The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	183.40	310.49	258.82	310.49
Radial Flange	104.19	207.00	147.04	207.00
Tangential Flange	65.84	207.00	92.92	207.00
Maximum Average	143.80	207.00	202.93	207.00
Bolting	196.35	200.00	5.63	200.00

Minimum Req'd Thickness, ambient, no Corrosion: 512.750 mm.

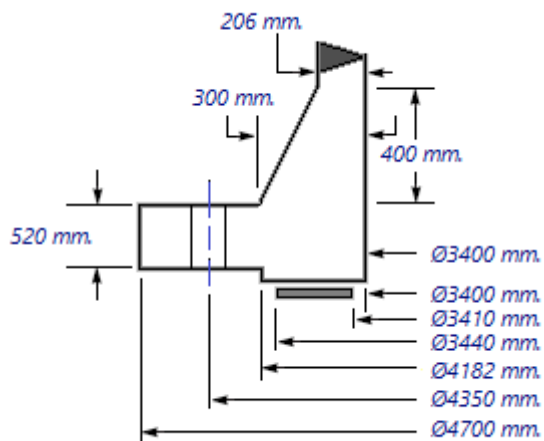
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Flange Input Data Values

Description: FLANGE :

Flange-2

Flange Type	Integral Weld Neck		
Design Pressure	P	184.73	bars
Design Temperature		285	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	3400.000	mm.
Flange Outside Diameter	A	4700.000	mm.
Flange Thickness	t	520.0000	mm.
Thickness of Hub at Small End	go	206.0000	mm.
Thickness of Hub at Large End	gl	300.0000	mm.
Length of Hub	h	400.0000	mm.
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	165.20	N./mm ²
Flange Allowable Stress At Ambient	Sfa	207.00	N./mm ²
Bolt Material	SA-540 B23		
Bolt Allowable Stress At Temperature	Sb	200.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	200.00	N./mm ²
Diameter of Bolt Circle	C	4350.000	mm.
Nominal Bolt Diameter	a	170.0000	mm.
Type of Threads	Special, Root Area		
Root Area of a Single Bolt		206.6800	cm ²
Number of Bolts		44	
Flange Face Outside Diameter	Fod	4182.000	mm.
Flange Face Inside Diameter	Fid	3400.000	mm.
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	3440.000	mm.
Gasket Inside Diameter	Gi	3410.000	mm.
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	52.40	N./mm ²
Gasket Thickness	tg	4.5000	mm.
Raised face depth	Fd	0.0000	mm.



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Hub Small End Required Thickness due to Internal Pressure:



$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$

$$= (185(3400+3) / (165-0.5 \cdot 185)) + (3)$$

$$= 204.7246 \text{ mm.}$$

Small End Hub MAWP:

$$= S \cdot \log(2 \cdot T + R) / R$$

$$= 165 \cdot \log(2 \cdot 203 + 1703) / 1703$$

$$= 186.029 \text{ bars}$$

Small End Hub MAPnc:

$$= S_a \cdot \log(2.0 \cdot T + R) / R$$

$$= 207 \cdot \log(2.0 \cdot 206 + 1700) / 1700$$

$$= 236.746 \text{ bars}$$

Corroded Flange ID,	$B_{cor} = B + 2 \cdot c_i$	3406.000 mm.
Corroded Large Hub,	$g1_{cor} = g1 - c_i$	297.000 mm.
Corroded Small Hub,	$g0_{cor} = g0 - c_i$	203.000 mm.
Code R Dimension,	$R = ((C - B_{cor}) / 2) - g1_{cor}$	175.000 mm.
Gasket Contact Width,	$N = (G_o - G_i) / 2$	15.000 mm.
Basic Gasket Width,	$b_o = N / 2$	7.500 mm.
Effective Gasket Width,	$b = C_b \cdot \sqrt{b_o}$	6.901 mm.
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	3426.198 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 3426^2 \cdot 185$$

$$= 17367958.000 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.9 \cdot \pi \cdot 3426 \cdot 3.75 \cdot 185$$

$$= 1049484.125 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 3406^2 \cdot 185$$

$$= 17163792.000 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 17367958 - 17163792$$

$$= 204167.172 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(17367958 + 1049484 + 0, 0)$$

$$= 18417442.000 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 52.4 \cdot 6.9 \cdot \pi \cdot 3426 + 0 \cdot 0 \cdot 0$$

$$= 396917.062 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a)$$

$$= \max((18417442 + 0 + 4 \cdot 0 / 3426) / 200, 396917 / 200)$$

$$= 9031.001 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 170 + 6 \cdot 520 / (3.75 + 0.5)$$

$$= 1074.118 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 4350 \cdot \sin(\pi / 44)$$



= 310.325 mm.

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{310}{(2 \cdot 170 + 520)}}, 1)$$

$$= 1.0000$$

Bolting Information for User Defined Bolts (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	9031.001	9093.919	
Circ. Spacing between the Bolts:	0.000	310.325	1074.118

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 9094 \cdot 200 / (52.4 \cdot \pi(3440 + 3410))$$

$$= 161.282 \text{ mm. [Note: Exceeds actual gasket width, 15.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm

Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 200 \cdot 9094$$

$$= 18545756.00 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 18417442 - 17367958$$

$$= 1049483.38 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (4350 - 3426) / 2$$

$$= 461.9011 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((4350 - 3406) / 2 + 462)$$

$$= 466.9505 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (4350 - 3406 - 297) / 2$$

$$= 323.4999 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	17163792.	323.4999	1.0000	5552594.
Face Pressure, Mt	204167.	466.9505	1.0000	95338.
Gasket Load, Mg	1049483.	461.9011	1.0000	484767.
Gasket Seating, Matm	18545756.	461.9011	1.0000	8566474.

Total Moment for Operation, Mo 6132700. Kg-m.

Total Moment for Gasket seating, Mg 8566474. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 831.516 mm.

Hub Ratio, h/h0 = HL / H0 0.481

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.463

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.852



Factor V per 4.16.5 0.326
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
= 4700/3406
= 1.380

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
= $(0.66845 + (5.7169(1.38^2 \cdot \log_{10}(1.38))) / (1.38^2 - 1)) / (1.38 - 1)$
= 6.191

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
= $(1.38^2(1 + 8.55246 \cdot \log_{10}(1.38)) - 1) / ((1.0472 + 1.9448 \cdot 1.38^2) \cdot (1.38 - 1))$
= 1.763

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
= $(1.38^2(1 + 8.55246 \cdot \log_{10}(1.38)) - 1) / (1.36136(1.38^2 - 1) \cdot (1.38 - 1))$
= 6.804

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
= $(1.38^2 + 1) / (1.38^2 - 1)$
= 3.212

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g_0^2 \cdot h_0 / V$
= $6.8 \cdot 203^2 \cdot 832 / 0.33$
= 714949760.000 mm.³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h₀
= 0.85/832
= 0.001 1/mm.

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / (T + t^3/d)$
= $(520 \cdot 0.001 + 1) / (1.76 + 520^3/714949760)$
= 1.066

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g_1^2)$
= $(1 \cdot 6132699 / 3406) / (1.07 \cdot 297^2)$
= 187.74 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g_1^2)$
= $(1 \cdot 8566474 / 3406) / (1.07 \cdot 297^2)$
= 262.25 N./mm²

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
= $(1.71 \cdot 6132699 / 3406) / (1.07 \cdot 520^2)$
= 104.77 N./mm²

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
= $(1.71 \cdot 8566474 / 3406) / (1.07 \cdot 520^2)$
= 146.35 N./mm²

Tangential Flange Stress, Operating [STo]:

= $(Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
= $(6.19 \cdot 6132699 / (520^2 \cdot 3406)) - 3.21 \cdot 105$
= 67.79 N./mm²



Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (6.19 \cdot 8566474 / (520^2 \cdot 3406)) - 3.21 \cdot 146$$

$$= 94.70 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (188 + \max(105.67.8)) / 2$$

$$= 146.26 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRa, STa)) / 2$$

$$= (262 + \max(146.94.7)) / 2$$

$$= 204.30 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 18417442 / 9094$$

$$= 198.61 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (396917 / 9094)$$

$$= 4.28 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	187.74	247.80	262.25	310.49
Radial Flange	104.77	165.20	146.35	207.00
Tangential Flange	67.79	165.20	94.70	207.00
Maximum Average	146.26	165.20	204.30	207.00
Bolting	198.61	200.00	4.28	200.00

Minimum Required Flange Thickness 515.214 mm.

Estimated M.A.W.P. (Operating) 186 bars

Estimated Finished Weight of Flange at given Thk. 42332.6 kg.

Estimated Unfinished Weight of Forging at given Thk 58970.6 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 8566474 / 1 \cdot 9807 \cdot 0.33 / (1.06 \cdot 204092 \cdot 203^2 \cdot 832 \cdot 0.3)$$

$$= 0.643 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6132699 / 1 \cdot 9807 \cdot 0.33 / (1.06 \cdot 187373 \cdot 203^2 \cdot 832 \cdot 0.3)$$

$$= 0.501 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 8566474 / 1 \cdot 9807 \cdot 0.33 / (1.07 \cdot 204092 \cdot 203^2 \cdot 832 \cdot 0.3)$$

$$= 0.638 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6132699 / 1 \cdot 9807 \cdot 0.33 / (1.07 \cdot 187373 \cdot 203^2 \cdot 832 \cdot 0.3)$$

$$= 0.497 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Warning:



The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed	
Longitudinal Hub	186.33	310.49	258.82	310.49
Radial Flange	105.86	207.00	147.04	207.00
Tangential Flange	66.89	207.00	92.92	207.00
Maximum Average	146.10	207.00	202.93	207.00
Bolting	198.61	200.00	4.28	200.00

Minimum Req'd Thickness, ambient, no Corrosion: 512.750 mm.

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**Internal Pressure Results Summary:****Element Thickness, Pressure, Diameter and Allowable Stress :**

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Left head		184.73	124	3	3501	165.2	1
Shell-1		184.73	206	3	3400	165.2	2
Shell-2		184.73	206	3	3400	165.2	3
Shell-3		184.73	206	3	3400	165.2	4
Shell-4		184.73	206	3	3400	165.2	5
Shell-5		184.73	206	3	3400	165.2	6
Shell-6		184.73	206	3	3400	165.2	7
Shell-7		184.73	206	3	3400	165.2	8
Shell-8		184.73	206	3	3400	165.2	9
Shell-10		184.73	206	3	3400	165.2	10
Shell-11		184.73	206	3	3400	165.2	11
Shell-12		184.73	206	3	3400	165.2	12
Flange-1		184.73	524	3	3400	165.2	13
Flange-2		184.73	524	3	3400	165.2	14
Shell-13		184.73	206	3	3400	165.2	15
Right head		184.73	124	3	3501	165.2	16

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Left head		184.72	187	241	105	103.839	1
Shell-1		184.72	186	237	206	204.502	2
Shell-2		184.72	186	237	206	204.502	3
Shell-3		184.72	186	237	206	204.502	4
Shell-4		184.72	186	237	206	204.502	5
Shell-5		184.72	186	237	206	204.502	6
Shell-6		184.72	186	237	206	204.502	7
Shell-7		184.72	186	237	206	204.502	8
Shell-8		184.72	186	237	206	204.502	9
Shell-10		184.72	186	237	206	204.502	10
Shell-11		184.72	186	237	206	204.502	11
Shell-12		184.72	186	237	206	204.502	12
Flange-1		184.72	186	188	520	515.214	13
Flange-2		184.72	186	186	520	515.214	14
Shell-13		184.72	186	237	206	204.502	15
Right head		184.72	187	241	105	103.839	16

Minimum

185

186

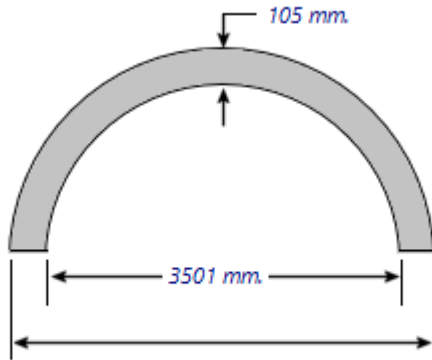
MAWP: 185 bars, limited by: DESIGN (user specified)

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Left head

Spherical Head Design Information:					
Design Pressure	P	184.730 bars	Int. Design Temperature		285 °C
Material		SA-387 11	Ext. Design Temperature		285 °C
Length		0.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	165.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	3501.000 mm.	Actual Stress		154.3 N./mm ²
Spec. Min. Thk t,e, or sa		105.0000 mm.	Spec. Nominal Thk.		124.000 mm.
Surface Area		220775.4 cm ²	SG of Contents		0.030
Element Volume		11236284.0 cm ³	Weight of Contents		336.9 Kgf
Empty Weight		19845.3 Kgf	Operating Weight		20182.1 Kgf
			Comp. All. Ope. Stress		93.1 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 43.763 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.881 mm.

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \left(\exp(0.5P/(S \cdot E)) - 1 \right) + ci + co$
 $= 3507/2 \left(\exp(0.5 \cdot 185/(165 \cdot 1)) - 1 \right) + 3 + 0$
 $= 101 + 3 + 0 = 104 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 2 \cdot 165 \cdot 1 \cdot \ln((2 \cdot 102 + 3507) / 3507) - 0.01$
 $= 187 - 0.01 = 187 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2 \cdot 207 \cdot 1 \cdot \ln((2 \cdot 105 + 3501) / 3501)$
 $= 241 \text{ bars}$



Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 185 \cdot 3507^2 / (1 (3711^2 - 3507^2))$$

$$= 154.308 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 185 / (1 \cdot \ln((2 \cdot 102 + 3507) / 3507))$$

$$= 163.371 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.131 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 41.501 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 220775 / 3.14159)$$

$$= 5301.886 \text{ mm.}$$

MDMT Calculations in the Spherical Portion:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
 This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	184.720 bars
Governing Thk.	105.000 mm.	Min. Des. Mtl. Temp.	-12 °C
Stress/Thk. Ratio	-1.000	Basic MDMT	-17 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

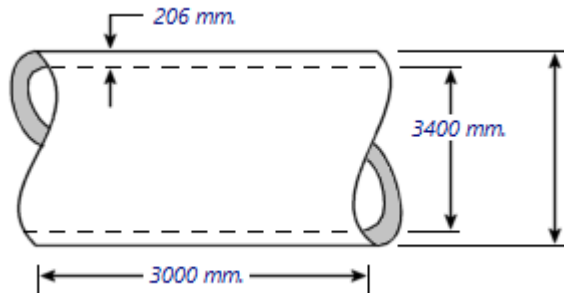
Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 43.763 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.881 mm.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	184.730 bars	Int. Design Temperature		285 °C
Material		SA-387 11	Ext. Design Temperature		285 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	165.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	3400.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t,e, or sa		206.0000 mm.	Spec. Nominal Thk.		206.000 mm.
Surface Area		359272.6 cm ²	SG of Contents		0.030
Element Volume		27242492.0 cm ³	Weight of Contents		816.7 Kg
Empty Weight		54261.8 Kg	Operating Weight		55078.5 Kg
			Comp. All. Ope. Stress		102.2 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3406 \cdot (\exp(185/(165 \cdot 1)) - 1) + 3 + 0$
 $= 202 + 3 + 0 = 205 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 165 \cdot 1 \cdot \ln((2 \cdot 203 + 3406) / 3406) - 0.01$
 $= 186 - 0.01 = 186 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 207 \cdot 1 \cdot \ln((2 \cdot 206 + 3400) / 3400)$
 $= 237 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 185 \cdot 3406 / (1 (3812 - 3406))$
 $= 154.9820 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 185 / (1 \cdot \ln((2 \cdot 203 + 3406) / 3406))$



= 164.0457 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 5.713 %

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.

Note:

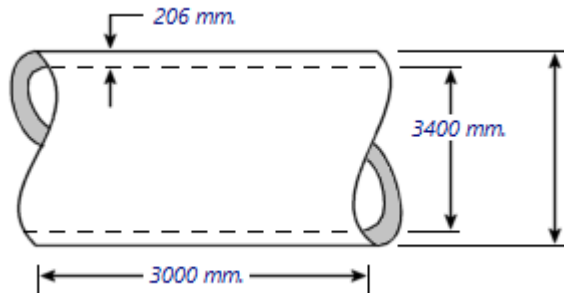
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	184.720 bars
Governing Thk.	206.000 mm.	Min. Des. Mtl. Temp.	-12 °C
Stress/Thk. Ratio	-1.000	Basic MDMT	-17 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	184.730 bars	Int. Design Temperature		285 °C
Material		SA-387 11	Ext. Design Temperature		285 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	165.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	3400.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t,e, or sa		206.0000 mm.	Spec. Nominal Thk.		206.000 mm.
Surface Area		359272.6 cm ²	SG of Contents		0.030
Element Volume		27242492.0 cm ³	Weight of Contents		816.8 Kg
Empty Weight		54261.8 Kg	Operating Weight		55078.6 Kg
			Comp. All. Ope. Stress		102.2 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3406 \cdot (\exp(185/(165 \cdot 1)) - 1) + 3 + 0$
 $= 202 + 3 + 0 = 205 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 165 \cdot 1 \cdot \ln((2 \cdot 203 + 3406) / 3406) - 0.01$
 $= 186 - 0.01 = 186 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 207 \cdot 1 \cdot \ln((2 \cdot 206 + 3400) / 3400)$
 $= 237 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 185 \cdot 3406 / (1 (3812 - 3406))$
 $= 154.9820 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 185 / (1 \cdot \ln((2 \cdot 203 + 3406) / 3406))$



= 164.0457 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 5.713 %

Minimum Design Metal Temperature Results:**Warning:**

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.

Note:

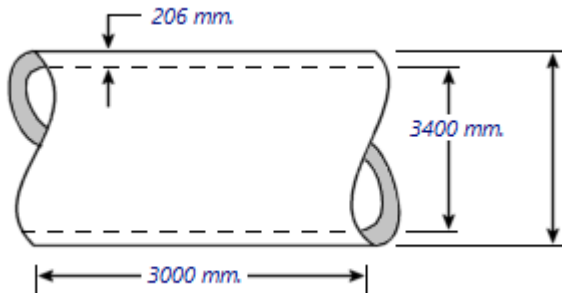
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	184.720 bars
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	206.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	184.730 bars	Int. Design Temperature		285 °C
Material		SA-387 11	Ext. Design Temperature		285 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	165.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	3400.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t,e, or sa		206.0000 mm.	Spec. Nominal Thk.		206.000 mm.
Surface Area		359272.6 cm ²	SG of Contents		0.030
Element Volume		27242492.0 cm ³	Weight of Contents		816.8 Kg
Empty Weight		54261.8 Kg	Operating Weight		55078.6 Kg
			Comp. All. Ope. Stress		102.2 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3406 \cdot (\exp(185/(165 \cdot 1)) - 1) + 3 + 0$
 $= 202 + 3 + 0 = 205 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 165 \cdot 1 \cdot \ln((2 \cdot 203 + 3406) / 3406) - 0.01$
 $= 186 - 0.01 = 186 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 207 \cdot 1 \cdot \ln((2 \cdot 206 + 3400) / 3400)$
 $= 237 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 185 \cdot 3406 / (1 (3812 - 3406))$
 $= 154.9820 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 185 / (1 \cdot \ln((2 \cdot 203 + 3406) / 3406))$



= 164.0457 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 5.713 %

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.

Note:

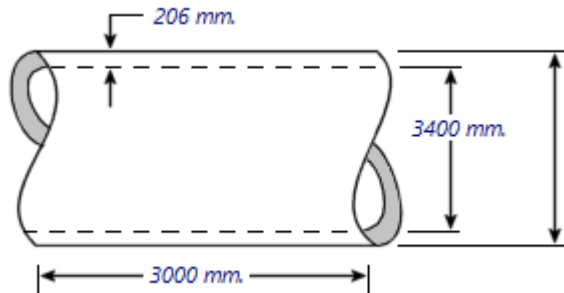
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	184.720 bars
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	206.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	184.730 bars	Int. Design Temperature		285 °C
Material		SA-387 11	Ext. Design Temperature		285 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	165.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	3400.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		206.0000 mm.	Spec. Nominal Thk.		206.000 mm.
Surface Area		359272.6 cm ²	SG of Contents		0.030
Element Volume		27242492.0 cm ³	Weight of Contents		816.7 Kg
Empty Weight		54261.8 Kg	Operating Weight		55078.5 Kg
			Comp. All. Ope. Stress		102.2 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3406 \cdot (\exp(185/(165 \cdot 1)) - 1) + 3 + 0$
 $= 202 + 3 + 0 = 205 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 165 \cdot 1 \cdot \ln((2 \cdot 203 + 3406) / 3406) - 0.01$
 $= 186 - 0.01 = 186 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 207 \cdot 1 \cdot \ln((2 \cdot 206 + 3400) / 3400)$
 $= 237 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 185 \cdot 3406 / (1 (3812 - 3406))$
 $= 154.9820 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 185 / (1 \cdot \ln((2 \cdot 203 + 3406) / 3406))$



= 164.0457 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 5.713 %

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.

Note:

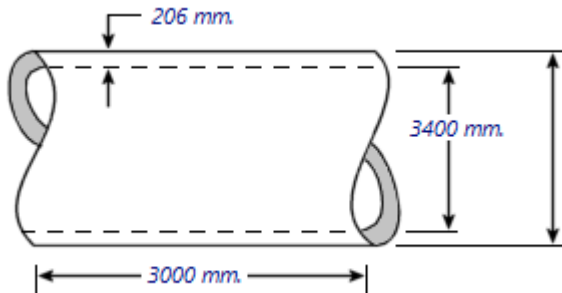
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	184.720 bars
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	206.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	184.730 bars	Int. Design Temperature		285 °C
Material		SA-387 11	Ext. Design Temperature		285 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	165.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	3400.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t,e, or sa		206.0000 mm.	Spec. Nominal Thk.		206.000 mm.
Surface Area		359272.6 cm ²	SG of Contents		0.030
Element Volume		27242492.0 cm ³	Weight of Contents		816.7 Kg
Empty Weight		54261.8 Kg	Operating Weight		55078.5 Kg
			Comp. All. Ope. Stress		102.2 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3406 \cdot (\exp(185/(165 \cdot 1)) - 1) + 3 + 0$
 $= 202 + 3 + 0 = 205 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 165 \cdot 1 \cdot \ln((2 \cdot 203 + 3406) / 3406) - 0.01$
 $= 186 - 0.01 = 186 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 207 \cdot 1 \cdot \ln((2 \cdot 206 + 3400) / 3400)$
 $= 237 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 185 \cdot 3406 / (1 (3812 - 3406))$
 $= 154.9820 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 185 / (1 \cdot \ln((2 \cdot 203 + 3406) / 3406))$



= 164.0457 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 5.713 %

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.

Note:

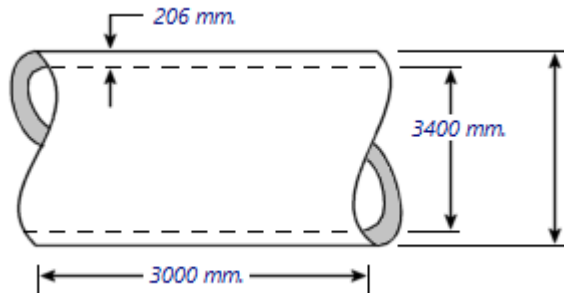
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	184.720 bars
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	206.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	184.730 bars	Int. Design Temperature	285 °C	
Material		SA-387 11	Ext. Design Temperature	285 °C	
Length		3000.000 mm.	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	165.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	3400.000 mm.	Actual Stress	155.0 N./mm ²	
Spec. Min. Thk t _e , or s _a		206.0000 mm.	Spec. Nominal Thk.	206.000 mm.	
Surface Area		359272.6 cm ²	SG of Contents	0.030	
Element Volume		27242492.0 cm ³	Weight of Contents	816.8 Kg	
Empty Weight		54261.8 Kg	Operating Weight	55078.6 Kg	
			Comp. All. Ope. Stress	102.2 N./mm ²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3406 \cdot (\exp(185/(165 \cdot 1)) - 1) + 3 + 0$
 $= 202 + 3 + 0 = 205 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 165 \cdot 1 \cdot \ln((2 \cdot 203 + 3406) / 3406) - 0.01$
 $= 186 - 0.01 = 186 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 207 \cdot 1 \cdot \ln((2 \cdot 206 + 3400) / 3400)$
 $= 237 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 185 \cdot 3406 / (1 (3812 - 3406))$
 $= 154.9820 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 185 / (1 \cdot \ln((2 \cdot 203 + 3406) / 3406))$



= 164.0457 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 5.713 %

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.

Note:

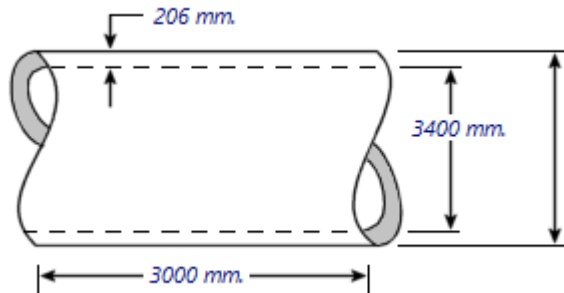
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	184.720 bars
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	206.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	184.730 bars	Int. Design Temperature		285 °C
Material		SA-387 11	Ext. Design Temperature		285 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	165.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	3400.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t,e, or sa		206.0000 mm.	Spec. Nominal Thk.		206.000 mm.
Surface Area		359272.6 cm ²	SG of Contents		0.030
Element Volume		27242492.0 cm ³	Weight of Contents		816.8 Kg
Empty Weight		54261.8 Kg	Operating Weight		55078.6 Kg
			Comp. All. Ope. Stress		102.2 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3406 \cdot (\exp(185/(165 \cdot 1)) - 1) + 3 + 0$
 $= 202 + 3 + 0 = 205 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 165 \cdot 1 \cdot \ln((2 \cdot 203 + 3406) / 3406) - 0.01$
 $= 186 - 0.01 = 186 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 207 \cdot 1 \cdot \ln((2 \cdot 206 + 3400) / 3400)$
 $= 237 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 185 \cdot 3406 / (1 (3812 - 3406))$
 $= 154.9820 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 185 / (1 \cdot \ln((2 \cdot 203 + 3406) / 3406))$



= 164.0457 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 5.713 %

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.

Note:

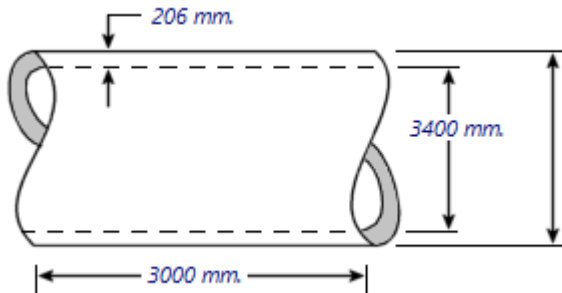
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	184.720 bars
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	206.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-8

Cylindrical Shell Design Information:					
Design Pressure	P	184.730 bars	Int. Design Temperature		285 °C
Material		SA-387 11	Ext. Design Temperature		285 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	165.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	3400.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t,e, or sa		206.0000 mm.	Spec. Nominal Thk.		206.000 mm.
Surface Area		359272.6 cm ²	SG of Contents		0.030
Element Volume		27242492.0 cm ³	Weight of Contents		816.8 Kg
Empty Weight		54261.8 Kg	Operating Weight		55078.6 Kg
			Comp. All. Ope. Stress		102.2 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3406 \cdot (\exp(185/(165 \cdot 1)) - 1) + 3 + 0$
 $= 202 + 3 + 0 = 205 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 165 \cdot 1 \cdot \ln((2 \cdot 203 + 3406) / 3406) - 0.01$
 $= 186 - 0.01 = 186 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 207 \cdot 1 \cdot \ln((2 \cdot 206 + 3400) / 3400)$
 $= 237 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 185 \cdot 3406 / (1 (3812 - 3406))$
 $= 154.9820 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 185 / (1 \cdot \ln((2 \cdot 203 + 3406) / 3406))$



= 164.0457 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 5.713 %

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.

Note:

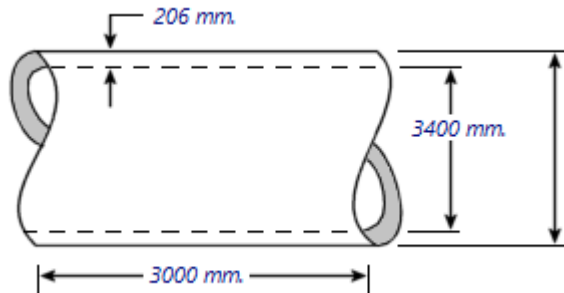
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	184.720 bars
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	206.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-10

Cylindrical Shell Design Information:			
Design Pressure P	184.730 bars	Int. Design Temperature	285 °C
Material	SA-387 11	Ext. Design Temperature	285 °C
Length	3000.000 mm.	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm.	Allowable Stress (ope) S	165.2 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	207.0 N./mm ²
Inside Diameter Di	3400.000 mm.	Actual Stress	155.0 N./mm ²
Spec. Min. Thk t _e , or s _a	206.0000 mm.	Spec. Nominal Thk.	206.000 mm.
Surface Area	359272.6 cm ²	SG of Contents	0.030
Element Volume	27242492.0 cm ³	Weight of Contents	816.8 Kg
Empty Weight	54261.8 Kg	Operating Weight	55078.6 Kg
		Comp. All. Ope. Stress	102.2 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3406 \cdot (\exp(185/(165 \cdot 1)) - 1) + 3 + 0$
 $= 202 + 3 + 0 = 205 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 165 \cdot 1 \cdot \ln((2 \cdot 203 + 3406) / 3406) - 0.01$
 $= 186 - 0.01 = 186 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 207 \cdot 1 \cdot \ln((2 \cdot 206 + 3400) / 3400)$
 $= 237 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 185 \cdot 3406 / (1 (3812 - 3406))$
 $= 154.9820 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 185 / (1 \cdot \ln((2 \cdot 203 + 3406) / 3406))$



= 164.0457 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 5.713 %

Minimum Design Metal Temperature Results:*Warning:*

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

Note:

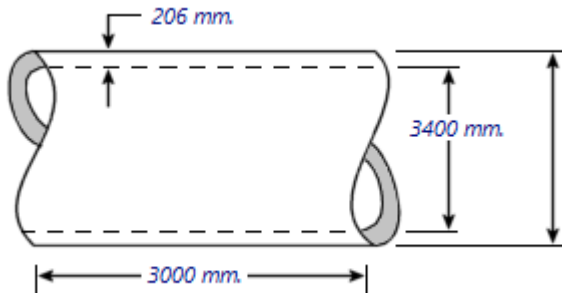
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	184.720 bars
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	206.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-11

Cylindrical Shell Design Information:					
Design Pressure	P	184.730 bars	Int. Design Temperature		285 °C
Material		SA-387 11	Ext. Design Temperature		285 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	165.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	3400.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t,e, or sa		206.0000 mm.	Spec. Nominal Thk.		206.000 mm.
Surface Area		359272.6 cm ²	SG of Contents		0.030
Element Volume		27242492.0 cm ³	Weight of Contents		816.8 Kg
Empty Weight		54261.8 Kg	Operating Weight		55078.6 Kg
			Comp. All. Ope. Stress		102.2 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3406 \cdot (\exp(185/(165 \cdot 1)) - 1) + 3 + 0$
 $= 202 + 3 + 0 = 205 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 165 \cdot 1 \cdot \ln((2 \cdot 203 + 3406) / 3406) - 0.01$
 $= 186 - 0.01 = 186 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 207 \cdot 1 \cdot \ln((2 \cdot 206 + 3400) / 3400)$
 $= 237 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 185 \cdot 3406 / (1 (3812 - 3406))$
 $= 154.9820 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 185 / (1 \cdot \ln((2 \cdot 203 + 3406) / 3406))$



= 164.0457 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 5.713 %

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.

Note:

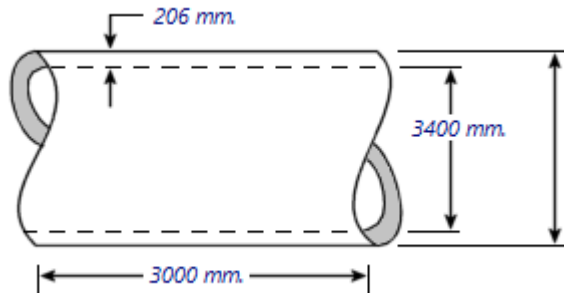
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	184.720 bars
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	206.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-12

Cylindrical Shell Design Information:					
Design Pressure	P	184.730 bars	Int. Design Temperature		285 °C
Material		SA-387 11	Ext. Design Temperature		285 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	165.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	3400.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t,e, or sa		206.0000 mm.	Spec. Nominal Thk.		206.000 mm.
Surface Area		359272.6 cm ²	SG of Contents		0.030
Element Volume		27242492.0 cm ³	Weight of Contents		816.8 Kg
Empty Weight		54261.8 Kg	Operating Weight		55078.6 Kg
			Comp. All. Ope. Stress		102.2 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3406 \cdot (\exp(185/(165 \cdot 1)) - 1) + 3 + 0$
 $= 202 + 3 + 0 = 205 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 165 \cdot 1 \cdot \ln((2 \cdot 203 + 3406) / 3406) - 0.01$
 $= 186 - 0.01 = 186 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 207 \cdot 1 \cdot \ln((2 \cdot 206 + 3400) / 3400)$
 $= 237 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 185 \cdot 3406 / (1 (3812 - 3406))$
 $= 154.9820 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 185 / (1 \cdot \ln((2 \cdot 203 + 3406) / 3406))$



= 164.0457 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 5.713 %

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.

Note:

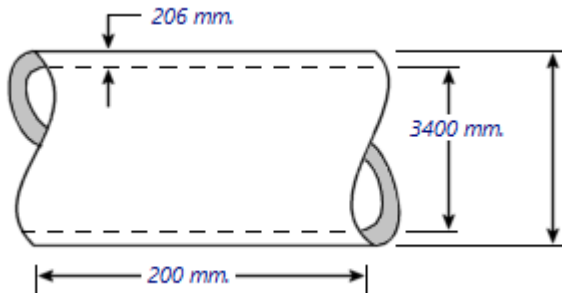
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	184.720 bars
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	206.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-13

Cylindrical Shell Design Information:			
Design Pressure P	184.730 bars	Int. Design Temperature	285 °C
Material	SA-336 F11	Ext. Design Temperature	285 °C
Length	200.000 mm.	External Pressure Chart	CS-3
		UNS Number	K11572
Int. Corr. All. c	3.0000 mm.	Allowable Stress (ope) S	165.2 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	207.0 N./mm ²
Inside Diameter Di	3400.000 mm.	Actual Stress	155.0 N./mm ²
Spec. Min. Thk t _e , or s _a	206.0000 mm.	Spec. Nominal Thk.	206.000 mm.
Surface Area	23951.5 cm ²	SG of Contents	0.030
Element Volume	1816166.1 cm ³	Weight of Contents	54.5 Kg
Empty Weight	3617.5 Kg	Operating Weight	3671.9 Kg
		Comp. All. Ope. Stress	102.2 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3406 \cdot (\exp(185/(165 \cdot 1)) - 1) + 3 + 0$
 $= 202 + 3 + 0 = 205 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 165 \cdot 1 \cdot \ln((2 \cdot 203 + 3406) / 3406) - 0.01$
 $= 186 - 0.01 = 186 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 207 \cdot 1 \cdot \ln((2 \cdot 206 + 3400) / 3400)$
 $= 237 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 185 \cdot 3406 / (1 (3812 - 3406))$
 $= 154.9820 \text{ N./mm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 185 / (1 \cdot \ln((2 \cdot 203 + 3406) / 3406))$



= 164.0457 N./mm²

Minimum Design Metal Temperature Results:

Warning:

*The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
 This component must be impact tested.*

Note:

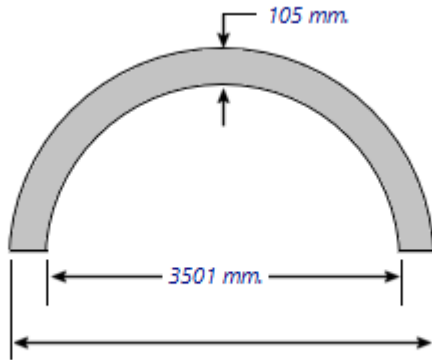
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	184.720 bars
		Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	206.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Right head

Spherical Head Design Information:					
Design Pressure	P	184.730 bars	Int. Design Temperature		285 °C
Material		SA-387 11	Ext. Design Temperature		285 °C
Length		0.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	165.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	3501.000 mm.	Actual Stress		154.3 N./mm ²
Spec. Min. Thk t,e, or sa		105.0000 mm.	Spec. Nominal Thk.		124.000 mm.
Surface Area		220775.4 cm ²	SG of Contents		0.030
Element Volume		11236284.0 cm ³	Weight of Contents		336.9 Kg
Empty Weight		19845.3 Kg	Operating Weight		20182.1 Kg
			Comp. All. Ope. Stress		102.2 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 43.763 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.881 mm.

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \left(\exp(0.5P/(S \cdot E)) - 1 \right) + ci + co$
 $= 3507/2 \left(\exp(0.5 \cdot 185/(165 \cdot 1)) - 1 \right) + 3 + 0$
 $= 101 + 3 + 0 = 104 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 2 \cdot 165 \cdot 1 \cdot \ln((2 \cdot 102 + 3507) / 3507) - 0.01$
 $= 187 - 0.01 = 187 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2 \cdot 207 \cdot 1 \cdot \ln((2 \cdot 105 + 3501) / 3501)$
 $= 241 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 185 \cdot 3507^2 / (1 (3711^2 - 3507^2))$$

$$= 154.308 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 185 / (1 \cdot \ln((2 \cdot 102 + 3507) / 3507))$$

$$= 163.371 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.131 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 41.501 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 220775 / 3.14159)$$

$$= 5301.886 \text{ mm.}$$

MDMT Calculations in the Spherical Portion:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	184.720 bars
Governing Thk.	105.000 mm.	Min. Des. Mtl. Temp.	-12 °C
Stress/Thk. Ratio	-1.000	Basic MDMT	-17 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	0 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 43.763 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.881 mm.

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Hydrostatic Test Pressure Results:

Test Pressure at high point in vessel per Equation 8.1
 $= \gamma_{St/s} \cdot MAWP(St/S)$
 $= 1.25 \cdot 185 (1.25)$
 $= 288.625 \text{ bars}$

Horizontal Test performed per: $\gamma_{St/s} \cdot MAWP(St/S)$

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Left head	255.6	294.5	0.868	288.97
Shell-1	256.6	294.5	0.871	288.96
Shell-2	256.6	294.5	0.871	288.96
Shell-3	256.6	294.5	0.871	288.96
Shell-4	256.6	294.5	0.871	288.96
Shell-5	256.6	294.5	0.871	288.96
Shell-6	256.6	294.5	0.871	288.96
Shell-7	256.6	294.5	0.871	288.96
Shell-8	256.6	294.5	0.871	288.96
Shell-10	256.6	294.5	0.871	288.96
Shell-11	256.6	294.5	0.871	288.96
Shell-12	256.6	294.5	0.871	288.96
Shell-13	256.6	294.5	0.871	288.96
Right head	255.6	294.5	0.868	288.97

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
A1_600	Nozzle	184.00	147.20	1.250
A2_450	Nozzle	184.00	147.20	1.250
B_600	Nozzle	184.00	147.20	1.250
T3_100	Nozzle	184.00	147.20	1.250
M1_600	Nozzle	184.00	147.20	1.250
M2_600	Nozzle	184.00	147.20	1.250
C_350	Nozzle	184.00	147.20	1.250
T1_100	Nozzle	184.00	147.20	1.250
T2_100	Nozzle	184.00	147.20	1.250

Minimum

1.250

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Left head	207.00	165.20	1.253
Shell-1	207.00	165.20	1.253
Shell-2	207.00	165.20	1.253
Shell-3	207.00	165.20	1.253
Shell-4	207.00	165.20	1.253
Shell-5	207.00	165.20	1.253
Shell-6	207.00	165.20	1.253
Shell-7	207.00	165.20	1.253
Shell-8	207.00	165.20	1.253
Shell-10	207.00	165.20	1.253
Shell-11	207.00	165.20	1.253
Shell-12	207.00	165.20	1.253
Flange-1	207.00	165.20	1.253
Flange-2	207.00	165.20	1.253
Shell-13	207.00	165.20	1.253



Right head	207.00	165.20	1.253
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Minimum 1.253

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
A1_600	220.99	261.20	0.846
A2_450	216.94	261.20	0.831
B_600	220.99	261.20	0.846
T3_100	186.83	261.20	0.715
M1_600	211.78	261.20	0.811
M2_600	211.78	261.20	0.811
C_350	207.63	261.20	0.795
T1_100	133.52	261.20	0.511
T2_100	186.83	261.20	0.715

Elements Suitable for Test Pressure.

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**External Pressure Calculation Results Summary:****ASME Code, Section VIII Division 2, 2025****External Pressure Calculations:**

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	3711	102	No Calc	No Calc
20	30	33583.5	3812	203	No Calc	No Calc
30	40	33583.5	3812	203	No Calc	No Calc
40	50	33583.5	3812	203	No Calc	No Calc
50	60	33583.5	3812	203	No Calc	No Calc
60	70	33583.5	3812	203	No Calc	No Calc
70	80	33583.5	3812	203	No Calc	No Calc
80	90	33583.5	3812	203	No Calc	No Calc
90	110	33583.5	3812	203	No Calc	No Calc
110	120	33583.5	3812	203	No Calc	No Calc
120	130	33583.5	3812	203	No Calc	No Calc
130	140	33583.5	3812	203	No Calc	No Calc
140	150	No Calc	...	517	No Calc	No Calc
150	160	No Calc	...	517	No Calc	No Calc
160	170	783.5	3812	203	No Calc	No Calc
170	180	No Calc	3711	102	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	105	No Calc	...	119
20	30	206	No Calc	...	101
30	40	206	No Calc	...	101
40	50	206	No Calc	...	101
50	60	206	No Calc	...	101
60	70	206	No Calc	...	101
70	80	206	No Calc	...	101
80	90	206	No Calc	...	101
90	110	206	No Calc	...	101
110	120	206	No Calc	...	101
120	130	206	No Calc	...	101
130	140	206	No Calc	...	101
140	150	520	515.214	...	No Calc
150	160	520	515.214	...	No Calc
160	170	206	No Calc	...	197
170	180	105	No Calc	...	119

Minimum

101

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	33583.5	No Calc	No Calc	No Calc
30	40	33583.5	No Calc	No Calc	No Calc
40	50	33583.5	No Calc	No Calc	No Calc
50	60	33583.5	No Calc	No Calc	No Calc
60	70	33583.5	No Calc	No Calc	No Calc
70	80	33583.5	No Calc	No Calc	No Calc
80	90	33583.5	No Calc	No Calc	No Calc
90	110	33583.5	No Calc	No Calc	No Calc



110	120	33583.5	No Calc	No Calc	No Calc
120	130	33583.5	No Calc	No Calc	No Calc
130	140	33583.5	No Calc	No Calc	No Calc
140	150	No Calc	No Calc	No Calc	No Calc
150	160	No Calc	No Calc	No Calc	No Calc
160	170	783.5	No Calc	No Calc	No Calc
170	180	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm³	Corroded Metal Wgt. kg.	Corroded ID Volume cm³	Extra due Misc % kg.
10	20	19845.3	11236284	19396.8	11294153	...
20	30	54261.8	27242492	53516.1	27338726	...
30	40	54261.8	27242492	53516.1	27338726	...
40	50	54261.8	27242492	53516.1	27338726	...
50	60	54261.8	27242492	53516.1	27338726	...
60	70	54261.8	27242492	53516.1	27338726	...
70	80	54261.8	27242492	53516.1	27338726	...
80	90	54261.8	27242492	53516.1	27338726	...
90	110	54261.8	27242492	53516.1	27338726	...
110	120	54261.8	27242492	53516.1	27338726	...
120	130	54261.8	27242492	53516.1	27338726	...
130	140	54261.8	27242492	53516.1	27338726	...
140	150	42589	8407496	42359.3	8420327	...
150	160	42589	8407496	42359.3	8420327	...
160	170	3617.45	1816166	3567.74	1822581	...
170	180	19845.3	11236284	19396.8	11294153	...
Total		725365	0.341E+09	715756	0.342E+09	0

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	336.883	-583.5	0.49997	...	Liquid 10
10	Insl	908.242	-875.25	...	...	Ins: 10
10	Nozl	1385.49	-987.017	...	1445.7	A1_600
10	Nozl	825.017	-902.358	...	-1500	A2_450
10	Nozl	1385.49	-1750.5	...	...	B_600
10	Nozl	273.192	-1436.75	...	1000	T3_100
10	Wght	5500	-1436.75	...	...	Cutback wt
20	Liqd	816.722	1500	...	...	Liquid 20
20	Insl	1299.33	1500	...	...	Ins: 10
20	Nozl	4749.41	1500	...	2004.8	M1_600
20	Wght	60	...	...	...	manway davit-1
30	Liqd	816.776	1500	...	...	Liquid 30
30	Insl	1299.33	1500	...	...	Ins: 10
30	Wght	30	1351.97	...	...	MTI 1 2 3 4 5 6
40	Sadl	4979.12	1455	2029	...	New Saddle
40	Pack	87535.1	1500	1700	...	Pack: [1 of 1]
40	Liqd	816.776	1500	...	...	Liquid 40
40	Insl	1299.33	1500	...	...	Ins: 10
50	Pack	87535.1	1500	1700	...	Pack: [1 of 1]
50	Liqd	816.722	1500	...	...	Liquid 50
50	Insl	1299.33	1500	...	...	Ins: 10
50	Wght	30	2455.97	...	...	MTI 7 8 9 10 11
60	Pack	87535.1	1500	1700	...	Pack: [1 of 1]
60	Liqd	816.722	1500	...	...	Liquid 60
60	Insl	1299.33	1500	...	...	Ins: 10
70	Pack	87535.1	1500	1700	...	Pack: [1 of 1]
70	Liqd	816.776	1500	...	...	Liquid 70
70	Insl	1299.33	1500	...	...	Ins: 10
70	Wght	300	...	...	...	clip weight
80	Pack	87535.1	1500	1700	...	Pack: [1 of 1]
80	Liqd	816.776	1500	...	...	Liquid 80
80	Insl	1299.33	1500	...	...	Ins: 10
80	Wght	10000	1311.98	...	...	Misc weight
90	Pack	87535.1	1500	1700	...	Pack: [1 of 1]
90	Liqd	816.776	1500	...	...	Liquid 90
90	Insl	1299.33	1500	...	...	Ins: 10
90	Wght	30	1713.97	...	...	MTI 13 14 15 16



110	Pack	87535.1	1500	1700	...	Pack: [1 of 1]
110	Liqd	816.776	1500	...	...	Liquid 110
110	Insl	1299.33	1500	...	...	Ins: 10
120	Sadl	4979.12	755	2029	...	New Saddle
120	Pack	87535.1	1500	1700	...	Pack: [1 of 1]
120	Liqd	816.776	1500	...	...	Liquid 120
120	Insl	1299.33	1500	...	...	Ins: 10
120	Nozl	4749.41	2300	...	2004.8	M2_600
120	Wght	35	1569.98	...	...	MTI 19 20 21 22
120	Wght	60	...	...	...	manway davit-2
130	Liqd	816.776	1500	...	...	Liquid 130
130	Insl	1299.33	1500	...	...	Ins: 10
140	Liqd	252.071	460	...	...	Liquid 140
140	Insl	431.306	460	...	...	Ins: 10
150	Liqd	252.071	460	...	...	Liquid 150
150	Insl	431.306	460	...	...	Ins: 10
160	Liqd	54.4517	100	...	...	Liquid 160
160	Insl	86.6223	100	...	...	Ins: 10
170	Liqd	336.883	583.5	0.49997	...	Liquid 170
170	Insl	908.242	875.25	...	...	Ins: 10
170	Nozl	1291.29	1750.5	...	...	C_350
170	Nozl	280.826	1581.69	-750	...	T1_100
170	Nozl	273.192	1172.28	1300	...	T2_100
170	Wght	5500	1172.28	...	...	Cutback-2 wt
40	Trun	3277.22	1455	...	...	Trunnions
120	Trun	3277.22	754.997	...	...	Trunnions

Total Weight of Each Detail Type:

Saddles	9958.2
Packing	700280.9
Liquid	10216.7
Insulation	17058.4
Nozzles	15213.3
Weights	21545.0
Trunnions	6554.4

Sum of the Detail Weights 780826.9 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	725365.6	725365.6	725365.6	725365.6	725365.6	725365.6
Saddles	9958.2	9958.2	9958.2	9958.2	9958.2	9958.2
Nozzles	15213.3	15213.3	15213.3	15213.3	15213.3	15213.3
Wld Weights	21545.0	21545.0	21545.0	21545.0	21545.0	21545.0
Trunnions	6554.4	6554.4	6554.4	6554.4	6554.4	6554.4
Packing	...	...	...	...	700280.9	700280.9 *
Insulation	...	...	...	17058.4	17058.4	17058.4
Ope. Liquid	...	...	...	...	...	10216.7
Test Liquid	...	340563.2	...	...	...	...
Totals	778636.6	1119199.6	778636.6	795694.9	1495975.9	1506192.6

Field Installation Options:

* Packing installed after lifting.

Insulation installed before lifting.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	778636.6 kg.
Shop Test Wt.	- Fabricated weight + water (full)	1119199.6 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	778636.6 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	795694.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	1495975.9 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	1506192.6 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	1496583.8 kg.



Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	220775	192533
20	30	359273	320442
30	40	359273	320442
40	50	359273	320442
50	60	359273	320442
60	70	359273	320442
70	80	359273	320442
80	90	359273	320442
90	110	359273	320442
110	120	359273	320442
120	130	359273	320442
130	140	359273	320442
140	150	210339	98696.3
150	160	210339	98696.3
160	170	23951.5	21362.8
170	180	220775	192533

Total 4838179.5 4128687.8 cm²

Total 5207.8 4444.1 ft²

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**Nozzle Flange MAWP Results: (bars & °C)**

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
M1_600	226.99	258.60	285	1500	GR 1.9	...	...	...	...
M2_600	226.99	258.60	285	1500	GR 1.9	...	...	...	...
T1_100	226.99	258.60	285	1500	GR 1.9	...	...	...	...
T2_100	226.99	258.60	285	1500	GR 1.9	...	...	...	...
Min Rating	226.991	258.600, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.97) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$
 $= \min(0.85, 0.91)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.613
Structure Height to Diameter ratio	9.388

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm.}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 2590 + 300 = 2890 \text{ mm.}$
 $= 9.48 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because $z (9.48 \text{ ft.}) < 15 \text{ ft.}$
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900)^{2/9.5}$
 $= 0.85$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: $[Kzt]$:



$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.85 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$$

$$= 16 \text{ psf } [78.1] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16 \cdot 0.85 \cdot 0.61 \cdot 64.7$$

$$= 539 \text{ lbs. } [245] \text{ Kgf}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Left head	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-1	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-2	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-3	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-4	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-5	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-6	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-7	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-8	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-10	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-11	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-12	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Flange-1	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Flange-2	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-13	2890.0	0.000	0.000	0.000	0.849	1.000	78.120
Right head	2890.0	0.000	0.000	0.000	0.849	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Cutback wt	0.00	2890.00	78.12	0.00
manway davit-	0.00	2890.00	78.12	0.00
MTI 1 2 3 4 5	0.00	2890.00	78.12	0.00
MTI 7 8 9 10	0.00	2890.00	78.12	0.00
clip weight	0.00	2890.00	78.12	0.00
Misc weight	0.00	2890.00	78.12	0.00
MTI 13 14 15	0.00	2890.00	78.12	0.00
MTI 19 20 21	0.00	2890.00	78.12	0.00
manway davit-	0.00	2890.00	78.12	0.00
Cutback-2 wt	0.00	2890.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	2890	4693.2	60068.4	78.12	269.072
20	30	2890	4814.4	144432	78.12	646.973
30	40	2890	4814.4	144432	78.12	646.973
40	50	2890	4814.4	144432	78.12	646.973
50	60	2890	4814.4	144432	78.12	646.973
60	70	2890	4814.4	144432	78.12	646.973
70	80	2890	4814.4	144432	78.12	646.973
80	90	2890	4814.4	144432	78.12	646.973
90	110	2890	4814.4	144432	78.12	646.973



110	120	2890	4814.4	144432	78.12	646.973
120	130	2890	4814.4	144432	78.12	646.973
130	140	2890	4814.4	144432	78.12	646.973
140	150	2890	4320	39744	78.12	178.031
150	160	2890	4320	39744	78.12	178.031
160	170	2890	4814.4	9628.8	78.12	43.1315
170	180	2890	4693.2	60068.4	78.12	269.072

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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**Earthquake Load Calculation Results:****Earthquake Load Calculation:****Input Values:**

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	0.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	C
Component Elevation Ratio z/h	1.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.3 * 0.15 = 0.19 \\
 SM1 &= Fv * S1 = 0.5 * 0.065 = 0.032 \\
 SDS &= 2/3 * Sms = 2/3 * 0.19 = 0.13 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.032 = 0.022
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn$ where $Ct = 0.020$, $x = 0.75$ and $hn = \text{Structural Height (ft.)}$
 $= 0.020(14.20.75)$
 $= 0.147 \text{ seconds}$

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.15, 0.03) \text{ per 12.8.2}$
 $= 0.030$

Loads on Building/Structure Supported Equipment:

$$\begin{aligned}
 &\text{Seismic Force per equation 13.3-1, [Fp]:} \\
 &= 0.4 \cdot Ap \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / Ie) \\
 &= 0.4 \cdot 1 \cdot 0.13 \cdot 1496234 (1 + 2 \cdot 1) / (3/1.25) \\
 &= 95310.125 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Maximum value of Fp per equation 13.3-2:} \\
 &= 1.6 \cdot Sds \cdot I \cdot W \\
 &= 1.6 \cdot 0.13 \cdot 1.25 \cdot 1496234 = 381240 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Minimum value of Fp per equation 13.3-3:} \\
 &= 0.3 \cdot 0.13 \cdot 1.25 \cdot 1496234 \\
 &= 71482.594 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Total Base Shear } V = Fp, [V]: \\
 &= 95310.125 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Vertical load per 12.4-4, [Ev]:} \\
 &= 0.2 \cdot Sds \cdot W \\
 &= 0.2 \cdot 0.13 \cdot 1496234 \\
 &= 38124.1 \text{ Kgf}
 \end{aligned}$$

**Earthquake Load Calculation:**

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1750.5	83124.1	5434.36
20	30	1700	83124.1	5277.59
30	40	1700	83124.1	5277.59
40	50	1700	83124.1	5277.59
40	50	1700	83124.1	5277.59
50	60	1700	83124.1	5277.59
60	70	1700	83124.1	5277.59
70	80	1700	83124.1	5277.59
80	90	1700	83124.1	5277.59
90	110	1700	83124.1	5277.59
110	120	1700	83124.1	5277.59
120	0	1700	83124.1	5277.59
120	130	1700	83124.1	5277.59
130	140	1700	83124.1	5277.59
140	150	1700	83124.1	5277.59
150	160	1700	83124.1	5277.59
160	170	1700	83124.1	5277.59
170	180	1750.5	83124.1	5434.36

The Earthquake Weight does not include the weight of the saddles.

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**Shop/Field Installation Options :**

Packing is installed in the Field after being lifted.
Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	17605.002 mm.
Center of Gravity of the Packing	18000.000 mm.
Center of Gravity of the Liquid	17525.652 mm.
Center of Gravity of the Insulation	17583.641 mm.
Center of Gravity of the Nozzles	13740.908 mm.
Center of Gravity of the Added Weights (Operating)	18226.291 mm.
Center of Gravity of the Added Weights (Empty)	18226.291 mm.
Center of Gravity of the Trunnions	17605.000 mm.
Center of Gravity of Bare Shell New and Cold	18694.041 mm.
Center of Gravity of Bare Shell Corroded	18709.748 mm.
Vessel CG in the Operating Condition	18287.064 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	18561.227 mm.
Vessel CG in the Test Condition	18246.062 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	35040.00 mm.
Total vessel weight (No Liquid) Twt	795694.81 Kg
Impact weight multiplication factor Imp	1.20
Design lifting weight, DWT = Imp * Twt	954833.75 Kg
Distance to the Lug on the Left End	7455.00 mm.
Distance to the Lug on the Right End	27755.00 mm.
Design Reaction force at the Left End Trunnion	432439.62 Kg
Design Reaction force at the Right End Trunnion	522394.09 Kg
CG distance during field lifting operation	18561.23 mm.
CG distance from the left end lug/trunnion	11106.23 mm.
CG distance from the nearer lifting lug/trunnion	9193.77 mm.
Compressive Allowable Increase Factor	1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	8.13	27600.01	125.66
Shear	1.20	28200.01	144.90 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kg	Shear Stress N./mm ²
0.00	1.5	0.0	88411.4	0.7
6000.00	734154.8	3.6	203344.8	0.9
15000.00	400370.0	2.0	23274.9	0.1
24000.01	1007706.4	4.9	182413.7	0.8
33000.02	335241.2	1.6	167980.0	0.7
35040.03	44541.6	0.4	8841.1	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.065 at elevation 27600 mm. - Must be <=1



Note: The rigging analysis is performed using a uniformly distributed load.

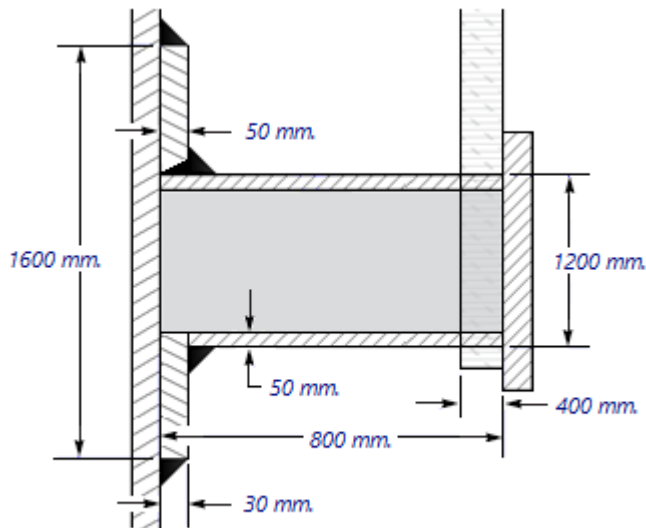
--- Plot data successfully generated---

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Trunnion Analysis: Left End of Vessel

Input Data:

Lifting Lug Material	SA-387 11
Lifting Lug Yield Stress	Yield 309.99 N./mm ²
Trunnion Outside Diameter	Tod 1200.000 mm.
Trunnion Thickness	Tthk 50.0000 mm.
Trunnion Length	Tl 800.0000 mm.
Reinforcing Pad Used	Yes
Pad Outside Diameter	Dp 1600.000 mm.
Pad Thickness	Thkp 50.0000 mm.
Pad Edge Fillet Weld Thk	Tfp 30.0000 mm.
Bail/Sling Width	Tt 400.0000 mm.
Sling Angle from Horizontal	90.0 degrees
Occasional Load Factor	Occfac 1.0
Impact Factor	Impfac 1.20
Number of Lugs in Group	2



Trunnion may be inserted or abutting

Trunnion Calculation Results:

Trunnion Lifting Load [Tlf]:
 = Resultant Load • Impact Factor
 = 180183 • 1.2
 = 216219.8 Kgf

Trunnion Section Modulus [Z]:

$$= \pi (Tod^4 - Tid^4) / (32 \cdot Tod)$$

$$= \pi (1200^4 - 1100^4) / (32 \cdot 1200)$$

 = 49864596.0 mm.³

Trunnion Bending Moment [Tmom]:

$$= (Tl - 0.5 \cdot Tt) / 9.807 \cdot Tlf$$

$$= (800 - 0.5 \cdot 400) / 9.807 \cdot 216220$$

 = 129734.5 Kg-m.

Trunnion Calculated Bending Stress [Sbc]:

$$= Tmom \cdot 9.807 / Z$$

$$= 129734 \cdot 9.807 / 49864596$$

 = 25.51 N./mm² [Passed] Sbc < min(0.66 • SY • Occfac, 0.75 • Sy) (204.60)

Trunnion Cross Section Area [Txa]:

$$= \pi / 4 (Tod^2 - Tid^2)$$

$$= \pi / 4 (1200^2 - 1100^2)$$



$$= 1806.4 \text{ cm}^2$$

Trunnion Calculated Shear Stress [Ssc]:

$$\begin{aligned} &= Tlf/Txsa \\ &= 216220/1806 \\ &= 11.74 \text{ N./mm}^2 \text{ [Passed] } Ssc < 0.4 \cdot Sy \cdot Occfac (124.00) \end{aligned}$$

Trunnion Unity Check [Uchk]:

$$\begin{aligned} &= Sbc/Sba + Ssc/Ssa \\ &= 25.5/205 + 11.7/124 \\ &= 0.219 \text{ [Passed]} \end{aligned}$$

Weld Section Modulus at Edge of Trunnion [Sw]:

$$\begin{aligned} &= \pi (T_{od}/2)^2 \\ &= \pi (1200/2)^2 \\ &= 1130972.4 \text{ mm}^2 \end{aligned}$$

Shear Stress in Weld due to Bending Moment [fs]:

$$\begin{aligned} &= T_{mom}/Sw \\ &= 129734/1130972 \\ &= 114.708 \text{ Kg/mm.} \end{aligned}$$

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:

$$\begin{aligned} &= \max(10\text{mm}, fs/(0.4 \cdot Sy)) \\ &= \max(10\text{mm}, 115/(0.4 \cdot 310)) \\ &= 10.000 \text{ mm.} \end{aligned}$$

Trunnion Bearing Contact Area [Tbca]:

$$\begin{aligned} &= T_{od} \cdot T_t \\ &= 1200 \cdot 400 \\ &= 4800.0 \text{ cm}^2 \end{aligned}$$

Trunnion Bearing Stress [Sbec]:

$$\begin{aligned} &= Tlf/Tbca \\ &= 216220/4800 \\ &= 4.42 \text{ N./mm}^2 \text{ [Passed] } Sbec < \min(0.75 \cdot SY \cdot Occfac, 0.9 \cdot Sy) (232.50) \end{aligned}$$

Summary of Results:

Trunnion Unity Check Value: 0.2194 Must be less than or equal to 1

Comparison of Stress Values versus Allowables: (N./mm²)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	25.51	204.60	Pass
Shear Stress	11.74	124.00	Pass
Bearing Stress	4.42	232.50	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

Input Echo, WRC107/537 Item 1, Description: Trunnion

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	3400.000 mm.
Vessel Thickness	Tv	206.000 mm.

Design Temperature	T1	37.8	C
Attachment Type	Type	Round	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm.
Nozzle Diameter	Dn	1200.000	mm.
Nozzle Thickness	Tn	50.000	mm.
Nozzle Cold S.I. Allowable	SNmc	207.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	207.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	50.000	mm.
Diameter of Reinforcing Pad	Dpad	1660.000	mm.
Design Internal Pressure	Dp	0.000	bars
Include Pressure Thrust		No	

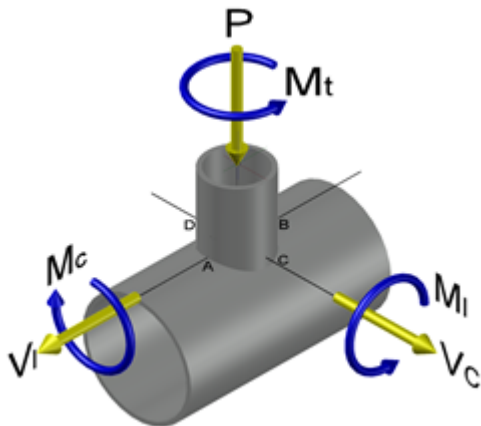
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	VL	0.0	Kgf
Circumferential Shear (SUS)	Vc	216219.8	Kgf
Circumferential Moment (SUS)	Mc	129734.5	Kg-m.
Longitudinal Moment (SUS)	ML	0.0	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))} \\
 &= 1200.0 + 2 \cdot 1.65 \cdot \sqrt{(1803.0 (206.0 - 0))} \\
 &= 3211.154 \text{ mm.}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	216219.8	Kgf
Longitudinal Shear	VL	0.0	Kgf
Circumferential Moment	MC	129734.5	Kg-m.
Longitudinal Moment	ML	0.0	Kg-m.



Torsional Moment MT 0.0 Kg-m.

Dimensionless Parameters used : Gamma = 7.14

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.287	4C	1.181	(A,B)
N(PHI) / (P/Rm)	0.287	3C	0.997	(C,D)
M(PHI) / (P)	0.287	2C1	0.068	(A,B)
M(PHI) / (P)	0.287	1C	0.093	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.287	3A	0.237	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.287	1A	0.093	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.287	3B	0.848	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.287	1B	0.040	(A,B,C,D)
N(x) / (P/Rm)	0.287	3C	0.997	(A,B)
N(x) / (P/Rm)	0.287	4C	1.181	(C,D)
M(x) / (P)	0.287	1C1	0.095	(A,B)
M(x) / (P)	0.287	2C	0.068	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.287	4A	0.447	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.287	2A	0.054	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.287	4B	0.240	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.287	2B	0.067	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Circ. Bend. MC		0.0	0.0	0.0	0.0	-20.7	20.7	20.7	-20.7
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-22.0	19.5	22.0	-19.5
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-2.3	-2.3	2.3	2.3
Long. Bend. MC		0.0	0.0	0.0	0.0	-12.0	12.0	12.0	-12.0
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-14.4	9.7	14.4	-9.7
Shear VC		4.4	4.4	-4.4	-4.4	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		4.4	4.4	-4.4	-4.4	0.0	0.0	0.0	0.0
Str. Int.		8.8	8.8	8.8	8.8	22.0	19.5	22.0	19.5

Dimensionless Parameters used : Gamma = 8.75

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.403	4C	1.250	(A,B)
N(PHI) / (P/Rm)	0.403	3C	0.875	(C,D)



M(PHI) / (P)	0.403	2C1	0.037	(A,B)
M(PHI) / (P)	0.403	1C !	0.060	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.403	3A	0.375	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.403	1A	0.084	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.403	3B	1.012	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.403	1B	0.026	(A,B,C,D)

N(x)	/ (P/Rm)	0.403	3C	0.875	(A,B)
N(x)	/ (P/Rm)	0.403	4C	1.250	(C,D)
M(x)	/ (P)	0.403	1C1	0.060	(A,B)
M(x)	/ (P)	0.403	2C	0.038	(C,D)
N(x)	/ (MC/(Rm**2 * Beta))	0.403	4A	0.794	(A,B,C,D)
M(x)	/ (MC/(Rm * Beta))	0.403	2A	0.044	(A,B,C,D)
N(x)	/ (ML/(Rm**2 * Beta))	0.403	4B	0.374	(A,B,C,D)
M(x)	/ (ML/(Rm * Beta))	0.403	2B	0.046	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Circ. Bend. MC		0.0	0.0	0.0	0.0	-20.8	20.8	20.8	-20.8
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-22.6	19.1	22.6	-19.1
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-3.7	-3.7	3.7	3.7
Long. Bend. MC		0.0	0.0	0.0	0.0	-10.8	10.8	10.8	-10.8
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-14.5	7.0	14.5	-7.0
Shear VC		3.9	3.9	-3.9	-3.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		3.9	3.9	-3.9	-3.9	0.0	0.0	0.0	0.0
Str. Int.		7.9	7.9	7.9	7.9	22.6	19.1	22.6	19.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-20.7	20.7	20.7	-20.7
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0



Long. P1 (SUS)	0.0	0.0	0.0	0.0	-2.3	-2.3	2.3	2.3
Long. Q (SUS)	0.0	0.0	0.0	0.0	-12.0	12.0	12.0	-12.0
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	4.4	4.4	-4.4	-4.4	0.0	0.0	0.0	0.0
Shear Q (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+P1 (SUS)	8.8	8.8	8.8	8.8	2.3	2.3	2.3	2.3
Pm+P1+Q (Total)	8.8	8.8	8.8	8.8	22.0	19.5	22.0	19.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	207.00	Passed
Pm+P1 (SUS)	8.79	310.49	Passed
Pm+P1+Q (TOTAL)	21.97	620.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. P1 (SUS)		0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-20.8	20.8	20.8	-20.8
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. P1 (SUS)		0.0	0.0	0.0	0.0	-3.7	-3.7	3.7	3.7
Long. Q (SUS)		0.0	0.0	0.0	0.0	-10.8	10.8	10.8	-10.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)		3.9	3.9	-3.9	-3.9	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+P1 (SUS)		7.9	7.9	7.9	7.9	3.7	3.7	3.7	3.7
Pm+P1+Q (Total)		7.9	7.9	7.9	7.9	22.6	19.1	22.6	19.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	207.00	Passed
Pm+P1 (SUS)	7.90	310.49	Passed
Pm+P1+Q (TOTAL)	22.60	620.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Ammonia Convertor (105-D)
PV Elite 28 U1 Licensee: SL User
FileName : 1-105-D
Trunnion Calcs: Left End



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$$= \pi/4 (1200^2 - 1100^2)$$

$$= 1806.4 \text{ cm}^2$$

Trunnion Calculated Shear Stress [Ssc]:

$$= Tlf/Txsa$$

$$= 261197/1806$$

$$= 14.18 \text{ N./mm}^2 \text{ [Passed] } Ssc < 0.4 \cdot Sy \cdot Occfac (124.00)$$

Trunnion Unity Check [Uchk]:

$$= Sbc/Sba + Ssc/Ssa$$

$$= 30.8/205 + 14.2/124$$

$$= 0.265 \text{ [Passed]}$$

Weld Section Modulus at Edge of Trunnion [Sw]:

$$= \pi (Tod/2)^2$$

$$= \pi (1200/2)^2$$

$$= 1130972.4 \text{ mm}^2$$

Shear Stress in Weld due to Bending Moment [fs]:

$$= Tmom/Sw$$

$$= 156721/1130972$$

$$= 138.569 \text{ Kgf/mm.}$$

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:

$$= \max(10 \text{ mm}, fs/(0.4 \cdot Sy))$$

$$= \max(10 \text{ mm}, 139/(0.4 \cdot 310))$$

$$= 10.959 \text{ mm.}$$

Trunnion Bearing Contact Area [Tbca]:

$$= Tod \cdot Tt$$

$$= 1200 \cdot 400$$

$$= 4800.0 \text{ cm}^2$$

Trunnion Bearing Stress [Sbec]:

$$= Tlf/Tbca$$

$$= 261197/4800$$

$$= 5.34 \text{ N./mm}^2 \text{ [Passed] } Sbec < \min(0.75 \cdot SY \cdot Occfac, 0.9 \cdot Sy) (232.50)$$

Summary of Results:

Trunnion Unity Check Value: 0.2650 Must be less than or equal to 1

Comparison of Stress Values versus Allowables: (N./mm²)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	30.82	204.60	Pass
Shear Stress	14.18	124.00	Pass
Bearing Stress	5.34	232.50	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).**Input Echo, WRC107/537 Item 1, Description: Trunnion**

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	3400.000 mm.
Vessel Thickness	Tv	206.000 mm.

Design Temperature	T1	37.8	C
Attachment Type	Type	Round	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm.
Nozzle Diameter	Dn	1200.000	mm.
Nozzle Thickness	Tn	50.000	mm.
Nozzle Cold S.I. Allowable	SNmc	207.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	207.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	50.000	mm.
Diameter of Reinforcing Pad	Dpad	1660.000	mm.
Design Internal Pressure	Dp	0.000	bars
Include Pressure Thrust	No		

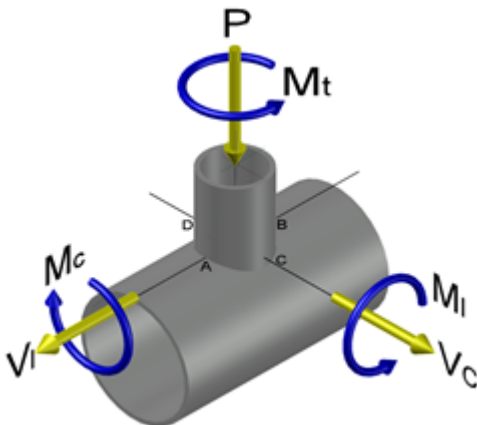
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	Vl	0.0	Kgf
Circumferential Shear (SUS)	Vc	261197.0	Kgf
Circumferential Moment (SUS)	Mc	156721.3	Kg-m.
Longitudinal Moment (SUS)	Ml	0.0	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1200.0 + 2 \cdot 1.65 \cdot \sqrt{(1803.0 (206.0 - 0))}$$

$$= 3211.154 \text{ mm.}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	261197.0	Kgf
Longitudinal Shear	VL	0.0	Kgf
Circumferential Moment	MC	156721.3	Kg-m.



Longitudinal Moment ML 0.0 Kg-m.
Torsional Moment MT 0.0 Kg-m.

Dimensionless Parameters used : Gamma = 7.14

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.287	4C	1.181	(A,B)
N(PHI) / (P/Rm)	0.287	3C	0.997	(C,D)
M(PHI) / (P)	0.287	2C1	0.068	(A,B)
M(PHI) / (P)	0.287	1C	0.093	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.287	3A	0.237	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.287	1A	0.093	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.287	3B	0.848	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.287	1B	0.040	(A,B,C,D)
N(x) / (P/Rm)	0.287	3C	0.997	(A,B)
N(x) / (P/Rm)	0.287	4C	1.181	(C,D)
M(x) / (P)	0.287	1C1	0.095	(A,B)
M(x) / (P)	0.287	2C	0.068	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.287	4A	0.447	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.287	2A	0.054	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.287	4B	0.240	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.287	2B	0.067	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Circ. Bend. MC		0.0	0.0	0.0	0.0	-25.1	25.1	25.1	-25.1
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-26.5	23.6	26.5	-23.6
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-2.8	-2.8	2.8	2.8
Long. Bend. MC		0.0	0.0	0.0	0.0	-14.6	14.6	14.6	-14.6
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-17.4	11.8	17.4	-11.8
Shear VC		5.3	5.3	-5.3	-5.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		5.3	5.3	-5.3	-5.3	0.0	0.0	0.0	0.0
Str. Int.		10.6	10.6	10.6	10.6	26.5	23.6	26.5	23.6

Dimensionless Parameters used : Gamma = 8.75

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.403	4C	1.250	(A,B)



N(PHI) / (P/Rm)	0.403	3C	0.875	(C,D)
M(PHI) / (P)	0.403	2C1	0.037	(A,B)
M(PHI) / (P)	0.403	1C !	0.060	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.403	3A	0.375	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.403	1A	0.084	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.403	3B	1.012	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.403	1B	0.026	(A,B,C,D)
N(x) / (P/Rm)	0.403	3C	0.875	(A,B)
N(x) / (P/Rm)	0.403	4C	1.250	(C,D)
M(x) / (P)	0.403	1C1	0.060	(A,B)
M(x) / (P)	0.403	2C !	0.038	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.403	4A	0.794	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.403	2A	0.044	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.403	4B	0.374	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.403	2B	0.046	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Circ. Bend. MC		0.0	0.0	0.0	0.0	-25.2	25.2	25.2	-25.2
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-27.3	23.0	27.3	-23.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-4.5	-4.5	4.5	4.5
Long. Bend. MC		0.0	0.0	0.0	0.0	-13.0	13.0	13.0	-13.0
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-17.6	8.5	17.6	-8.5
Shear VC		4.8	4.8	-4.8	-4.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		4.8	4.8	-4.8	-4.8	0.0	0.0	0.0	0.0
Str. Int.		9.5	9.5	9.5	9.5	27.3	23.0	27.3	23.0

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-25.1	25.1	25.1	-25.1



Long. Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)	0.0	0.0	0.0	0.0	-2.8	-2.8	2.8	2.8
Long. Q (SUS)	0.0	0.0	0.0	0.0	-14.6	14.6	14.6	-14.6
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	5.3	5.3	-5.3	-5.3	0.0	0.0	0.0	0.0
Shear Q (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)	10.6	10.6	10.6	10.6	2.8	2.8	2.8	2.8
Pm+Pl+Q (Total)	10.6	10.6	10.6	10.6	26.5	23.6	26.5	23.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	207.00	Passed
Pm+Pl (SUS)	10.62	310.49	Passed
Pm+Pl+Q (TOTAL)	26.54	620.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-25.2	25.2	25.2	-25.2
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		0.0	0.0	0.0	0.0	-4.5	-4.5	4.5	4.5
Long. Q (SUS)		0.0	0.0	0.0	0.0	-13.0	13.0	13.0	-13.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		4.8	4.8	-4.8	-4.8	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		9.5	9.5	9.5	9.5	4.5	4.5	4.5	4.5
Pm+Pl+Q (Total)		9.5	9.5	9.5	9.5	27.3	23.0	27.3	23.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	207.00	Passed
Pm+Pl (SUS)	9.54	310.49	Passed
Pm+Pl+Q (TOTAL)	27.30	620.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

Ammonia Convertor (105-D)
PV Elite 28 U1 Licensee: SL User
FileName : 1-105-D
Trunnion Calcs: Right End



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**ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:**

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

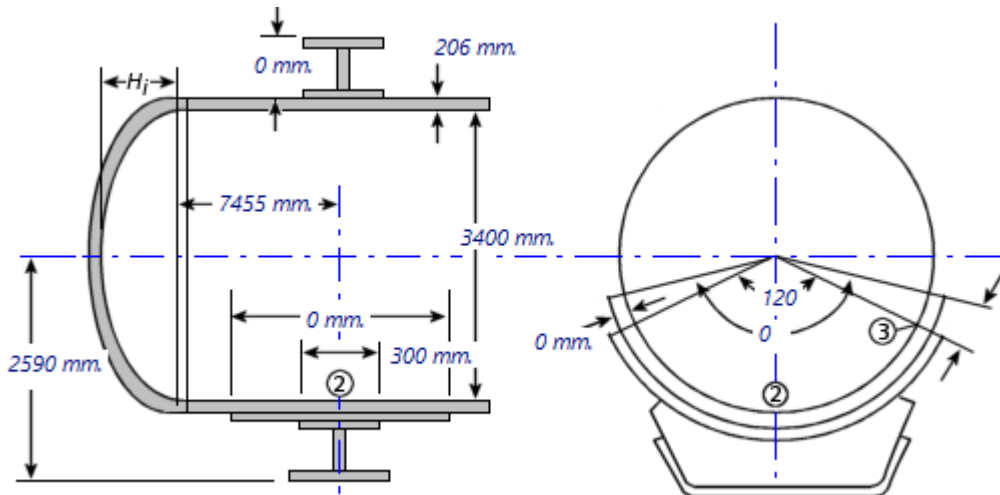
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(300 + 1.56 \cdot \sqrt{1805 \cdot 203}, 2 \cdot 7455)$$

$$= 1244.1718 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1804.50	mm.
Shell Thickness used in this Case	t	206.000	mm.
Stiffened Vessel Length per 4.15.6	L	35044.50	mm.
Distance from Saddle to Vessel tangent	a1 or a	7455.00	mm.
Saddle Width	b1 or b	300.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1804.50	mm.
Shell Allowable Stress used in Calculation		165.20	N./mm ²
Head Allowable Stress used in Calculation		165.20	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2590.00	mm.
Coefficient of Friction	mu	0.00	

**Results for Horizontal Vessel Analysis:**

Saddle Force Q, Operating Case	827612.62	Kgf
Pressure used in Analysis	P	184.725 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	79.20	165.20
Long. Stress at Bottom of Midspan	85.01	165.20
Long. Stress at Top of Saddles	154.07	165.20
Long. Stress at Bottom of Saddles	42.22	165.20
Tangential Shear in Shell	13.95	132.08

Circ. Stress at Horn of Saddle	23.65	206.50
Circ. Compressive Stress in Shell	2.44	165.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (8054/2 + 0) \cdot 2590/3346$$

$$= 9350.6 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(1564, 0) \cdot 2590/20300$$

$$= 199.6 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(95310, 0) \cdot 2590/20300$$

$$= 12160.3 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (95310/2 + 0) \cdot 2590/3346$$

$$= 110652.8 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 716960 + \max(200, 9351, 12160, 110653)$$

$$= 827612.6 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(4.01 \cdot 1.2)^2)$$

$$= 15340.670 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	832591.75	Kgf
Transverse Shear Load	47655.05	Kgf
Longitudinal Shear Load	95310.10	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(3400/2 + 206 + 26)\sin(120/2), 2 \cdot 1805)$$

$$= 3346.323 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$$

$$= -827613 \cdot 7455 [1 - (1 - 7455/35044 + (1805^2 - 1804^2)/(2 \cdot 7455 \cdot 35044))/(1 + (4 \cdot 1804)/(3 \cdot 35044))]$$

$$= -1624600.9 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/(1 + (4\text{hm}^2)/(3L)) - 4a/L)$$

$$= 827613 \cdot 35044/4((1 + 2(1805^2 - 1804^2)/(35044^2))/(1 + (4 \cdot 1804)/(3 \cdot 35044)) - 4 \cdot 7455/35044)$$

$$= 615150.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 t)$$



$$= 185 \cdot 1805 / (2 \cdot 203) - 615150 / (\pi \cdot 1805^2 \cdot 203)$$

$$= 79.20 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 185 \cdot 1805 / (2 \cdot 203) + 615150 / (\pi \cdot 1805^2 \cdot 203)$$

$$= 85.01 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 185 \cdot 1805 / (2 \cdot 203) - 1624600 / (0.11 \cdot \pi \cdot 1805^2 \cdot 203)$$

$$= 154.07 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 185 \cdot 1805 / (2 \cdot 203) + 1624600 / (0.19 \cdot \pi \cdot 1805^2 \cdot 203)$$

$$= 42.22 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 827613 (35044 - 2 \cdot 7455) / (35044 + (4 \cdot 1804 / 3))$$

$$= 444949.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 444949 / (1805 \cdot 203)$$

$$= 13.95 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1805 \cdot 203)}$$

$$= 472.086 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 827613 \cdot 0.1 / (203 (300 + 472 + 472))$$

$$= -2.44 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -827613 / (4 \cdot 203 (300 + 472 + 472)) - 3 \cdot 0.053 \cdot 827613 / (2 \cdot 203^2)$$

$$= -23.65 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(300 + 1.56 \cdot \sqrt{(1805 \cdot 203)}, 2 \cdot 7455)$$

$$= 1244.17 \text{ mm.}$$

Distance between Saddle Supports [Ls]:

$$= 20300.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.13139\text{E-}04 \cdot 20300 (285 - 21.1)$$

$$= 70.384 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3556.0000	mm.
Baseplate Thickness	Bpthk	110.0000	mm.
Baseplate Width	Bpwid	1012.0001	mm.
Number of Ribs (inc. outside ribs)	Nribs	9	
Rib Thickness	Ribtk	30.0000	mm.
Web Thickness	Webtk	30.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	247.9	N./mm ²
Height of Web at Center	Hw,c	547.0	mm.
Friction Coefficient	mu	0.000	

**Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)**

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1703 \cdot 203)} + 350)^3) / 12$$

$$= 88341.156 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}} (C1 - Y)^2$$

$$= 88341 + 2572 (327 - 102)^2$$

$$= 1399118.250 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1267.2	203.0	101.5	2572.5	26110710.0	0.140E+07
Wearplate	350.0	26.0	216.0	91.0	1965599.8	0.113E+05
Web	30.0	548.0	503.0	164.4	8269324.0	0.919E+05
BasePlate	1012.0	110.0	832.0	1113.2	92618256.0	0.285E+07
Totals	...	...	...	3941.1	128963888.0	0.435E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 128963888 / 3941$$

$$= 327.229 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 827613$$

$$= 168437.1094 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 12.0695 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 148.7688 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 2590 - 1703 \cdot \sin(120 / 2) / 1.05$$

$$= 1181.630 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 199034.3438 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 14.6831 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 165.2987 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(827613 + 4979) \cdot 1012) / (4 \cdot 3556 \cdot 165)}$$

$$= 102.680 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120 / 2) (1700 + 206 + 26)$$

$$= 3346.322 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$



$$= 3346 / (9 - 1)$$

$$= 418.290 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 418 \cdot 1012 / 2$$

$$= 2116.549 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 832592 / (3556 \cdot 1012)$$

$$= 23.136 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 2117 \cdot 23.1$$

$$= 48968.645 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 81 + 62.7$$

$$= 143.744 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 48969 / 144$$

$$= 33.409 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	30.0	300.0	...	90.0	...	0.675E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 300 - 30$$

$$= 270.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 144$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 300 / 2$$

$$= 150.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(Total \text{ Inertia} / Total \text{ Area})}$$

$$= \sqrt{(6750 / 144)}$$

$$= 68.526 \text{ mm.}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 2590 - \cos(120 / 2) (1700 + 206 + 26) - 110$$

$$= 1514.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 248)}$$

$$= 126.168$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1514 / 68.5$$



$$= 22.094$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 95310 / (2 \cdot 3556) \cdot 418 \cdot 1514 / 2$$

$$= 4243.552 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($22.1 < 126$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.1)^2 / (2 \cdot 126^2)) 248 /$$

$$(5/3 + 3 \cdot (22.1) / (8 \cdot 126) - (22.1^3) / (8 \cdot 126^3))$$

$$= 141 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 33.4 / 141 + (4244 \cdot 150 / 67500000) / 165$$

$$= 0.796$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	30.0	270.0	0.0	81.0	0.0	0.674E+04
Web	418.3	30.0	0.0	125.5	0.0	94.1
Totals	...	...	...	206.5	...	0.684E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 206$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300 / 2$$

$$= 150.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2590 - \cos(1932 / 2) (110 + 0 + 0) - 0$$

$$= 548.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (6837 / 206) }$$

$$= 57.544 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 548 / 57.5$$

$$= 9.523$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 95310 / (2 \cdot 3556)$$

$$= 13.401 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 13.4 \cdot 418 \cdot 548$$

$$= 3071.951 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 3072 / 455824$$

$$= 66.090 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($9.52 < 126$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$



$$= (1 - (9.52)^2 / (2 \cdot 126^2)) \cdot 248 / (5/3 + 3 \cdot (9.52) / (8 \cdot 126) - (9.52^3) / (8 \cdot 126^3))$$

$$= 145.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 23.3 / 146 + (3072 \cdot 150 / 6837) / 165$$

$$= 0.559$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	16
Total Number of Bolts in Tension/Baseplate	Nbt	8
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.0 mm.
User Defined Area in a Single Bolt	UserBoltArea	31.3100 cm ²
BasePlate Allowable Stress	S	165.14 N./mm ²
Area Available in a Single Bolt	BlArea	28.4051 cm ²
Saddle Load QO (Weight)	QO	721939.0 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	12160.3 Kgf
Maximum Transverse Force	Ft	47655.1 Kgf
Maximum Longitudinal Force	Fl	95310.1 Kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
= 328951.1 Kgf

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
= max(Ff, Fl) / (Bolt Area • Number of Bolts)
= 328951 / (28.4 • 16)
= 71.0 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= Ft / (Bolt Area • Number of Bolts)
= 47655 / (28.4 • 16)
= 10.3 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= Fl / (BoltShearAllowable • Nbolts)
= 95310 / (82.8 • 16)
= 7.0553 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 2590 • 47655 + 0
= 123429.04 Kg-m.

Eccentricity (e):
= Rmom / QO
= 123429 / 721939
= 170.97 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 7.06, 0]
= 7.0553 cm²

**MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: A**

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 18 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 18 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (159 \cdot 1) / (26 - 3 - 0)$$

$$= 1.000$$

Warning: Impact Testing Is Mandatory!

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(300 + 1.56 \cdot \sqrt{1805 \cdot 203}, 2 \cdot 6065)$$

$$= 1244.1718 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1804.50	mm.
Shell Thickness used in this Case	t	206.000	mm.
Stiffened Vessel Length per 4.15.6	L	35044.50	mm.
Distance from Saddle to Vessel tangent	a1 or a	6065.00	mm.
Saddle Width	b1 or b	300.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		165.20	N./mm ²
Head Allowable Stress used in Calculation		165.20	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2590.00	mm.
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		928051.38	Kgf
Pressure used in Analysis	P	184.725	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	70.09	165.20
Long. Stress at Bottom of Midspan	94.13	165.20
Long. Stress at Top of Saddles	123.35	165.20
Long. Stress at Bottom of Saddles	59.25	165.20
Tangential Shear in Shell	19.02	132.08
Circ. Stress at Horn of Saddle	26.52	206.50
Circ. Compressive Stress in Shell	2.74	165.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (8054/2 + 0) \cdot 2590/3346$$

$$= 9350.6 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(1564, 0) \cdot 2590/20300$$

$$= 199.6 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(95310, 0) \cdot 2590/20300$$

$$= 12160.3 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (95310/2 + 0) \cdot 2590/3346$$

$$= 110652.8 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 817399 + \max(200, 9351, 12160, 110653)$$

$$= 928051.4 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(4.01 \cdot 1.2)^2)$$

$$= 15340.670 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	933030.44	Kgf
Transverse Shear Load	47655.05	Kgf
Longitudinal Shear Load	95310.10	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(3400/2 + 206 + 26) \sin(120/2), 2 \cdot 1805)$$

$$= 3346.323 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -928051 \cdot 6065 [1 - (1 - 6065/35044 + (1805^2 - 0^2) / (2 \cdot 6065 \cdot 35044)) / (1 + (4 \cdot 0) / (3 \cdot 35044))]$$

$$= -931025.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 \cdot ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 928051 \cdot 35044/4 \cdot ((1 + 2(1805^2 - 0^2)/(35044^2)) / (1 + (4 \cdot 0) / (3 \cdot 35044)) - 4 \cdot 6065/35044)$$

$$= 2545309.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 185 \cdot 1805/(2 \cdot 203) - 2545309/(\pi \cdot 1805^2 \cdot 203)$$

$$= 70.09 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 185 \cdot 1805/(2 \cdot 203) + 2545309/(\pi \cdot 1805^2 \cdot 203)$$

$$= 94.13 \text{ N./mm}^2$$



Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 185 \cdot 1805 / (2 \cdot 203) - 931025 / (0.11 \cdot \pi \cdot 1805^2 \cdot 203)$$

$$= 123.35 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 185 \cdot 1805 / (2 \cdot 203) + 931025 / (0.19 \cdot \pi \cdot 1805^2 \cdot 203)$$

$$= 59.25 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 928051 (35044 - 2 \cdot 6065) / (35044 + (4 \cdot 0 / 3))$$

$$= 606823.7 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 606824 / (1805 \cdot 203)$$

$$= 19.02 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1805 \cdot 203)}$$

$$= 472.086 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 928051 \cdot 0.1 / (203(300 + 472 + 472))$$

$$= -2.74 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -928051 / (4 \cdot 203(300 + 472 + 472)) - 3 \cdot 0.053 \cdot 928051 / (2 \cdot 203^2)$$

$$= -26.52 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(300 + 1.56 \cdot \sqrt{(1805 \cdot 203)}, 2 \cdot 6065)$$

$$= 1244.17 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3556.0000	mm.
Baseplate Thickness	Bpthk	110.0000	mm.
Baseplate Width	Bpwid	1012.0001	mm.
Number of Ribs (inc. outside ribs)	Nribs	9	
Rib Thickness	Ribtk	30.0000	mm.
Web Thickness	Webtk	30.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	247.9	N./mm ²
Height of Web at Center	Hw,c	547.0	mm.
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1703 \cdot 203)} + 350) 203^3) / 12$$

$$= 88341.156 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 88341 + 2572 (327 - 102)^2$$

$$= 1399118.250 \text{ cm}^4$$



	mm.	mm.	mm.	cm ²	mm. ³	cm ⁴
Shell	1267.2	203.0	101.5	2572.5	26110710.0	0.140E+07
Wearplate	350.0	26.0	216.0	91.0	1965599.8	0.113E+05
Web	30.0	548.0	503.0	164.4	8269324.0	0.919E+05
BasePlate	1012.0	110.0	832.0	1113.2	92618256.0	0.285E+07
Totals	...	...	...	3941.1	128963888.0	0.435E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 128963888/3941$$

$$= 327.229 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 928051$$

$$= 188878.5625 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 13.5343 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 148.7688 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2590 - 1703 \cdot \sin(120/2) / 1.05$$

$$= 1181.630 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 223189.0781 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 16.4650 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 165.2987 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(928051 + 4979) \cdot 1012) / (4 \cdot 3556 \cdot 165)}$$

$$= 108.697 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1700 + 206 + 26)$$

$$= 3346.322 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 3346 / (9 - 1)$$

$$= 418.290 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 418 \cdot 1012/2$$

$$= 2116.549 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 933030 / (3556 \cdot 1012)$$

$$= 25.927 \text{ Kg/cm}^2$$

Axial Load [P]:



$$= A_p \cdot B_p$$

$$= 2117 \cdot 25.9$$

$$= 54875.922 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 81 + 62.7$$

$$= 143.744 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 54876/144$$

$$= 37.439 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	30.0	300.0	...	90.0	...	0.675E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300 - 30$$

$$= 270.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/144$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300/2$$

$$= 150.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(6750/144)}$$

$$= 68.526 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2590 - \cos(120/2) (1700 + 206 + 26) - 110$$

$$= 1514.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/248)}$$

$$= 126.168$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1514/68.5$$

$$= 22.094$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 95310 / (2 \cdot 3556) \cdot 418 \cdot 1514/2$$

$$= 4243.552 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($22.1 < 126$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.1)^2 / (2 \cdot 126^2)) 248 / (5/3 + 3 \cdot (22.1) / (8 \cdot 126) - (22.1^3) / (8 \cdot 126^3))$$

$$= 141 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)



$$\begin{aligned}
 &= Sc/Sca + (Rm \cdot C1 / I)/Sba \\
 &= 37.4/141 + (4244 \cdot 150/67500000)/165 \\
 &= 0.825
 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	30.0	270.0	0.0	81.0	0.0	0.674E+04
Web	418.3	30.0	0.0	125.5	0.0	94.1
Totals	...	...	...	206.5	...	0.684E+04

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0/206 \\
 &= 0.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 300/2 \\
 &= 150.000 \text{ mm.}
 \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk} \\
 &= 2590 - \cos(1932/2)(110 + 0 + 0) - 0 \\
 &= 548.000 \text{ mm.}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(6837/206)} \\
 &= 57.544 \text{ mm.}
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 548/57.5 \\
 &= 9.523
 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Baseplate Length}) \\
 &= 95310/(2 \cdot 3556) \\
 &= 13.401 \text{ Kg/mm.}
 \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 13.4 \cdot 418 \cdot 548 \\
 &= 3071.951 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 3072/455824 \\
 &= 66.090 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($9.52 < 126$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\
 &= (1 - (9.52)^2 / (2 \cdot 126^2)) 248 / \\
 &\quad (5/3 + 3 \cdot (9.52) / (8 \cdot 126) - (9.52^3) / (8 \cdot 126^3)) \\
 &= 145.9 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Mbase,c \cdot C1/I)/Sba \\
 &= 26.1/146 + (3072 \cdot 150/6837)/165 \\
 &= 0.579
 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	16
Total Number of Bolts in Tension/Baseplate	Nbt	8
Bolt Design Temperature		21.1 °C



Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.0 mm.
User Defined Area in a Single Bolt	UserBoltArea	31.3100 cm ²
BasePlate Allowable Stress	S	165.14 N./mm ²
Area Available in a Single Bolt	BltArea	28.4051 cm ²
Saddle Load QO (Weight)	QO	822377.7 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	12160.3 Kgf
Maximum Transverse Force	Ft	47655.1 Kgf
Maximum Longitudinal Force	Fl	95310.1 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 95310 / (28.4 \cdot 16)$$

$$= 20.6 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 47655 / (28.4 \cdot 16)$$

$$= 10.3 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 95310 / (82.8 \cdot 16)$$

$$= 7.0553 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2590 \cdot 47655 + 0$$

$$= 123429.04 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 123429 / 822378$$

$$= 150.09 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 7.06, 0]$$

$$= 7.0553 \text{ cm}^2$$

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: A

This element is Post Weld Heat Treated PWHT

$$\text{Computed Minimum Design Metal Temperature} : 18 \text{ } ^\circ\text{C}$$

$$\text{User entered Minimum Design Temperature} : -12 \text{ } ^\circ\text{C}$$

$$\text{The Curve for this Material is} : A$$

$$\text{From Figure 3.8M- with PWHT, the MDMT is} : 18 \text{ } ^\circ\text{C}$$

$$\text{Temperature reduction per Graph 3.13} : 0 \text{ } ^\circ\text{C}$$

$$\text{The computed stress ratio is} : 1.000$$

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$

$$= (159 \cdot 1) / (26 - 3 - 0)$$

$$= 1.000$$



Warning: Impact Testing Is Mandatory!

Review notes about nozzle loadings and supports in the Warnings report.

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**ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:**

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

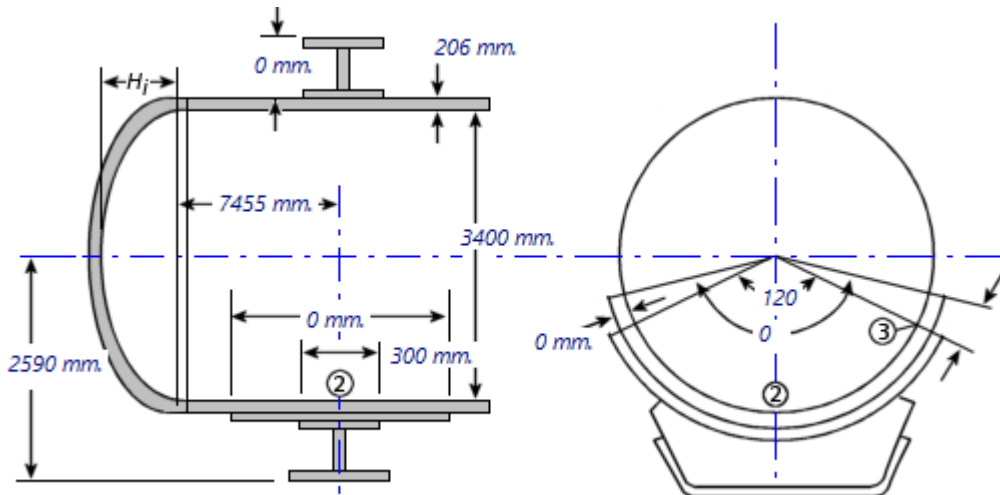
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(300 + 1.56 \cdot \sqrt{1805 \cdot 203}, 2 \cdot 7455)$$

$$= 1244.1718 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1804.50	mm.
Shell Thickness used in this Case	t	206.000	mm.
Stiffened Vessel Length per 4.15.6	L	35044.50	mm.
Distance from Saddle to Vessel tangent	a1 or a	7455.00	mm.
Saddle Width	b1 or b	300.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1804.50	mm.
Shell Allowable Stress used in Calculation		294.49	N./mm ²
Head Allowable Stress used in Calculation		294.49	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2590.00	mm.
Coefficient of Friction	mu	0.00	

**Results for Horizontal Vessel Analysis:**

Saddle Force Q, Test Case, no Ext. Forces		869046.44	Kgf
Pressure used in Analysis	P	288.802	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	125.32	294.49
Long. Stress at Bottom of Midspan	131.42	294.49
Long. Stress at Top of Saddles	203.93	294.49
Long. Stress at Bottom of Saddles	86.49	294.49
Tangential Shear in Shell	14.64	165.23



Circ. Stress at Horn of Saddle	24.83	441.74
Circ. Compressive Stress in Shell	2.57	294.49

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (4027/2 + 0) \cdot 2590/3346$$

$$= 4675.3 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(782, 0) \cdot 2590/20300$$

$$= 99.8 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 864371 + \max(99.8, 4675, 0, 0)$$

$$= 869046.4 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (4.01 \cdot 1.2)^2)$$

$$= 15340.670 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	874025.56	Kgf
Transverse Shear Load	2013.51	Kgf
Longitudinal Shear Load	782.17	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(3400/2 + 206 + 26) \sin(120/2), 2 \cdot 1805)$$

$$= 3346.323 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -869046 \cdot 7455 [1 - (1 - 7455/35044 + (1805^2 - 1804^2) / (2 \cdot 7455 \cdot 35044)) / (1 + (4 \cdot 1804)/(3 \cdot 35044))]$$

$$= -1705935.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 869046 \cdot 35044 / 4 ((1 + 2(1805^2 - 1804^2)/(35044^2)) / (1 + (4 \cdot 1804)/(3 \cdot 35044)) - 4 \cdot 7455/35044)$$

$$= 645947.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 289 \cdot 1805 / (2 \cdot 203) - 645947 / (\pi \cdot 1805^2 \cdot 203)$$

$$= 125.32 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 289 \cdot 1805 / (2 \cdot 203) + 645947 / (\pi \cdot 1805^2 \cdot 203)$$

$$= 131.42 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$



$$= 289 \cdot 1805 / (2 \cdot 203) - 1705935 / (0.11 \cdot \pi \cdot 1805^2 \cdot 203)$$

$$= 203.93 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 289 \cdot 1805 / (2 \cdot 203) + 1705935 / (0.19 \cdot \pi \cdot 1805^2 \cdot 203)$$

$$= 86.49 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 869046 (35044 - 2 \cdot 7455) / (35044 + (4 \cdot 1804 / 3))$$

$$= 467225.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 467225 / (1805 \cdot 203)$$

$$= 14.64 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1805 \cdot 203)}$$

$$= 472.086 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 869046 \cdot 0.1 / (203(300 + 472 + 472))$$

$$= -2.57 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -869046 / (4 \cdot 203(300 + 472 + 472)) - 3 \cdot 0.053 \cdot 869046 / (2 \cdot 203^2)$$

$$= -24.83 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(300 + 1.56 \cdot \sqrt{(1805 \cdot 203)}, 2 \cdot 7455)$$

$$= 1244.17 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3556.0000	mm.
Baseplate Thickness	Bpthk	110.0000	mm.
Baseplate Width	Bpwid	1012.0001	mm.
Number of Ribs (inc. outside ribs)	Nribs	9	
Rib Thickness	Ribtk	30.0000	mm.
Web Thickness	Webtk	30.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	N./mm ²
Height of Web at Center	Hw,c	547.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1703 \cdot 203)} + 350)^3) / 12$$

$$= 88341.156 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 88341 + 2572(327 - 102)^2$$

$$= 1399118.250 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1267.2	203.0	101.5	2572.5	26110710.0	0.140E+07
Wearplate	350.0	26.0	216.0	91.0	1965599.8	0.113E+05
Web	30.0	548.0	503.0	164.4	8269324.0	0.919E+05



BasePlate	1012.0	110.0	832.0	1113.2	92618256.0	0.285E+07
Totals	...	...	...	3941.1	128963888.0	0.435E+07

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 128963888 / 3941 \\
 &= 327.229 \text{ mm.}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 120 / 2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 869046 \\
 &= 176869.7969 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 12.6738 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 185.9966 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 2590 - 1703 \cdot \sin(120/2) / 1.05 \\
 &= 1181.630 \text{ mm.}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 208998.8594 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 15.4181 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 206.6629 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt[3]{(3(869046 + 4979) \cdot 1012) / (4 \cdot 3556 \cdot 207)} \\
 &= 94.088 \text{ mm.}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (1700 + 206 + 26) \\
 &= 3346.322 \text{ mm.}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nrings} - 1) \\
 &= 3346 / (9 - 1) \\
 &= 418.290 \text{ mm.}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 418 \cdot 1012 / 2 \\
 &= 2116.549 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 874026 / (3556 \cdot 1012) \\
 &= 24.287 \text{ Kgf/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 2117 \cdot 24.3 \\
 &= 51405.566 \text{ Kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:



$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 81 + 62.7 \\
 &= 143.744 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P/A_r \\
 &= 51406/144 \\
 &= 35.071 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	30.0	300.0	...	90.0	...	0.675E+04

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 300 - 30 \\
 &= 270.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0/144 \\
 &= 0.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 300/2 \\
 &= 150.000 \text{ mm.}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(6750/144)} \\
 &= 68.526 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 2590 - \cos(120/2) (1700 + 206 + 26) - 110 \\
 &= 1514.000 \text{ mm.}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 199943008/310)} \\
 &= 112.838
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 1514/68.5 \\
 &= 22.094
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2 \\
 &= 782 / (2 \cdot 3556) \cdot 418 \cdot 1514/2 \\
 &= 34.825 \text{ Kg-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($22.1 < 113$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (22.1)^2 / (2 \cdot 113^2)) 310 / \\
 &\quad (5/3 + 3 \cdot (22.1) / (8 \cdot 113) - (22.1)^3 / (8 \cdot 113^3)) \\
 &= 174.8 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba} \\
 &= 35.1/175 + (34.8 \cdot 150/67500000) / 207 \\
 &= 0.204
 \end{aligned}$$

Check of Inside Ribs:



Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	30.0	270.0	0.0	81.0	0.0	0.674E+04
Web	418.3	30.0	0.0	125.5	0.0	94.1
Totals	...	...	...	206.5	...	0.684E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/206$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300/2$$

$$= 150.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 2590 - \cos(1932/2)(110 + 0 + 0) - 0$$

$$= 548.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(6837/206)}$$

$$= 57.544 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 548/57.5$$

$$= 9.523$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 782/(2 \cdot 3556)$$

$$= 0.110 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.11 \cdot 418 \cdot 548$$

$$= 25.210 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 25.2/455824$$

$$= 0.542 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($9.52 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.52)^2 / (2 \cdot 113^2)) 310 /$$

$$(5/3 + 3 \cdot (9.52) / (8 \cdot 113) - (9.52^3) / (8 \cdot 113^3))$$

$$= 181.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 24.4/182 + (25.2 \cdot 150/6837) / 207$$

$$= 0.137$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	16
Total Number of Bolts in Tension/Baseplate	Nbt	8
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.0 mm.
User Defined Area in a Single Bolt	UserBoltArea	31.3100 cm ²
BasePlate Allowable Stress	S	207.00 N./mm ²



Area Available in a Single Bolt	BltArea	28.4051 cm ²
Saddle Load QO (Weight)	QO	869350.3 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	99.8 Kgf
Maximum Transverse Force	Ft	2013.5 Kgf
Maximum Longitudinal Force	Fl	782.2 Kgf
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 $= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 782 / (28.4 \cdot 16)$
 $= 0.2 \text{ N./mm}^2$. Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 2014 / (28.4 \cdot 16)$
 $= 0.4 \text{ N./mm}^2$. Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= Fl / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 782 / (103 \cdot 16)$
 $= 0.0465 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 2590 \cdot 2014 + 0$
 $= 5215.10 \text{ Kg-m.}$

Eccentricity (e):
 $= Rmom / QO$
 $= 5215 / 869350$
 $= 6.00 \text{ mm.}$ < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0, 0.046, 0]$
 $= 0.0465 \text{ cm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 $= \min(300 + 1.56 \cdot \sqrt{1805 \cdot 203}, 2 \cdot 6065)$
 $= 1244.1718 \text{ mm.}$

Input and Calculated Values:

Vessel Mean Radius	Rm	1804.50 mm.
Shell Thickness used in this Case	t	206.000 mm.
Stiffened Vessel Length per 4.15.6	L	35044.50 mm.
Distance from Saddle to Vessel tangent	a1 or a	6065.00 mm.
Saddle Width	b1 or b	300.00 mm.
Saddle Bearing Angle	delta or theta	120.00 degrees



Shell Allowable Stress used in Calculation	294.49	N./mm ²
Head Allowable Stress used in Calculation	294.49	N./mm ²
Circumferential Efficiency in Plane of Saddle	1.00	
Circumferential Efficiency at Mid-Span	1.00	
Distance from Saddle Base to Centerline	B	2590.00 mm.
Coefficient of Friction	mu	0.40

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	966885.00	Kgf
Pressure used in Analysis	P	288.802 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	115.84	294.49
Long. Stress at Bottom of Midspan	140.89	294.49
Long. Stress at Top of Saddles	171.33	294.49
Long. Stress at Bottom of Saddles	104.55	294.49
Tangential Shear in Shell	19.81	165.23
Circ. Stress at Horn of Saddle	27.63	441.74
Circ. Compressive Stress in Shell	2.85	294.49

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (4027/2 + 0) \cdot 2590/3346$$

$$= 4675.3 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(782, 0) \cdot 2590/20300$$

$$= 99.8 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 962210 + \max(99.8, 4675, 0, 0)$$

$$= 966885.0 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (4.01 \cdot 1.2)^2)$$

$$= 15340.670 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	971864.19	Kgf
Transverse Shear Load	2013.51	Kgf
Longitudinal Shear Load	782.17	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2 (ShellID/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2 (3400/2 + 206 + 26) \sin(120/2), 2 \cdot 1805)$$

$$= 3346.323 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$



$$= -966885 \cdot 6065 [1 - (1 - 6065/35044 + (1805^2 - 0^2) / (2 \cdot 6065 \cdot 35044)) / (1 + (4 \cdot 0) / (3 \cdot 35044))] \\ = -969983.5 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\ = 966885 \cdot 35044 / 4 ((1 + 2(1805^2 - 0^2) / (35044^2)) / (1 + (4 \cdot 0) / (3 \cdot 35044)) - 4 \cdot 6065 / 35044) \\ = 2651815.8 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\ = 289 \cdot 1805 / (2 \cdot 203) - 2651815 / (\pi \cdot 1805^2 \cdot 203) \\ = 115.84 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ = 289 \cdot 1805 / (2 \cdot 203) + 2651815 / (\pi \cdot 1805^2 \cdot 203) \\ = 140.89 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = 289 \cdot 1805 / (2 \cdot 203) - 969984 / (0.11 \cdot \pi \cdot 1805^2 \cdot 203) \\ = 171.33 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = 289 \cdot 1805 / (2 \cdot 203) + 969984 / (0.19 \cdot \pi \cdot 1805^2 \cdot 203) \\ = 104.55 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ = 966885 (35044 - 2 \cdot 6065) / (35044 + (4 \cdot 0 / 3)) \\ = 632215.8 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t) \\ = 1.17 \cdot 632216 / (1805 \cdot 203) \\ = 19.81 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)} \\ = 0.78 \cdot \sqrt{(1805 \cdot 203)} \\ = 472.086 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\ = -0.76 \cdot 966885 \cdot 0.1 / (203(300 + 472 + 472)) \\ = -2.85 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2) \\ = -966885 / (4 \cdot 203(300 + 472 + 472)) - 3 \cdot 0.053 \cdot 966885 / (2 \cdot 203^2) \\ = -27.63 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ = \min(300 + 1.56 \cdot \sqrt{(1805 \cdot 203)}, 2 \cdot 6065) \\ = 1244.17 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3556.0000	mm.
Baseplate Thickness	Bpthk	110.0000	mm.
Baseplate Width	Bpwid	1012.0001	mm.
Number of Ribs (inc. outside ribs)	Nribs	9	
Rib Thickness	Ribtk	30.0000	mm.
Web Thickness	Webtk	30.0000	mm.



Web Location	Webloc	Center
Saddle Yield Stress	Sy	310.0 N./mm ²
Height of Web at Center	Hw,c	547.0 mm.
Friction Coefficient	mu	0.400

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1703 \cdot 203)} + 350)^3) / 12$$

$$= 88341.156 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 88341 + 2572 (327 - 102)^2$$

$$= 1399118.250 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1267.2	203.0	101.5	2572.5	26110710.0	0.140E+07
Wearplate	350.0	26.0	216.0	91.0	1965599.8	0.113E+05
Web	30.0	548.0	503.0	164.4	8269324.0	0.919E+05
BasePlate	1012.0	110.0	832.0	1113.2	92618256.0	0.285E+07
Totals	...	...	...	3941.1	128963888.0	0.435E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 128963888 / 3941$$

$$= 327.229 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 966885$$

$$= 196782.0625 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 14.1006 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 185.9966 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2590 - 1703 \cdot \sin(120/2) / 1.05$$

$$= 1181.630 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 232528.2656 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 17.1539 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 206.6629 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3 \cdot (966885 + 4979) \cdot 1012) / (4 \cdot 3556 \cdot 207)}$$

$$= 99.215 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1700 + 206 + 26)$$



$$= 3346.322 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 3346 / (9 - 1)$$

$$= 418.290 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 418 \cdot 1012 / 2$$

$$= 2116.549 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 971864 / (3556 \cdot 1012)$$

$$= 27.006 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 2117 \cdot 27$$

$$= 57159.914 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 81 + 62.7$$

$$= 143.744 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 57160 / 144$$

$$= 38.997 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	30.0	300.0	...	90.0	...	0.675E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300 - 30$$

$$= 270.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 144$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300 / 2$$

$$= 150.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(6750 / 144)}$$

$$= 68.526 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2590 - \cos(120/2) (1700 + 206 + 26) - 110$$

$$= 1514.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 310)}$$

$$= 112.838$$



Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1514/68.5$$

$$= 22.094$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 782 / (2 \cdot 3556) \cdot 418 \cdot 1514 / 2$$

$$= 34.825 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (22.1 < 113) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.1)^2 / (2 \cdot 113^2)) 310 / (5/3 + 3 \cdot (22.1) / (8 \cdot 113) - (22.1)^3 / (8 \cdot 113^3))$$

$$= 174.8 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 39/175 + (34.8 \cdot 150 / 67500000) / 207$$

$$= 0.227$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	30.0	270.0	0.0	81.0	0.0	0.674E+04
Web	418.3	30.0	0.0	125.5	0.0	94.1
Totals	...	...	...	206.5	...	0.684E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/206$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300/2$$

$$= 150.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2590 - \cos(1932/2) (110 + 0 + 0) - 0$$

$$= 548.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (6837/206) }$$

$$= 57.544 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 548/57.5$$

$$= 9.523$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 782 / (2 \cdot 3556)$$

$$= 0.110 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.11 \cdot 418 \cdot 548$$

$$= 25.210 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 25.2/455824$$



$$= 0.542 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($9.52 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.52)^2 / (2 \cdot 113^2)) 310 /$$

$$(5/3 + 3 \cdot (9.52) / (8 \cdot 113) - (9.52)^3 / (8 \cdot 113^3))$$

$$= 181.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 27.2 / 182 + (25.2 \cdot 150 / 6837) / 207$$

$$= 0.152$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	16
Total Number of Bolts in Tension/Baseplate	Nbt	8
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.0 mm.
User Defined Area in a Single Bolt	UserBoltArea	31.3100 cm ²
BasePlate Allowable Stress	S	207.00 N./mm ²
Area Available in a Single Bolt	BltArea	28.4051 cm ²
Saddle Load Q0 (Weight)	Q0	967188.9 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	99.8 Kgf
Maximum Transverse Force	Ft	2013.5 Kgf
Maximum Longitudinal Force	Fl	782.2 Kgf
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 782 / (28.4 \cdot 16)$$

$$= 0.2 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2014 / (28.4 \cdot 16)$$

$$= 0.4 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 782 / (103 \cdot 16)$$

$$= 0.0465 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2590 \cdot 2014 + 0$$

$$= 5215.10 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 5215 / 967189$$

$$= 5.39 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.046, 0]$$

$$= 0.0465 \text{ cm}^2$$



Review notes about nozzle loadings and supports in the Warnings report.

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**Input, Nozzle Desc: A1600****From: 10**

Pressure for Reinforcement Calculations	P	184.725	bars
Temperature for Internal Pressure	Temp	285	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.20	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	3501.00	mm.
Head Finished (Minimum) Thickness	t	105.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1445.7000	mm.
User Entered Minimum Design Metal Temperature		-12.22	°C

Type of Element Connected to the Parent : Nozzle

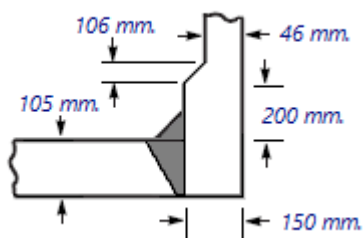
Material		SA-182 F11 +IT	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.20	N./mm ²
Allowable Stress At Ambient	Sna	184.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	46.0000	mm.
Hub Height of Integral Nozzle	h	200.0000	mm.
Height of Beveled Transition	L'	106.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	800.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	102.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: A1600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	1.811	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 616 \cdot (\exp(185/(147 \cdot 1)) - 1) + 3 + 0 \\ &= 41.2 + 3 + 0 = 44.2 \text{ mm. Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 147 \cdot 1 \cdot \ln((2 \cdot 43 + 616) / 616) - 0 \\ &= 192 - 0 = 192 \text{ bars} \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
Min. required shell thickness, internal p + c : 103.864 mm.

Nozzle neck minimum required thickness : 11.340 mm.

Minimum nozzle neck thickness 45.999 mm., is greater than required thickness 11.340 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 72.0,	Allowable	: 147.2 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 342.0 N./mm ²	Passed
Occasional	: 61.8,	Allowable	: 195.8 N./mm ²	Passed
Shear	: 7.0,	Allowable	: 103.0 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[147/165, 1] \\ &= 0.891 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 150 - 3 \\ &= 147.000 \text{ mm.} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 46 - 3 \\ &= 43.000 \text{ mm.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 3501/2 + 3 \\ &= 1753.500 \text{ mm.} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{1753 \cdot 102}, 2 \cdot 308) \\ &= 422.915 \text{ mm.} \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned} &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min(((102 + 0)/147)^{0.35}, 1) \\ &= \min(0.88, 1) \end{aligned}$$



$$= 0.880$$

Intermediate value [Cp]:
 $= 9999999.000$

Effective Nozzle Wall Length Outside the Vessel [LH]:
 $= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$
 $= \min(102 + 0 + 0.88 \cdot \sqrt{(308 \cdot 147)}, 800 + 102)$
 $= 289.172 \text{ mm.}$

Effective Vessel Thickness [teff]:
 $= t + ((A_5 \cdot frp) / LR)$
 $= 102 + ((0 \cdot 1) / 423)$
 $= 102.000 \text{ mm.}$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
 $= t \cdot LR$
 $= 102 \cdot 423$
 $= 431.373 \text{ cm}^2$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
 $= t_n \cdot LH$
 $= 147 \cdot 289$
 $= 425.084 \text{ cm}^2$

Area Contributed by the Outside Fillet Weld [A41]:
 $= 0.5 \cdot Leg^2$
 $= 0.5 \cdot 10^2$
 $= 0.500 \text{ cm}^2$

The total area contributed by A1 through A43 [AT]:
 $= A1 + frn(A2 + A3) + A41 + A42 + A43$
 $= 431 + 0.89(425 + 0) + 0.5 + 0 + 0$
 $= 810.643 \text{ cm}^2$

Nozzle Radius for Force Calculation [Rxn]:
 $= t_n / \ln[1 + t_n/R_n]$
 $= 147 / \ln[1 + 147/308]$
 $= 376.530 \text{ mm.}$

Shell Radius for Force Calculation [Rxs]:
 $= teff / \ln[1 + teff/Reff]$
 $= 102 / \ln[1 + 102/1753]$
 $= 1804.020 \text{ mm.}$

Allowable Local Primary Membrane Stress [Sallow]:
 $= 1.5 \cdot S \cdot E$
 $= 1.5 \cdot 165 \cdot 1$
 $= 247.8 \text{ N./mm}^2$

Force acting on the Nozzle [fN]:
 $= P \cdot Rxn \cdot LH$
 $= 185 \cdot 377 \cdot 289$
 $= 205106.2 \text{ Kg}$

Force acting on the Shell [fS]:
 $= P \cdot Rxs \cdot (LR + t_n)/2$
 $= 185 \cdot 1804 \cdot (423 + 147)/2$
 $= 968375.6 \text{ Kg}$

Discontinuity Force from Internal Pressure [fY]:
 $= (P \cdot Rxs \cdot R_{nc})/2$
 $= (185 \cdot 1804 \cdot 308)/2$
 $= 523001.0 \text{ Kg}$

Area Resisting Internal Pressure [Ap]:
 $= (fS + fY + fN)/P$



$$= (968376 + 523001 + 205106) / 185$$

$$= 9005.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/(2 \cdot \text{teff}))$$

$$= 248 / (2 \cdot 9006/811 - 1804/(2 \cdot 102))$$

$$= 185.2 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot \text{S}[t/\text{Rxs}]$$

$$= 2 \cdot 165 [102/1804]$$

$$= 186.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(185, 187)$$

$$= 185.246 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / \text{AT}$$

$$= (205106 + 968376 + 523001) / 811$$

$$= 205.234 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / (2 \cdot \text{teff})$$

$$= 185 \cdot 1804 / (2 \cdot 102)$$

$$= 163.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 205 - 163, 163)$$

$$= 247.1 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.80	N./mm ²
Local Primary Membrane Stress	PL	247.10	N./mm ²
Maximum Allowable Working Pressure	Pmax	185.25	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (308 + 147) / 308$$

$$= 1.478 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (308 + 147)$$

$$= 714.398 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, \text{ L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 147.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot \text{Sn}(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(523001 \cdot 1.48, 1.5 \cdot 147(463+0), \pi/4 \cdot 185 \cdot 308^2 \cdot 1.48^2)$$

$$= 306022.531 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 306023 / (714(0.49 \cdot 7.07 + 0.6 \cdot 102 + 0.49 \cdot 0))$$

$$= 64.964 < \text{or} = \text{to } 165.199 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A**



Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

The Drop for this Nozzle is : 60.1132 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 965.1132 mm.

Input Echo, WRC107/537 Item 1, Description: A1600 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	3501.000	mm.
Vessel Thickness	Tv	105.000	mm.
Design Temperature	T1	285.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	207.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	165.20	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	609.600	mm.
Nozzle Thickness	Tn	150.000	mm.
Nozzle Material		SA-182 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	184.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	147.20	N./mm ²
Design Internal Pressure	Dp	184.725	bars
Include Pressure Thrust		No	

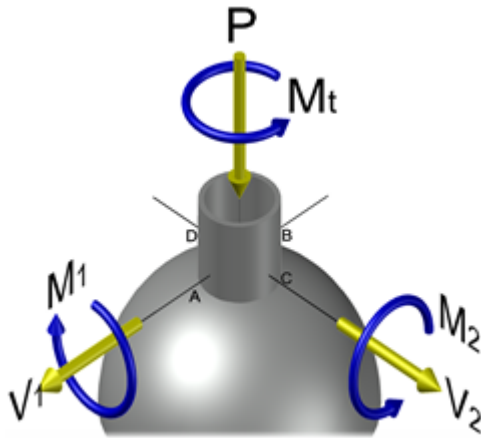
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-7645.4	Kgf
Longitudinal Shear (SUS)	(V1) V1	6116.3	Kgf
Circumferential Shear (SUS)	(Vc) V2	6116.3	Kgf
Circumferential Moment (SUS)	(Mc) M1	11010.0	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	11010.0	Kg-m.
Torsional Moment (SUS)	Mt	12845.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 909.6 + 2 \cdot 1.65 \cdot \sqrt{(1804.5 (105.0 - 3.0))}$$

$$= 2325.369 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	-7645.4	Kgf
Circumferential Shear	(VC)	V2	6116.3	Kgf
Longitudinal Shear	(VL)	V1	6116.3	Kgf
Circumferential Moment	(MC)	M1	11010.0	Kg-m.
Longitudinal Moment	(ML)	M2	11010.0	Kg-m.
Torsional Moment		MT	12845.0	Kg-m.

$$\text{Dimensionless Param: } U = 1.06 \text{ TAU} = 5.00 (2.59) \text{ RHO} = 0.69$$

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03828	(A,B,C,D)
$M(x) / P$	SP 1	0.03786	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04749	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.06571	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04749	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.06571	(A,B,C,D)
$N(y) * T / P$	SP 1	0.06643	(A,B,C,D)
$M(y) / P$	SP 1	0.01180	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.06576	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.02379	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.06576	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.02379	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Rad.	Bend. P	1.6	-1.6	1.6	-1.6	1.6	-1.6	1.6	-1.6
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-9.5	9.5	9.5	-9.5
Rad.	Memb. ML	-1.1	-1.1	1.1	1.1	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-9.5	9.5	9.5	-9.5	0.0	0.0	0.0	0.0



Tot. Rad. Str.	-8.8	7.0	12.6	-9.8	-8.8	7.0	12.6	-9.8
Tang. Memb. P	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Tang. Bend. P	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5
Tang. Memb. MC	0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Tang. Bend. MC	0.0	0.0	0.0	0.0	-3.5	3.5	3.5	-3.5
Tang. Memb. ML	-1.6	-1.6	1.6	1.6	0.0	0.0	0.0	0.0
Tang. Bend. ML	-3.5	3.5	3.5	-3.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-4.1	1.8	6.0	-1.9	-4.1	1.8	6.0	-1.9
Shear VC	0.4	0.4	-0.4	-0.4	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear MT	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Tot. Shear	1.4	1.4	0.5	0.5	0.5	0.5	1.4	1.4
Str. Int.	9.1	7.4	12.6	9.8	8.8	7.1	12.9	10.0

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		163.4	163.4	163.4	163.4	163.4	163.4	163.4	163.4
Rad. Pl (SUS)		-0.9	-0.9	1.4	1.4	-0.9	-0.9	1.4	1.4
Rad. Q (SUS)		-7.9	7.9	11.2	-11.2	-7.9	7.9	11.2	-11.2
Long. Pm (SUS)		163.4	163.4	163.4	163.4	163.4	163.4	163.4	163.4
Long. Pl (SUS)		-1.1	-1.1	2.1	2.1	-1.1	-1.1	2.1	2.1
Long. Q (SUS)		-2.9	2.9	4.0	-4.0	-2.9	2.9	4.0	-4.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.4	0.4	-0.4	-0.4	-0.4	-0.4	0.4	0.4
Shear Q (SUS)		1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pm (SUS)		163.4	163.4	163.4	163.4	163.4	163.4	163.4	163.4
Pm+Pl (SUS)		162.8	162.8	165.6	165.6	162.8	162.8	165.6	165.6
Pm+Pl+Q (Total)		159.7	170.7	176.0	161.5	159.4	170.4	176.2	161.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	163.37	165.20	Passed
Pm+Pl (SUS)	165.64	247.80	Passed
Pm+Pl+Q (TOTAL)	176.24	495.62	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: A2450

From: 10

Pressure for Reinforcement Calculations	P	184.725	bars
Temperature for Internal Pressure	Temp	285	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.20	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	3501.00	mm.
Head Finished (Minimum) Thickness	t	105.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.22	°C

Type of Element Connected to the Parent : Nozzle

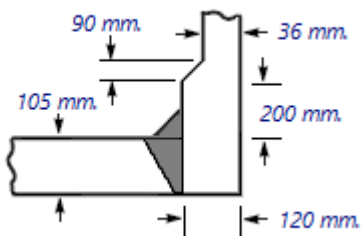
Material		SA-182 F11 +IT	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.20	N./mm ²
Allowable Stress At Ambient	Sna	184.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	36.0000	mm.
Hub Height of Integral Nozzle	h	200.0000	mm.
Height of Beveled Transition	L'	90.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	800.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	102.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: A2450

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	18.000	in.
Actual Thickness Used in Calculation	1.417	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 463 \cdot (\exp(185/(147 \cdot 1)) - 1) + 3 + 0 \\ &= 31 + 3 + 0 = 34 \text{ mm. Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq} \\ &= 147 \cdot 1 \cdot \ln((2 \cdot 33 + 463)/463) - 0 \\ &= 196 - 0 = 196 \text{ bars} \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
Min. required shell thickness, internal p + c : 103.864 mm.

Nozzle neck minimum required thickness : 11.340 mm.

Minimum nozzle neck thickness 35.992 mm., is greater than required thickness 11.340 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 70.8,	Allowable	: 147.2 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 343.2 N./mm ²	Passed
Occasional	: 60.5,	Allowable	: 195.8 N./mm ²	Passed
Shear	: 7.4,	Allowable	: 103.0 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[147/165, 1] \\ &= 0.891 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 120 - 3 \\ &= 117.000 \text{ mm.} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 36 - 3 \\ &= 33.000 \text{ mm.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 3501/2 + 3 \\ &= 1753.500 \text{ mm.} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{1753 \cdot 102}, 2 \cdot 232) \\ &= 422.915 \text{ mm.} \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned} &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min(((102 + 0)/117)^{0.35}, 1) \\ &= \min(0.95, 1) \end{aligned}$$



$$= 0.953$$

Intermediate value [Cp]:
 $= 9999999.000$

Effective Nozzle Wall Length Outside the Vessel [LH]:
 $= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$
 $= \min(102 + 0 + 0.95 \cdot \sqrt{(232 \cdot 117)}, 800 + 102)$
 $= 258.894 \text{ mm.}$

Effective Vessel Thickness [teff]:
 $= t + ((A_5 \cdot frp) / LR)$
 $= 102 + ((0 \cdot 1) / 423)$
 $= 102.000 \text{ mm.}$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
 $= t \cdot LR$
 $= 102 \cdot 423$
 $= 431.373 \text{ cm}^2$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
 $= t_n \cdot LH$
 $= 117 \cdot 259$
 $= 302.906 \text{ cm}^2$

Area Contributed by the Outside Fillet Weld [A41]:
 $= 0.5 \cdot Leg^2$
 $= 0.5 \cdot 10^2$
 $= 0.500 \text{ cm}^2$

The total area contributed by A1 through A43 [AT]:
 $= A1 + frn(A2 + A3) + A41 + A42 + A43$
 $= 431 + 0.89(303 + 0) + 0.5 + 0 + 0$
 $= 701.777 \text{ cm}^2$

Nozzle Radius for Force Calculation [Rxn]:
 $= t_n / \ln[1 + t_n / R_n]$
 $= 117 / \ln[1 + 117 / 232]$
 $= 286.124 \text{ mm.}$

Shell Radius for Force Calculation [Rxs]:
 $= teff / \ln[1 + teff / R_{eff}]$
 $= 102 / \ln[1 + 102 / 1753]$
 $= 1804.020 \text{ mm.}$

Allowable Local Primary Membrane Stress [Sallow]:
 $= 1.5 \cdot S \cdot E$
 $= 1.5 \cdot 165 \cdot 1$
 $= 247.8 \text{ N./mm}^2$

Force acting on the Nozzle [fN]:
 $= P \cdot Rxn \cdot LH$
 $= 185 \cdot 286 \cdot 259$
 $= 139540.3 \text{ Kg}$

Force acting on the Shell [fS]:
 $= P \cdot Rxs \cdot (LR + t_n) / 2$
 $= 185 \cdot 1804 \cdot (423 + 117) / 2$
 $= 917401.0 \text{ Kg}$

Discontinuity Force from Internal Pressure [fY]:
 $= (P \cdot Rxs \cdot R_{nc}) / 2$
 $= (185 \cdot 1804 \cdot 232) / 2$
 $= 393525.1 \text{ Kg}$

Area Resisting Internal Pressure [Ap]:
 $= (fS + fY + fN) / P$



$$= (917401 + 393525 + 139540) / 185$$
$$= 7699.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/(2 \cdot \text{teff}))$$
$$= 248 / (2 \cdot 7700/702 - 1804/(2 \cdot 102))$$
$$= 189.1 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot \text{S}[t/\text{Rxs}]$$
$$= 2 \cdot 165 [102/1804]$$
$$= 186.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$
$$= \min(189, 187)$$
$$= 186.797 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / \text{AT}$$
$$= (139540 + 917401 + 393525) / 702$$
$$= 202.692 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / (2 \cdot \text{teff})$$
$$= 185 \cdot 1804 / (2 \cdot 102)$$
$$= 163.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$
$$= \max(2 \cdot 203 - 163, 163)$$
$$= 242.0 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.80	N./mm ²
Local Primary Membrane Stress	PL	242.02	N./mm ²
Maximum Allowable Working Pressure	Pmax	186.80	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (\text{Rnc} + \text{tn}) / \text{Rnc}$$
$$= (232 + 117) / 232$$
$$= 1.505 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (\text{Rn} + \text{tn})$$
$$= \pi/2 \cdot (232 + 117)$$
$$= 547.580 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$\text{tc}, \text{L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 117.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi/4 \cdot P \cdot \text{Rn}^2 \cdot k_y^2)$$
$$= \min(393525 \cdot 1.51, 1.5 \cdot 147(312+0), \pi/4 \cdot 185 \cdot 232^2 \cdot 1.51^2)$$
$$= 179790.734 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T}))$$
$$= 179791 / (548 (0.49 \cdot 7.07 + 0.6 \cdot 102 + 0.49 \cdot 0))$$
$$= 49.794 < \text{or} = \text{to } 165.199 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A



Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

The Drop for this Nozzle is : 35.0618 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 940.0618 mm.

Input Echo, WRC107/537 Item 1, Description: A2450 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	3501.000	mm.
Vessel Thickness	Tv	105.000	mm.
Design Temperature	T1	285.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	207.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	165.20	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	457.200	mm.
Nozzle Thickness	Tn	120.000	mm.
Nozzle Material		SA-182 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	184.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	147.20	N./mm ²
Design Internal Pressure	Dp	184.725	bars
Include Pressure Thrust		No	

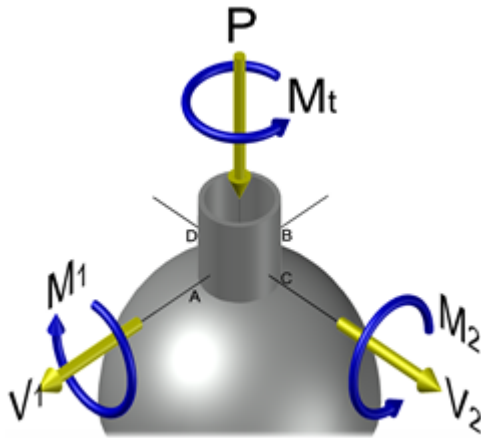
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-5098.9	Kgf
Longitudinal Shear (SUS)	(V1) V1	4079.1	Kgf
Circumferential Shear (SUS)	(Vc) V2	4079.1	Kgf
Circumferential Moment (SUS)	(Mc) M1	4893.1	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	4893.1	Kg-m.
Torsional Moment (SUS)	Mt	5708.6	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 697.2 + 2 \cdot 1.65 \cdot \sqrt{(1804.5 (105.0 - 3.0))}$$

$$= 2112.969 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	-5098.9	Kgf
Circumferential Shear	(VC)	V2	4079.1	Kgf
Longitudinal Shear	(VL)	V1	4079.1	Kgf
Circumferential Moment	(MC)	M1	4893.1	Kg-m.
Longitudinal Moment	(ML)	M2	4893.1	Kg-m.
Torsional Moment		MT	5708.6	Kg-m.

Dimensionless Param: U = 0.81 TAU = 5.00 (2.48) RHO = 0.87

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.04860	(A,B,C,D)
$M(x) / P$	SP 1	0.05125	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.07360	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.10235	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.07360	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.10235	(A,B,C,D)
$N(y) * T / P$	SP 1	0.09372	(A,B,C,D)
$M(y) / P$	SP 1	0.01766	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.08757	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.04272	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.08757	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.04272	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Rad.	Bend. P	1.5	-1.5	1.5	-1.5	1.5	-1.5	1.5	-1.5
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-6.6	6.6	6.6	-6.6
Rad.	Memb. ML	-0.8	-0.8	0.8	0.8	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-6.6	6.6	6.6	-6.6	0.0	0.0	0.0	0.0



Tot. Rad. Str.	-5.7	4.6	9.1	-7.1	-5.7	4.6	9.1	-7.1
Tang. Memb. P	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Tang. Bend. P	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Tang. Bend. MC	0.0	0.0	0.0	0.0	-2.8	2.8	2.8	-2.8
Tang. Memb. ML	-0.9	-0.9	0.9	0.9	0.0	0.0	0.0	0.0
Tang. Bend. ML	-2.8	2.8	2.8	-2.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-2.7	1.8	4.7	-1.9	-2.7	1.8	4.7	-1.9
Shear VC	0.4	0.4	-0.4	-0.4	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear MT	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Tot. Shear	1.1	1.1	0.4	0.4	0.4	0.4	1.1	1.1
Str. Int.	6.0	4.9	9.1	7.1	5.7	4.6	9.4	7.3

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		163.4	163.4	163.4	163.4	163.4	163.4	163.4	163.4
Rad. Pl (SUS)		-0.6	-0.6	1.0	1.0	-0.6	-0.6	1.0	1.0
Rad. Q (SUS)		-5.1	5.1	8.1	-8.1	-5.1	5.1	8.1	-8.1
Long. Pm (SUS)		163.4	163.4	163.4	163.4	163.4	163.4	163.4	163.4
Long. Pl (SUS)		-0.5	-0.5	1.4	1.4	-0.5	-0.5	1.4	1.4
Long. Q (SUS)		-2.2	2.2	3.3	-3.3	-2.2	2.2	3.3	-3.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.4	0.4	-0.4	-0.4	-0.4	-0.4	0.4	0.4
Shear Q (SUS)		0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Pm (SUS)		163.4	163.4	163.4	163.4	163.4	163.4	163.4	163.4
Pm+Pl (SUS)		163.2	163.2	165.0	165.0	163.2	163.2	165.0	165.0
Pm+Pl+Q (Total)		161.0	168.3	172.5	161.5	160.7	168.0	172.7	161.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	163.37	165.20	Passed
Pm+Pl (SUS)	164.98	247.80	Passed
Pm+Pl+Q (TOTAL)	172.72	495.62	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: B600

From: 10

Pressure for Reinforcement Calculations	P	184.725	bars
Temperature for Internal Pressure	Temp	285	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.20	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	3501.00	mm.
Head Finished (Minimum) Thickness	t	105.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.22	°C

Type of Element Connected to the Parent : Nozzle

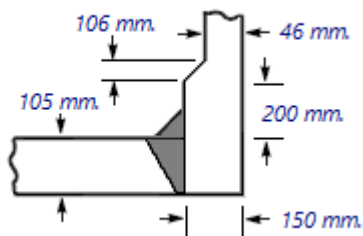
Material		SA-182 F11 +IT	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.20	N./mm ²
Allowable Stress At Ambient	Sna	184.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	46.0000	mm.
Hub Height of Integral Nozzle	h	200.0000	mm.
Height of Beveled Transition	L'	106.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	800.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	102.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	1.811	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 616 \cdot (\exp(185/(147 \cdot 1)) - 1) + 3 + 0 \\ &= 41.2 + 3 + 0 = 44.2 \text{ mm. Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 147 \cdot 1 \cdot \ln((2 \cdot 43 + 616) / 616) - 0 \\ &= 192 - 0 = 192 \text{ bars} \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
Min. required shell thickness, internal p + c : 103.864 mm.

Nozzle neck minimum required thickness : 11.340 mm.

Minimum nozzle neck thickness 45.999 mm., is greater than required thickness 11.340 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 72.0,	Allowable	: 147.2 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 342.0 N./mm ²	Passed
Occasional	: 61.8,	Allowable	: 195.8 N./mm ²	Passed
Shear	: 7.0,	Allowable	: 103.0 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[147/165, 1] \\ &= 0.891 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 150 - 3 \\ &= 147.000 \text{ mm.} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 46 - 3 \\ &= 43.000 \text{ mm.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 3501/2 + 3 \\ &= 1753.500 \text{ mm.} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{1753 \cdot 102}, 2 \cdot 308) \\ &= 422.915 \text{ mm.} \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned} &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min(((102 + 0)/147)^{0.35}, 1) \\ &= \min(0.88, 1) \end{aligned}$$

$$= 0.880$$

Intermediate value [Cp]:
 $= 9999999.000$

Effective Nozzle Wall Length Outside the Vessel [LH]:
 $= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$
 $= \min(102 + 0 + 0.88 \cdot \sqrt{(308 \cdot 147)}, 800 + 102)$
 $= 289.172 \text{ mm.}$

Effective Vessel Thickness [teff]:
 $= t + ((A_5 \cdot f_{rp}) / LR)$
 $= 102 + ((0 \cdot 1) / 423)$
 $= 102.000 \text{ mm.}$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
 $= t \cdot LR$
 $= 102 \cdot 423$
 $= 431.373 \text{ cm}^2$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
 $= t_n \cdot LH$
 $= 147 \cdot 289$
 $= 425.084 \text{ cm}^2$

Area Contributed by the Outside Fillet Weld [A41]:
 $= 0.5 \cdot \text{Leg}^2$
 $= 0.5 \cdot 10^2$
 $= 0.500 \text{ cm}^2$

The total area contributed by A1 through A43 [AT]:
 $= A1 + f_{rn}(A2 + A3) + A41 + A42 + A43$
 $= 431 + 0.89(425 + 0) + 0.5 + 0 + 0$
 $= 810.643 \text{ cm}^2$

Nozzle Radius for Force Calculation [Rxn]:
 $= t_n / \ln[1 + t_n / R_n]$
 $= 147 / \ln[1 + 147 / 308]$
 $= 376.530 \text{ mm.}$

Shell Radius for Force Calculation [Rxs]:
 $= t_{eff} / \ln[1 + t_{eff} / R_{eff}]$
 $= 102 / \ln[1 + 102 / 1753]$
 $= 1804.020 \text{ mm.}$

Allowable Local Primary Membrane Stress [Sallow]:
 $= 1.5 \cdot S \cdot E$
 $= 1.5 \cdot 165 \cdot 1$
 $= 247.8 \text{ N./mm}^2$

Force acting on the Nozzle [fN]:
 $= P \cdot R_{xn} \cdot LH$
 $= 185 \cdot 377 \cdot 289$
 $= 205106.2 \text{ Kg}$

Force acting on the Shell [fS]:
 $= P \cdot R_{xs} \cdot (LR + t_n) / 2$
 $= 185 \cdot 1804 \cdot (423 + 147) / 2$
 $= 968375.6 \text{ Kg}$

Discontinuity Force from Internal Pressure [fY]:
 $= (P \cdot R_{xs} \cdot R_{nc}) / 2$
 $= (185 \cdot 1804 \cdot 308) / 2$
 $= 523001.0 \text{ Kg}$

Area Resisting Internal Pressure [Ap]:
 $= (f_S + f_Y + f_N) / P$



$$= (968376 + 523001 + 205106) / 185$$

$$= 9005.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap/AT} - \text{Rxs} / (2 \cdot \text{teff}))$$

$$= 248 / (2 \cdot 9006/811 - 1804 / (2 \cdot 102))$$

$$= 185.2 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot \text{S[t/Rxs]}$$

$$= 2 \cdot 165 [102/1804]$$

$$= 186.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(185, 187)$$

$$= 185.246 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / \text{AT}$$

$$= (205106 + 968376 + 523001) / 811$$

$$= 205.234 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / (2 \cdot \text{teff})$$

$$= 185 \cdot 1804 / (2 \cdot 102)$$

$$= 163.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 205 - 163, 163)$$

$$= 247.1 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.80	N./mm ²
Local Primary Membrane Stress	PL	247.10	N./mm ²
Maximum Allowable Working Pressure	Pmax	185.25	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (308 + 147) / 308$$

$$= 1.478 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (308 + 147)$$

$$= 714.398 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 147.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot \text{Sn}(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(523001 \cdot 1.48, 1.5 \cdot 147(463+0), \pi/4 \cdot 185 \cdot 308^2 \cdot 1.48^2)$$

$$= 306022.531 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (LT(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 306023 / (714(0.49 \cdot 7.07 + 0.6 \cdot 102 + 0.49 \cdot 0))$$

$$= 64.964 < \text{or } = \text{to } 165.199 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A



Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

The Drop for this Nozzle is : 60.1132 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 965.1132 mm.

Input Echo, WRC107/537 Item 1, Description: B600 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	3501.000	mm.
Vessel Thickness	Tv	105.000	mm.
Design Temperature	T1	285.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	207.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	165.20	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	609.600	mm.
Nozzle Thickness	Tn	150.000	mm.
Nozzle Material		SA-182 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	184.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	147.20	N./mm ²
Design Internal Pressure	Dp	184.725	bars
Include Pressure Thrust		No	

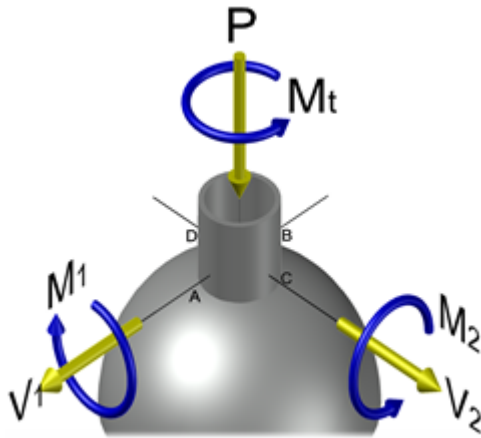
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-7645.4	Kgf
Longitudinal Shear (SUS)	(V1) V1	6116.3	Kgf
Circumferential Shear (SUS)	(Vc) V2	6116.3	Kgf
Circumferential Moment (SUS)	(Mc) M1	11010.0	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	11010.0	Kg-m.
Torsional Moment (SUS)	Mt	12845.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 909.6 + 2 \cdot 1.65 \cdot \sqrt{(1804.5 (105.0 - 3.0))}$$

$$= 2325.369 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	-7645.4	Kgf
Circumferential Shear	(VC)	V2	6116.3	Kgf
Longitudinal Shear	(VL)	V1	6116.3	Kgf
Circumferential Moment	(MC)	M1	11010.0	Kg-m.
Longitudinal Moment	(ML)	M2	11010.0	Kg-m.
Torsional Moment		MT	12845.0	Kg-m.

Dimensionless Param: U = 1.06 TAU = 5.00 (2.59) RHO = 0.69

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03828	(A,B,C,D)
$M(x) / P$	SP 1	0.03786	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04749	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.06571	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04749	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.06571	(A,B,C,D)
$N(y) * T / P$	SP 1	0.06643	(A,B,C,D)
$M(y) / P$	SP 1	0.01180	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.06576	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.02379	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.06576	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.02379	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Rad.	Bend. P	1.6	-1.6	1.6	-1.6	1.6	-1.6	1.6	-1.6
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-9.5	9.5	9.5	-9.5
Rad.	Memb. ML	-1.1	-1.1	1.1	1.1	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-9.5	9.5	9.5	-9.5	0.0	0.0	0.0	0.0



Tot. Rad. Str.	-8.8	7.0	12.6	-9.8	-8.8	7.0	12.6	-9.8
Tang. Memb. P	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Tang. Bend. P	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5
Tang. Memb. MC	0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Tang. Bend. MC	0.0	0.0	0.0	0.0	-3.5	3.5	3.5	-3.5
Tang. Memb. ML	-1.6	-1.6	1.6	1.6	0.0	0.0	0.0	0.0
Tang. Bend. ML	-3.5	3.5	3.5	-3.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-4.1	1.8	6.0	-1.9	-4.1	1.8	6.0	-1.9
Shear VC	0.4	0.4	-0.4	-0.4	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear MT	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Tot. Shear	1.4	1.4	0.5	0.5	0.5	0.5	1.4	1.4
Str. Int.	9.1	7.4	12.6	9.8	8.8	7.1	12.9	10.0

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		163.4	163.4	163.4	163.4	163.4	163.4	163.4	163.4
Rad. Pl (SUS)		-0.9	-0.9	1.4	1.4	-0.9	-0.9	1.4	1.4
Rad. Q (SUS)		-7.9	7.9	11.2	-11.2	-7.9	7.9	11.2	-11.2
Long. Pm (SUS)		163.4	163.4	163.4	163.4	163.4	163.4	163.4	163.4
Long. Pl (SUS)		-1.1	-1.1	2.1	2.1	-1.1	-1.1	2.1	2.1
Long. Q (SUS)		-2.9	2.9	4.0	-4.0	-2.9	2.9	4.0	-4.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.4	0.4	-0.4	-0.4	-0.4	-0.4	0.4	0.4
Shear Q (SUS)		1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pm (SUS)		163.4	163.4	163.4	163.4	163.4	163.4	163.4	163.4
Pm+Pl (SUS)		162.8	162.8	165.6	165.6	162.8	162.8	165.6	165.6
Pm+Pl+Q (Total)		159.7	170.7	176.0	161.5	159.4	170.4	176.2	161.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	163.37	165.20	Passed
Pm+Pl (SUS)	165.64	247.80	Passed
Pm+Pl+Q (TOTAL)	176.24	495.62	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.



Input, Nozzle Desc: T3100

From: 10

Pressure for Reinforcement Calculations	P	184.725	bars
Temperature for Internal Pressure	Temp	285	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.20	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	3501.00	mm.
Head Finished (Minimum) Thickness	t	105.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		-12.22	°C

Type of Element Connected to the Parent : Nozzle

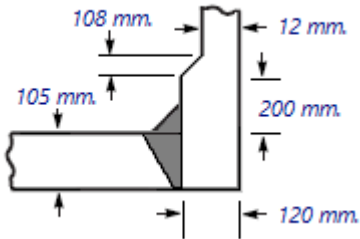
Material		SA-182 F11 +IT	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.20	N./mm ²
Allowable Stress At Ambient	Sna	184.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	200.0000	mm.
Height of Beveled Transition	L'	108.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	800.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	102.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: T3100

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	4.000	in.
Actual Thickness Used in Calculation	0.472	in.



Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 108 \cdot (\exp(185/(147 \cdot 1)) - 1) + 3 + 0$$

$$= 7.19 + 3 + 0 = 10.2 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147 \cdot 1 \cdot \ln((2 \cdot 9 + 108) / 108) - 0$$

$$= 228 - 0 = 228 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147/165, 1]$$

$$= 0.891$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 120 - 3$$

$$= 117.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12 - 3$$

$$= 9.000 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 350/2 + 3$$

$$= 175.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1753 \cdot 102}, 2 \cdot 53.8)$$

$$= 107.600 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((102 + 0)/117)^{0.35}, 1)$$

$$= \min(0.95, 1)$$

$$= 0.953$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t)$$



$$= \min(102 + 0 + 0.95 \sqrt{(53.8 \cdot 117)}, 800 + 102)$$

$$= 177.619 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 102 + ((0 \cdot 1) / 108)$$

$$= 102.000 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 102 \cdot 108$$

$$= 109.752 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 117 \cdot 178$$

$$= 207.814 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 110 + 0.89(208 + 0) + 0.5 + 0 + 0$$

$$= 295.424 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 117 / \ln[1 + 117/53.8]$$

$$= 101.279 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 102 / \ln[1 + 102/1753]$$

$$= 1804.020 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165 \cdot 1$$

$$= 247.8 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 185 \cdot 101 \cdot 178$$

$$= 33887.0 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + t_n) / 2$$

$$= 185 \cdot 1804 \cdot (108 + 117) / 2$$

$$= 381631.0 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot Rxs \cdot R_{nc}) / 2$$

$$= (185 \cdot 1804 \cdot 53.8) / 2$$

$$= 91414.7 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (381631 + 91415 + 33887) / 185$$

$$= 2691.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/(2 \cdot t_{eff}))$$

$$= 248 / (2 \cdot 2691/295 - 1804/(2 \cdot 102))$$

$$= 264.3 \text{ bars}$$



Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$

$$= 2 \cdot 165 [102/1804]$$

$$= 186.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(264, 187)$$

$$= 186.797 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (33887 + 381631 + 91415) / 295$$

$$= 168.280 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 185 \cdot 1804 / (2 \cdot 102)$$

$$= 163.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 168 - 163, 163)$$

$$= 173.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.80	N./mm ²
Local Primary Membrane Stress	PL	173.19	N./mm ²
Maximum Allowable Working Pressure	Pmax	186.80	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (53.8 + 117) / 53.8$$

$$= 3.175 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (53.8 + 117)$$

$$= 268.292 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 117.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(91415 \cdot 3.17, 1.5 \cdot 147(212+0), \pi/4 \cdot 185 \cdot 53.8^2 \cdot 3.17^2)$$

$$= 43160.676 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (LT(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 43161 / (268(0.49 \cdot 7.07 + 0.6 \cdot 102 + 0.49 \cdot 0))$$

$$= 24.397 < \text{ or } = \text{ to } 165.199 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-18 °C
User entered Minimum Design Temperature	:	-12 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	3 °C
Temperature reduction per Graph 3.13	:	21 °C



The computed stress ratio is : 0.631

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (5.68 \cdot 1) / (12 - 3 - 0)$$

$$= 0.631$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..

This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -18 °C

The Drop for this Nozzle is : 8.3526 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 913.3526 mm.

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Input, Nozzle Desc: M1600

From: 20

Pressure for Reinforcement Calculations	P	184.725	bars
Temperature for Internal Pressure	Temp	285	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.20	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	3400.00	mm.
Shell Finished (Minimum) Thickness	t	206.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1500.00	mm.
User Entered Minimum Design Metal Temperature		-12.22	°C

Type of Element Connected to the Parent : Nozzle

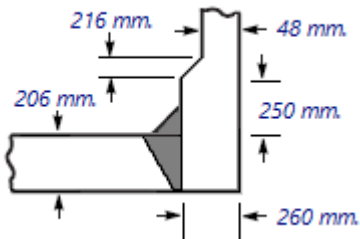
Material		SA-182 F11 +IT	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.20	N./mm ²
Allowable Stress At Ambient	Sna	184.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	48.0000	mm.
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	250.0000	mm.
Height of Beveled Transition	L'	216.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		260.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	800.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	200.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: M1600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	1.890	in.



Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 616 \cdot (\exp(185/(147 \cdot 1)) - 1) + 3 + 0$$

$$= 41.2 + 3 + 0 = 44.2 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147 \cdot 1 \cdot \ln((2 \cdot 45 + 616) / 616) - 0$$

$$= 201 - 0 = 201 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 204.719 mm.

Nozzle neck minimum required thickness : 11.340 mm.

Minimum nozzle neck thickness 48.006 mm., is greater than required thickness 11.340 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147/165, 1]$$

$$= 0.891$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 260 - 3$$

$$= 257.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 48 - 3$$

$$= 45.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 3406/203$$

$$= 16.778 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$



$$= 3400/2 + 3$$

$$= 1703.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1703 \cdot 203}, 2 \cdot 308)$$

$$= 587.970 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 203, 0] + \sqrt{308 \cdot 257}$$

$$= 281.255 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 800$$

$$= 800.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(203 + 0)$$

$$= 1624.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[281, 800, 1624] + 203$$

$$= 484.255 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 203 + ((0 \cdot 1) / 588)$$

$$= 203.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 308 + 257) / (\sqrt{(3406 + 203) \cdot 203}))$$

$$= 1.019$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 203 \cdot 588 \cdot \max((1.02/5)^{0.85}, 1)$$

$$= 1193.580 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 1164 + 75.5$$

$$= 1239.742 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 1194 + 0.89(1240 + 0) + 0.5 + 0 + 0$$

$$= 2298.748 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 257 / \ln[1 + 257/308]$$

$$= 423.379 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 203 / \ln[1 + 203/1703]$$

$$= 1802.595 \text{ mm.}$$



Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165 \cdot 1$$

$$= 247.8 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 185 \cdot 423 \cdot 484$$

$$= 386212.0 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 185 \cdot 1803 (588 + 257)$$

$$= 2869206.8 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 185 \cdot 1803 \cdot 308$$

$$= 1045175.1 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (2869206 + 1045175 + 386212) / 185$$

$$= 22830.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 248 / (2 \cdot 22830 / 2299 - 1803 / 203)$$

$$= 225.6 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 165 [203 / 1803]$$

$$= 186.0 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(226, 186)$$

$$= 186.029 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (386212 + 2869206 + 1045175) / 2299$$

$$= 183.470 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 185 \cdot 1803 / 203$$

$$= 164.0 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 183 - 164, 164)$$

$$= 202.9 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.80	N./mm ²
Local Primary Membrane Stress	PL	202.90	N./mm ²
Maximum Allowable Working Pressure	Pmax	186.03	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (308 + 257) / 308$$

$$= 1.835 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:



$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (308 + 257)$$

$$= 887.186 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :
tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 257.000, 6.000)$ Passed

Weld Load Value [fwelds]:
= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(1045175 \cdot 1.83, 1.5 \cdot 147(1240+0), \pi/4 \cdot 185 \cdot 308^2 \cdot 1.83^2)$
= 471956.062 Kgf

Weld Stress Value [τ]:
= $fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $471956 / (887(0.49 \cdot 7.07 + 0.6 \cdot 200 + 0.49 \cdot 0))$
= 42.254 $\leq$ or = to 165.199 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -7 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 8 °C
 Temperature reduction per Graph 3.13 : 15 °C
 The computed stress ratio is : 0.722

The computed thickness ratio [sr]:
 = $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 = $(32.5 \cdot 1) / (48 - 3 - 0)$
 = 0.722

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

Warning:
The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -7 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -34 °C

The Drop for this Nozzle is : 96.5658 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 1102.5658 mm.

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Input, Nozzle Desc: M2600

From: 120

Pressure for Reinforcement Calculations	P	184.725	bars
Temperature for Internal Pressure	Temp	285	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.20	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	3400.00	mm.
Shell Finished (Minimum) Thickness	t	206.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		29300.00	mm.
User Entered Minimum Design Metal Temperature		-12.22	°C

Type of Element Connected to the Parent : Nozzle

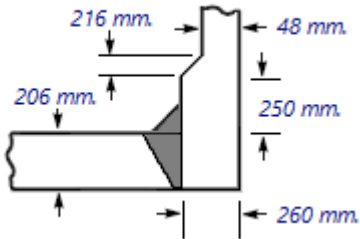
Material		SA-182 F11 +IT	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.20	N./mm ²
Allowable Stress At Ambient	Sna	184.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	48.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	250.0000	mm.
Height of Beveled Transition	L'	216.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		260.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	800.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	200.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: M2600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	1.890	in.



Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 616 \cdot (\exp(185/(147 \cdot 1)) - 1) + 3 + 0$$

$$= 41.2 + 3 + 0 = 44.2 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147 \cdot 1 \cdot \ln((2 \cdot 45 + 616) / 616) - 0$$

$$= 201 - 0 = 201 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 204.719 mm.

Nozzle neck minimum required thickness : 11.340 mm.

Minimum nozzle neck thickness 48.006 mm., is greater than required thickness 11.340 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147/165, 1]$$

$$= 0.891$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 260 - 3$$

$$= 257.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 48 - 3$$

$$= 45.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 3406/203$$

$$= 16.778 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$



$$= 3400/2 + 3$$

$$= 1703.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(1703 \cdot 203)}, 2 \cdot 308)$$

$$= 587.970 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 203, 0] + \sqrt{(308 \cdot 257)}$$

$$= 281.255 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 800$$

$$= 800.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(203 + 0)$$

$$= 1624.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[281, 800, 1624] + 203$$

$$= 484.255 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot \text{frp}) / LR)$$

$$= 203 + ((0 \cdot 1) / 588)$$

$$= 203.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{((D_i + \text{teff}) \cdot \text{teff})}))$$

$$= \min(12, (2 \cdot 308 + 257) / (\sqrt{((3406 + 203) \cdot 203)}))$$

$$= 1.019$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 203 \cdot 588 \cdot \max((1.02/5)^{0.85}, 1)$$

$$= 1193.580 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 1164 + 75.5$$

$$= 1239.742 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 1194 + 0.89(1240 + 0) + 0.5 + 0 + 0$$

$$= 2298.748 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 257 / \ln[1 + 257/308]$$

$$= 423.379 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= \text{teff} / \ln[1 + \text{teff}/\text{Reff}]$$

$$= 203 / \ln[1 + 203/1703]$$

$$= 1802.595 \text{ mm.}$$



Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165 \cdot 1$$

$$= 247.8 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 185 \cdot 423 \cdot 484$$

$$= 386212.0 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 185 \cdot 1803 (588 + 257)$$

$$= 2869206.5 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 185 \cdot 1803 \cdot 308$$

$$= 1045175.1 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (2869206 + 1045175 + 386212) / 185$$

$$= 22830.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 248 / (2 \cdot 22830 / 2299 - 1803 / 203)$$

$$= 225.6 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 165 [203 / 1803]$$

$$= 186.0 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(226, 186)$$

$$= 186.029 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (386212 + 2869206 + 1045175) / 2299$$

$$= 183.470 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 185 \cdot 1803 / 203$$

$$= 164.0 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 183 - 164, 164)$$

$$= 202.9 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.80	N./mm ²
Local Primary Membrane Stress	PL	202.90	N./mm ²
Maximum Allowable Working Pressure	Pmax	186.03	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (308 + 257) / 308$$

$$= 1.835 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:



$$\begin{aligned}
 &= \pi/2 \cdot (R_n + t_n) \\
 &= \pi/2 \cdot (308 + 257) \\
 &= 887.186 \text{ mm.}
 \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :
 tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 257.000, 6.000)$ Passed

Weld Load Value [fwelds]:
 $= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 $= \min(1045175 \cdot 1.83, 1.5 \cdot 147(1240+0), \pi/4 \cdot 185 \cdot 308^2 \cdot 1.83^2)$
 $= 471955.938 \text{ Kgf}$

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 471956 / (887(0.49 \cdot 7.07 + 0.6 \cdot 200 + 0.49 \cdot 0))$
 $= 42.254 < \text{or} = \text{to } 165.199 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : 14 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 29 °C
 Temperature reduction per Graph 3.13 : 15 °C
 The computed stress ratio is : 0.722

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (32.5 \cdot 1) / (48 - 3 - 0)$
 $= 0.722$

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

Warning:
The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : 14 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -34 °C

The Drop for this Nozzle is : 96.5658 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 1102.5658 mm.

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**Input, Nozzle Desc: C350****From: 170**

Pressure for Reinforcement Calculations	P	184.725	bars
Temperature for Internal Pressure	Temp	285	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.20	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	3501.00	mm.
Head Finished (Minimum) Thickness	t	105.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.22	°C

Type of Element Connected to the Parent : Nozzle

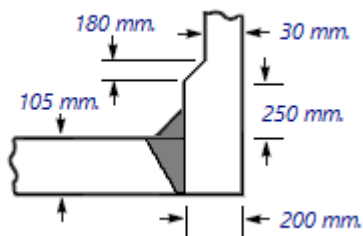
Material		SA-182 F11 +IT	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.20	N./mm ²
Allowable Stress At Ambient	Sna	184.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	30.0000	mm.
Hub Height of Integral Nozzle	h	250.0000	mm.
Height of Beveled Transition	L'	180.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	800.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	102.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: C350

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	14.000	in.
Actual Thickness Used in Calculation	1.181	in.





Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 362 \cdot (\exp(185/(147 \cdot 1)) - 1) + 3 + 0 \\ &= 24.2 + 3 + 0 = 27.2 \text{ mm. Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 147 \cdot 1 \cdot \ln((2 \cdot 27 + 362) / 362) - 0 \\ &= 205 - 0 = 205 \text{ bars} \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
Min. required shell thickness, internal p + c : 103.864 mm.

Nozzle neck minimum required thickness : 11.340 mm.

Minimum nozzle neck thickness 29.997 mm., is greater than required thickness 11.340 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	62.8,	Allowable	:	147.2 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	351.2 N./mm ²	Passed
Occasional	:	57.6,	Allowable	:	195.8 N./mm ²	Passed
Shear	:	4.6,	Allowable	:	103.0 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[147/165, 1] \\ &= 0.891 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 200 - 3 \\ &= 197.000 \text{ mm.} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 30 - 3 \\ &= 27.000 \text{ mm.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 3501/2 + 3 \\ &= 1753.500 \text{ mm.} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{1753 \cdot 102}, 2 \cdot 181) \\ &= 361.600 \text{ mm.} \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned} &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min(((102 + 0)/197)^{0.35}, 1) \\ &= \min(0.79, 1) \end{aligned}$$



$$= 0.794$$

Intermediate value [Cp]:
 $= 9999999.000$

Effective Nozzle Wall Length Outside the Vessel [LH]:
 $= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$
 $= \min(102 + 0 + 0.79 \cdot \sqrt{(181 \cdot 197)}, 800 + 102)$
 $= 251.892 \text{ mm.}$

Effective Vessel Thickness [teff]:
 $= t + ((A_5 \cdot frp) / LR)$
 $= 102 + ((0 \cdot 1) / 362)$
 $= 102.000 \text{ mm.}$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
 $= t \cdot LR$
 $= 102 \cdot 362$
 $= 368.832 \text{ cm}^2$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
 $= t_n \cdot LH$
 $= 197 \cdot 252$
 $= 496.228 \text{ cm}^2$

Area Contributed by the Outside Fillet Weld [A41]:
 $= 0.5 \cdot Leg_1^2$
 $= 0.5 \cdot 10^2$
 $= 0.500 \text{ cm}^2$

The total area contributed by A1 through A43 [AT]:
 $= A_1 + frn(A_2 + A_3) + A_{41} + A_{42} + A_{43}$
 $= 369 + 0.89(496 + 0) + 0.5 + 0 + 0$
 $= 811.494 \text{ cm}^2$

Nozzle Radius for Force Calculation [Rxn]:
 $= t_n / \ln[1 + t_n / R_n]$
 $= 197 / \ln[1 + 197 / 181]$
 $= 267.309 \text{ mm.}$

Shell Radius for Force Calculation [Rxs]:
 $= teff / \ln[1 + teff / Reff]$
 $= 102 / \ln[1 + 102 / 1753]$
 $= 1804.020 \text{ mm.}$

Allowable Local Primary Membrane Stress [Sallow]:
 $= 1.5 \cdot S \cdot E$
 $= 1.5 \cdot 165 \cdot 1$
 $= 247.8 \text{ N./mm}^2$

Force acting on the Nozzle [fN]:
 $= P \cdot Rxn \cdot LH$
 $= 185 \cdot 267 \cdot 252$
 $= 126838.7 \text{ Kgf}$

Force acting on the Shell [fS]:
 $= P \cdot Rxs \cdot (LR + t_n) / 2$
 $= 185 \cdot 1804 \cdot (362 + 197) / 2$
 $= 949149.9 \text{ Kgf}$

Discontinuity Force from Internal Pressure [fY]:
 $= (P \cdot Rxs \cdot R_{nc}) / 2$
 $= (185 \cdot 1804 \cdot 181) / 2$
 $= 307207.8 \text{ Kgf}$

Area Resisting Internal Pressure [Ap]:
 $= (fS + fY + fN) / P$



$$= (949150 + 307208 + 126839) / 185$$

$$= 7342.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/(2 \cdot \text{teff}))$$

$$= 248 / (2 \cdot 7343/811 - 1804/(2 \cdot 102))$$

$$= 267.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot \text{S}[t/\text{Rxs}]$$

$$= 2 \cdot 165 [102/1804]$$

$$= 186.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(268, 187)$$

$$= 186.797 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / \text{AT}$$

$$= (126839 + 949150 + 307208) / 811$$

$$= 167.158 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / (2 \cdot \text{teff})$$

$$= 185 \cdot 1804 / (2 \cdot 102)$$

$$= 163.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 167 - 163, 163)$$

$$= 170.9 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.80	N./mm ²
Local Primary Membrane Stress	PL	170.95	N./mm ²
Maximum Allowable Working Pressure	Pmax	186.80	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (181 + 197) / 181$$

$$= 2.090 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (181 + 197)$$

$$= 593.447 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 197.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot ky, 1.5 \cdot \text{Sn}(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot ky^2)$$

$$= \min(307208 \cdot 2.09, 1.5 \cdot 147(573+0), \pi/4 \cdot 185 \cdot 181^2 \cdot 2.09^2)$$

$$= 211172.031 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 211172 / (593(0.49 \cdot 7.07 + 0.6 \cdot 102 + 0.49 \cdot 0))$$

$$= 53.965 < \text{or } = \text{to } 165.199 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A



Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

The Drop for this Nozzle is : 41.2552 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 946.2552 mm.

Input Echo, WRC107/537 Item 1, Description: C350 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	3501.000	mm.
Vessel Thickness	Tv	105.000	mm.
Design Temperature	T1	285.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	207.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	165.20	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	355.600	mm.
Nozzle Thickness	Tn	200.000	mm.
Nozzle Material		SA-182 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	184.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	147.20	N./mm ²
Design Internal Pressure	Dp	184.725	bars
Include Pressure Thrust		No	

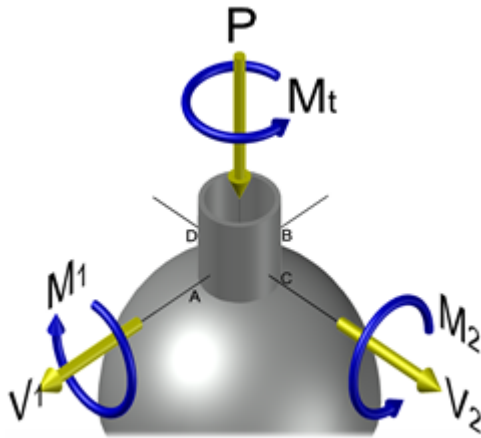
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-2293.6	Kgf
Longitudinal Shear (SUS)	(V1) V1	1529.1	Kgf
Circumferential Shear (SUS)	(Vc) V2	1529.1	Kgf
Circumferential Moment (SUS)	(Mc) M1	1284.5	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	1284.5	Kg-m.
Torsional Moment (SUS)	Mt	1835.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 755.6 + 2 \cdot 1.65 \cdot \sqrt{(1804.5 (105.0 - 3.0))}$$

$$= 2171.369 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	-2293.6	Kgf
Circumferential Shear	(VC)	V2	1529.1	Kgf
Longitudinal Shear	(VL)	V1	1529.1	Kgf
Circumferential Moment	(MC)	M1	1284.5	Kg-m.
Longitudinal Moment	(ML)	M2	1284.5	Kg-m.
Torsional Moment		MT	1835.0	Kg-m.

Dimensionless Param: U = 0.88 TAU = 5.00 (1.42) RHO = 0.52

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.04620	(A,B,C,D)
$M(x) / P$	SP 1	0.04962	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.05838	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.09716	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.05838	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.09716	(A,B,C,D)
$N(y) * T / P$	SP 1	0.07647	(A,B,C,D)
$M(y) / P$	SP 1	0.01495	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.07607	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.03307	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.07607	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.03307	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Rad.	Bend. P	0.6	-0.6	0.6	-0.6	0.6	-0.6	0.6	-0.6
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-1.6	1.6	1.6	-1.6
Rad.	Memb. ML	-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-1.6	1.6	1.6	-1.6	0.0	0.0	0.0	0.0



Tot. Rad. Str.	-1.1	0.9	2.6	-2.0	-1.1	0.9	2.6	-2.0
Tang. Memb. P	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Tang. Bend. P	0.2	-0.2	0.2	-0.2	0.2	-0.2	0.2	-0.2
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang. Bend. MC	0.0	0.0	0.0	0.0	-0.6	0.6	0.6	-0.6
Tang. Memb. ML	-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Tang. Bend. ML	-0.6	0.6	0.6	-0.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-0.4	0.3	1.1	-0.4	-0.4	0.3	1.1	-0.4
Shear VC	0.1	0.1	-0.1	-0.1	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear MT	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Tot. Shear	0.3	0.3	0.1	0.1	0.1	0.1	0.3	0.3
Str. Int.	1.2	1.1	2.6	2.0	1.1	0.9	2.6	2.1

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		163.4	163.4	163.4	163.4	163.4	163.4	163.4	163.4
Rad. Pl (SUS)		-0.1	-0.1	0.3	0.3	-0.1	-0.1	0.3	0.3
Rad. Q (SUS)		-1.0	1.0	2.3	-2.3	-1.0	1.0	2.3	-2.3
Long. Pm (SUS)		163.4	163.4	163.4	163.4	163.4	163.4	163.4	163.4
Long. Pl (SUS)		-0.0	-0.0	0.4	0.4	-0.0	-0.0	0.4	0.4
Long. Q (SUS)		-0.4	0.4	0.8	-0.8	-0.4	0.4	0.8	-0.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.1	0.1	-0.1	-0.1	-0.1	-0.1	0.1	0.1
Shear Q (SUS)		0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Pm (SUS)		163.4	163.4	163.4	163.4	163.4	163.4	163.4	163.4
Pm+Pl (SUS)		163.4	163.4	163.8	163.8	163.4	163.4	163.8	163.8
Pm+Pl+Q (Total)		163.1	164.4	165.9	163.0	163.0	164.3	166.0	163.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	163.37	165.20	Passed
Pm+Pl (SUS)	163.83	247.80	Passed
Pm+Pl+Q (TOTAL)	165.99	495.62	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

**Input, Nozzle Desc: T1100****From: 170**

Pressure for Reinforcement Calculations	P	184.727	bars
Temperature for Internal Pressure	Temp	285	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.20	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	3501.00	mm.
Head Finished (Minimum) Thickness	t	105.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	750.0000	mm.
User Entered Minimum Design Metal Temperature		-12.22	°C

Type of Element Connected to the Parent : Nozzle

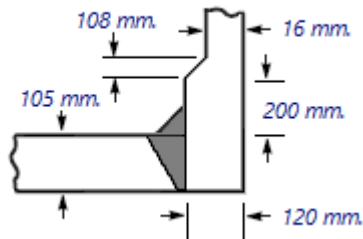
Material		SA-182 F11 +IT	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.20	N./mm ²
Allowable Stress At Ambient	Sna	184.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm.
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	200.0000	mm.
Height of Beveled Transition	L'	108.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	800.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	102.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		1500	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: T1100

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	4.000	in.
Actual Thickness Used in Calculation	0.630	in.

**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 108 \cdot (\exp(185/(147 \cdot 1)) - 1) + 3 + 0$$

$$= 7.19 + 3 + 0 = 10.2 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147 \cdot 1 \cdot \ln((2 \cdot 13 + 108) / 108) - 0$$

$$= 319 - 0 = 319 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147/165, 1]$$

$$= 0.891$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 120 - 3$$

$$= 117.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16 - 3$$

$$= 13.000 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 3501/2 + 3$$

$$= 1753.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1753 \cdot 102}, 2 \cdot 53.8)$$

$$= 107.600 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((102 + 0)/117)^{0.35}, 1)$$

$$= \min(0.95, 1)$$

$$= 0.953$$



Intermediate value [Cp]:
= 9999999.000

Effective Nozzle Wall Length Outside the Vessel [LH]:
= $\min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$
= $\min(102 + 0 + 0.95 \cdot \sqrt{(53.8 \cdot 117)}, 800 + 102)$
= 177.619 mm.

Effective Vessel Thickness [teff]:
= $t + ((A5 \cdot frp) / LR)$
= $102 + ((0 \cdot 1) / 108)$
= 102.000 mm.

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
= $t \cdot LR$
= $102 \cdot 108$
= 109.752 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
= $t_n \cdot LH$
= $117 \cdot 178$
= 207.814 cm²

Area Contributed by the Outside Fillet Weld [A41]:
= $0.5 \cdot Leg^2$
= $0.5 \cdot 10^2$
= 0.500 cm²

The total area contributed by A1 through A43 [AT]:
= $A1 + frn(A2 + A3) + A41 + A42 + A43$
= $110 + 0.89(208 + 0) + 0.5 + 0 + 0$
= 295.424 cm²

Nozzle Radius for Force Calculation [Rxn]:
= $t_n / \ln[1 + t_n/R_n]$
= $117 / \ln[1 + 117/53.8]$
= 101.279 mm.

Shell Radius for Force Calculation [Rxs]:
= $teff / \ln[1 + teff/Reff]$
= $102 / \ln[1 + 102/1753]$
= 1804.020 mm.

Allowable Local Primary Membrane Stress [Sallow]:
= $1.5 \cdot S \cdot E$
= $1.5 \cdot 165 \cdot 1$
= 247.8 N./mm²

Force acting on the Nozzle [fN]:
= $P \cdot Rxn \cdot LH$
= $185 \cdot 101 \cdot 178$
= 33887.4 Kgf

Force acting on the Shell [fS]:
= $P \cdot Rxs \cdot (LR + t_n) / 2$
= $185 \cdot 1804 \cdot (108 + 117) / 2$
= 381635.6 Kgf

Discontinuity Force from Internal Pressure [fY]:
= $(P \cdot Rxs \cdot R_{nc}) / 2$
= $(185 \cdot 1804 \cdot 59.5) / 2$
= 101133.4 Kgf

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:



$$= (fS + fY + fN) / P$$

$$= (381636 + 101133 + 33887) / 185$$

$$= 2742.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$

$$= 248 / (2 \cdot 2743 / 295 - 1804 / (2 \cdot 102))$$

$$= 254.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / Rxs]$$

$$= 2 \cdot 165 [102 / 1804]$$

$$= 186.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(255, 187)$$

$$= 186.797 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (33887 + 381636 + 101133) / 295$$

$$= 171.508 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 185 \cdot 1804 / (2 \cdot 102)$$

$$= 163.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 172 - 163, 163)$$

$$= 179.6 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.80	N./mm ²
Local Primary Membrane Stress	PL	179.65	N./mm ²
Maximum Allowable Working Pressure	Pmax	186.80	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (59.5 + 117) / 59.5$$

$$= 2.966 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi / 2 \cdot \sqrt{((Rnc + Tn)^2 + (Rn + Tn)^2) / 2}$$

$$= \pi / 2 \cdot \sqrt{((59.5 + 117)^2 + (53.8 + 117)^2) / 2}$$

$$= 272.821 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 117.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(101133 \cdot 2.97, 1.5 \cdot 147(212 + 0), \pi / 4 \cdot 185 \cdot 53.8^2 \cdot 2.97^2)$$

$$= 37666.387 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 37666 / (273(0.49 \cdot 7.07 + 0.6 \cdot 102 + 0.49 \cdot 0))$$

$$= 20.938 < \text{ or } = \text{ to } 165.199 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

**Nozzle Neck to Flange Weld, Curve: B**

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-61 °C
User entered Minimum Design Temperature	:	-12 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	-13 °C
Temperature reduction per Graph 3.13	:	48 °C
The computed stress ratio is	:	0.437

The computed thickness ratio [sr]:

= (tr • E*)/(tnom - ci - cext)

= (5.68 • 1)/(16 - 3 - 0)

= 0.437

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A**Warning:**

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..

This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -61 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -34 °C

The Drop for this Nozzle is : 92.9419 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1008.4178 mm.

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**Input, Nozzle Desc: T2100****From: 170**

Pressure for Reinforcement Calculations	P	184.721	bars
Temperature for Internal Pressure	Temp	285	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.20	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	3501.00	mm.
Head Finished (Minimum) Thickness	t	105.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1300.0000	mm.
User Entered Minimum Design Metal Temperature		-12.22	°C

Type of Element Connected to the Parent : Nozzle

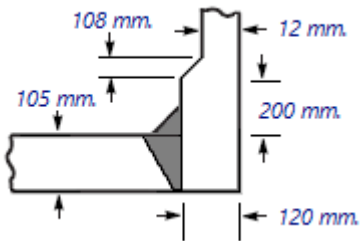
Material		SA-182 F11 +IT	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.20	N./mm ²
Allowable Stress At Ambient	Sna	184.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	200.0000	mm.
Height of Beveled Transition	L'	108.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	800.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	102.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		1500	
Flange Grade		GR 1.9	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: T2100

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	4.000	in.
Actual Thickness Used in Calculation	0.472	in.



Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 108 \cdot (\exp(185/(147 \cdot 1)) - 1) + 3 + 0$$

$$= 7.19 + 3 + 0 = 10.2 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147 \cdot 1 \cdot \ln((2 \cdot 9 + 108) / 108) - 0$$

$$= 228 - 0 = 228 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147/165, 1]$$

$$= 0.891$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 120 - 3$$

$$= 117.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12 - 3$$

$$= 9.000 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 3501/2 + 3$$

$$= 1753.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1753 \cdot 102}, 2 \cdot 53.8)$$

$$= 107.600 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((102 + 0)/117)^{0.35}, 1)$$

$$= \min(0.95, 1)$$

$$= 0.953$$



Intermediate value [Cp]:
= 9999999.000

Effective Nozzle Wall Length Outside the Vessel [LH]:
= $\min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$
= $\min(102 + 0 + 0.95 \cdot \sqrt{(53.8 \cdot 117)}, 800 + 102)$
= 177.619 mm.

Effective Vessel Thickness [teff]:
= $t + ((A5 \cdot frp) / LR)$
= $102 + ((0 \cdot 1) / 108)$
= 102.000 mm.

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
= $t \cdot LR$
= $102 \cdot 108$
= 109.752 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
= $t_n \cdot LH$
= $117 \cdot 178$
= 207.814 cm²

Area Contributed by the Outside Fillet Weld [A41]:
= $0.5 \cdot Leg^2$
= $0.5 \cdot 10^2$
= 0.500 cm²

The total area contributed by A1 through A43 [AT]:
= $A1 + frn(A2 + A3) + A41 + A42 + A43$
= $110 + 0.89(208 + 0) + 0.5 + 0 + 0$
= 295.424 cm²

Nozzle Radius for Force Calculation [Rxn]:
= $t_n / \ln[1 + t_n/R_n]$
= $117 / \ln[1 + 117/53.8]$
= 101.279 mm.

Shell Radius for Force Calculation [Rxs]:
= $teff / \ln[1 + teff/Reff]$
= $102 / \ln[1 + 102/1753]$
= 1804.020 mm.

Allowable Local Primary Membrane Stress [Sallow]:
= $1.5 \cdot S \cdot E$
= $1.5 \cdot 165 \cdot 1$
= 247.8 N./mm²

Force acting on the Nozzle [fN]:
= $P \cdot Rxn \cdot LH$
= $185 \cdot 101 \cdot 178$
= 33886.3 Kgf

Force acting on the Shell [fS]:
= $P \cdot Rxs \cdot (LR + t_n) / 2$
= $185 \cdot 1804 \cdot (108 + 117) / 2$
= 381623.1 Kgf

Discontinuity Force from Internal Pressure [fY]:
= $(P \cdot Rxs \cdot R_{nc}) / 2$
= $(185 \cdot 1804 \cdot 80.2) / 2$
= 136215.1 Kgf

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:



$$= (fS + fY + fN) / P$$

$$= (381623 + 136215 + 33886) / 185$$

$$= 2928.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$

$$= 248 / (2 \cdot 2929 / 295 - 1804 / (2 \cdot 102))$$

$$= 225.6 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / Rxs]$$

$$= 2 \cdot 165 [102 / 1804]$$

$$= 186.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(226, 187)$$

$$= 186.797 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (33886 + 381623 + 136215) / 295$$

$$= 183.149 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 185 \cdot 1804 / (2 \cdot 102)$$

$$= 163.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 183 - 163, 163)$$

$$= 202.9 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.80	N./mm ²
Local Primary Membrane Stress	PL	202.94	N./mm ²
Maximum Allowable Working Pressure	Pmax	186.80	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (80.2 + 117) / 80.2$$

$$= 2.459 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi / 2 \cdot \sqrt{((Rnc + Tn)^2 + (Rn + Tn)^2) / 2}$$

$$= \pi / 2 \cdot \sqrt{((80.2 + 117)^2 + (53.8 + 117)^2) / 2}$$

$$= 289.742 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 117.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(136215 \cdot 2.46, 1.5 \cdot 147(212 + 0), \pi / 4 \cdot 185 \cdot 53.8^2 \cdot 2.46^2)$$

$$= 25902.369 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 25902 / (290(0.49 \cdot 7.07 + 0.6 \cdot 102 + 0.49 \cdot 0))$$

$$= 13.558 < \text{ or } = \text{ to } 165.199 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

**Nozzle Neck to Flange Weld, Curve: B**

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-39 °C
User entered Minimum Design Temperature	:	-12 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	-18 °C
Temperature reduction per Graph 3.13	:	21 °C
The computed stress ratio is	:	0.631

The computed thickness ratio [sr]:

= (tr • E*)/(tnom - ci - cext)

= (5.68 • 1)/(12 - 3 - 0)

= 0.631

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A**Warning:**

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -39 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -34 °C

The Drop for this Nozzle is : 223.0753 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1174.7549 mm.

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**Noted issues for the current model:**

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: One or more nozzles do not meet proper Spacing Requirements!**

Please review all reports carefully!

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: